MOVING CHARGES AND MAGNETISM

Chapter Four
Class - XII

Manish Joshi

D.A.V. Lohaghat

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Learning Objectives:

- Concept of magnetic field, Oersted's experiment.

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Oersted's Experiment

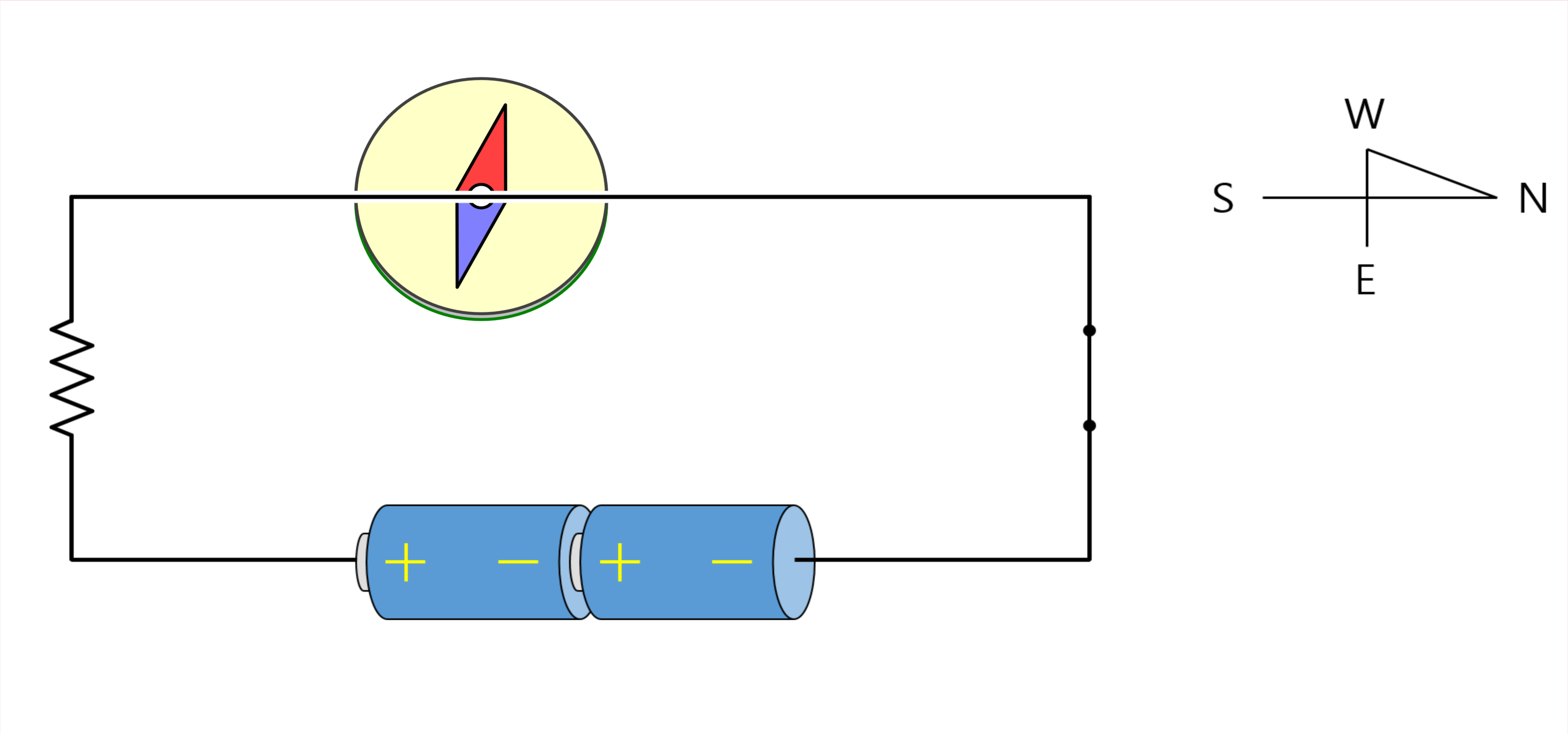

Figure: Oersted's Experiment Setup.
Source : Java Lab

- Danish physicist **Hans Christian Oersted accidentally discovered** that a **current in a straight wire caused** a noticeable **deflection** in a nearby **magnetic compass needle**.

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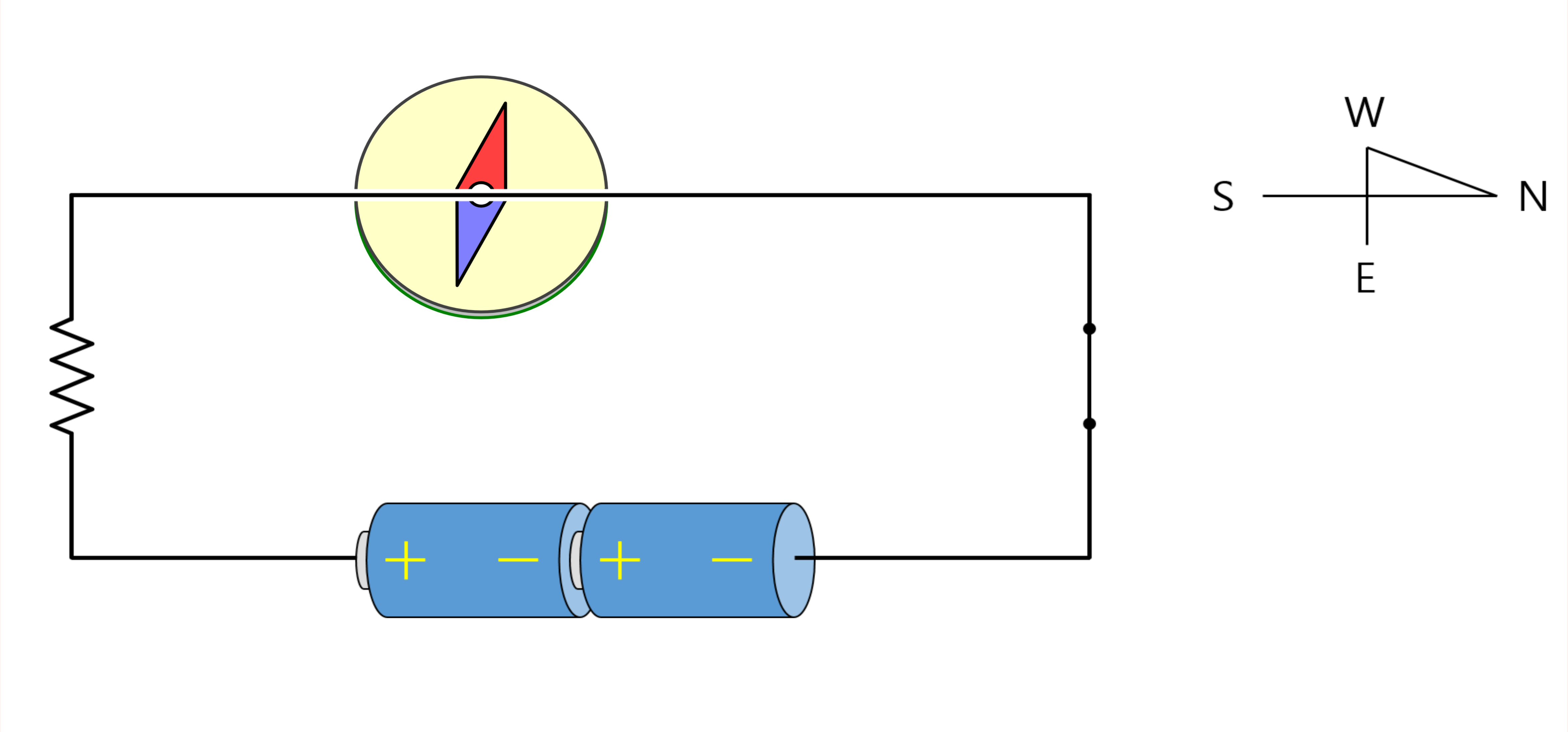

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- **Oersted experimented** with this phenomenon several times, **revealing a close relationship between electrical and magnetic phenomena**.

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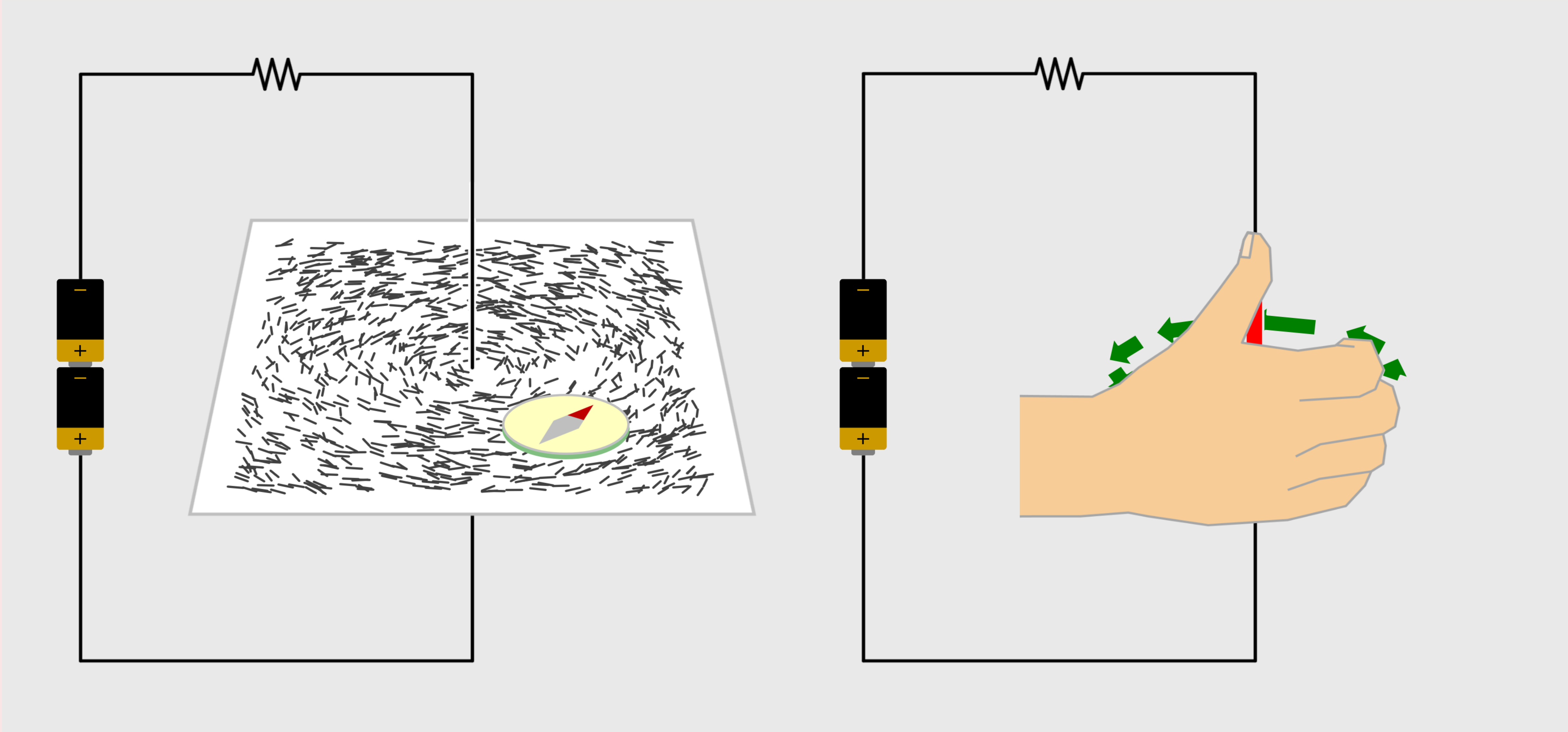

Figure: Magnetic Field around a Wire.
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Observations and Results:

- He found that the **alignment of the needle is tangential to an imaginary circle** which has the straight wire as its center and has its plane perpendicular to the wire.

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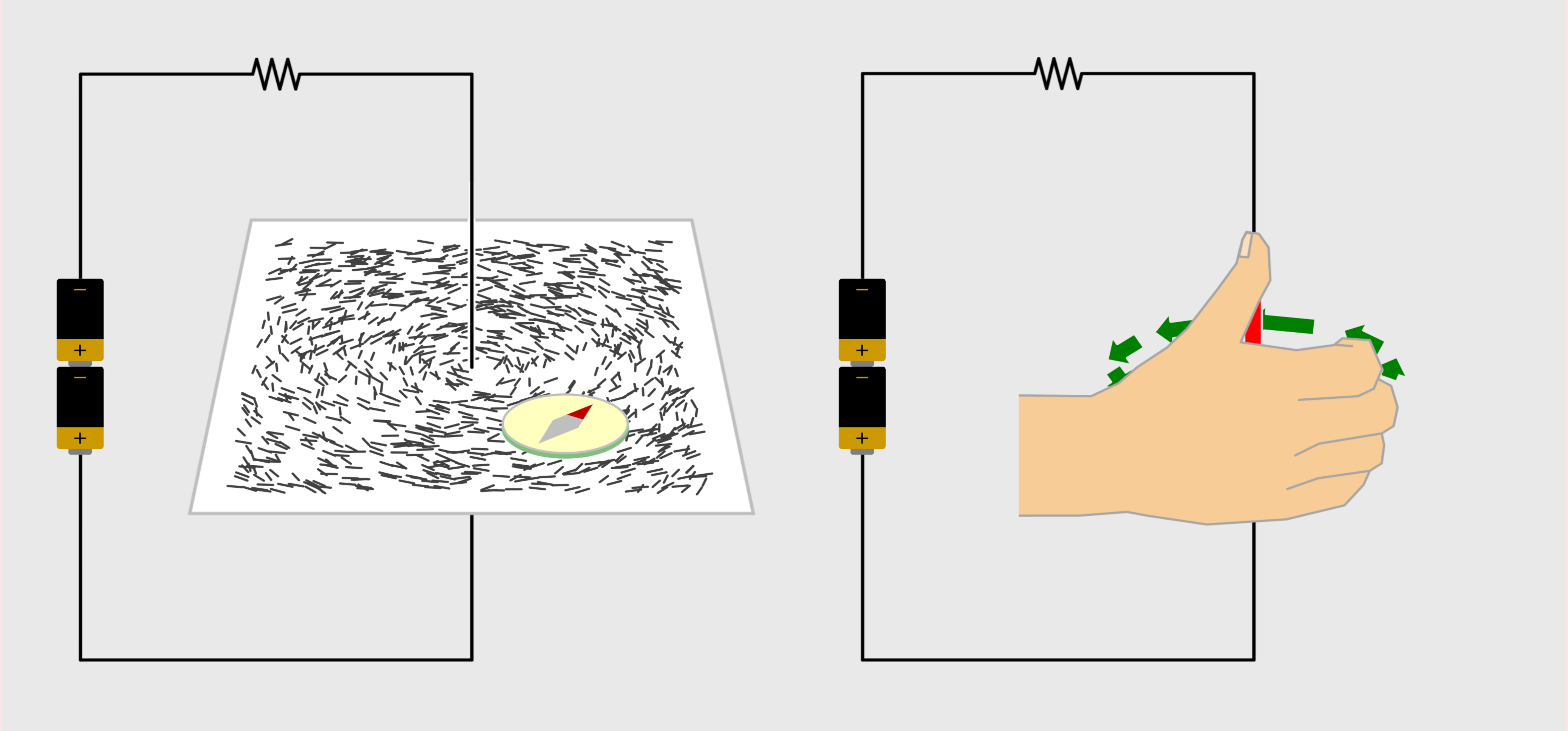

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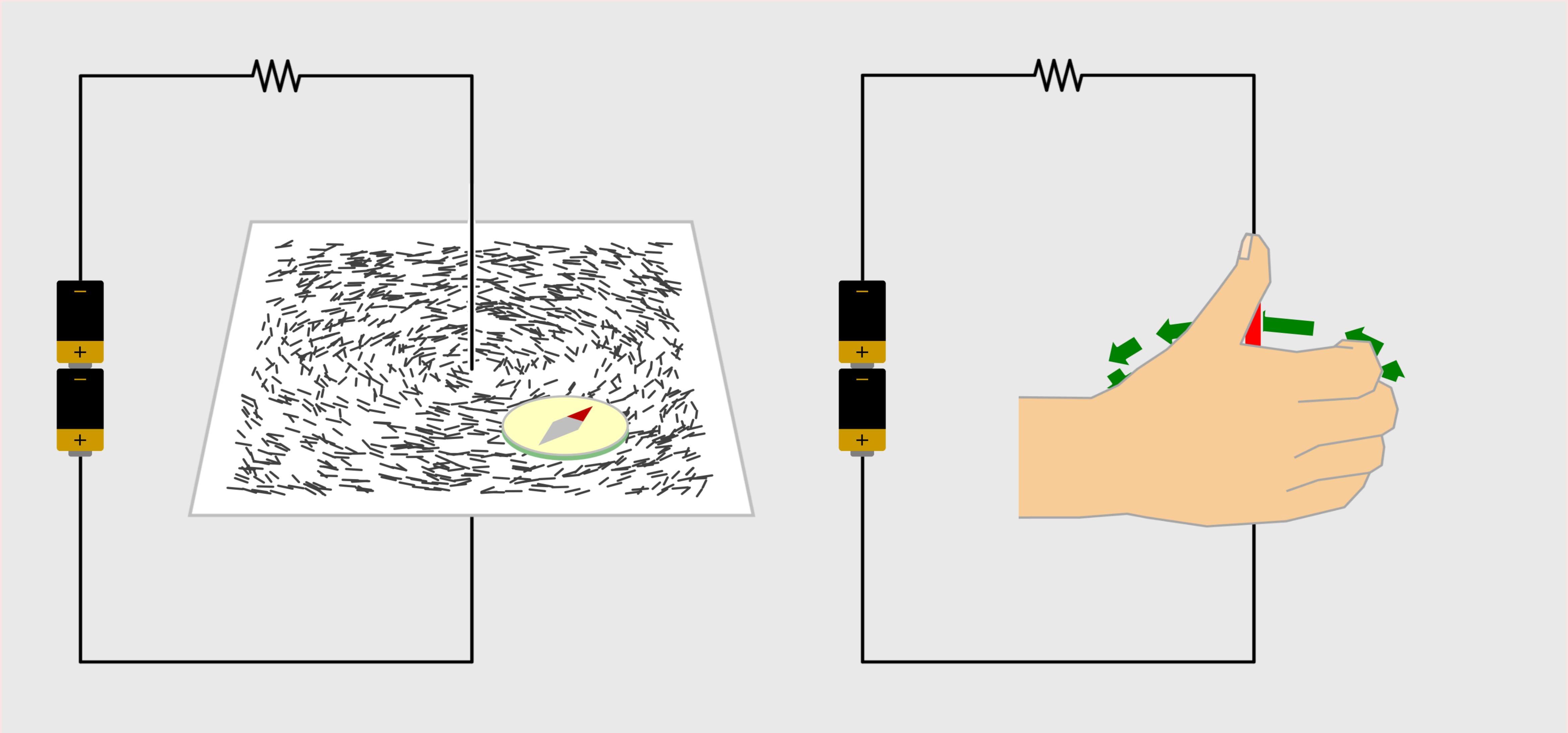

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- **Reversing** the direction of the **current reverses** the **orientation of the needle**.
- The **deflection increases on increasing the current** or **bringing the needle closer to the wire**.

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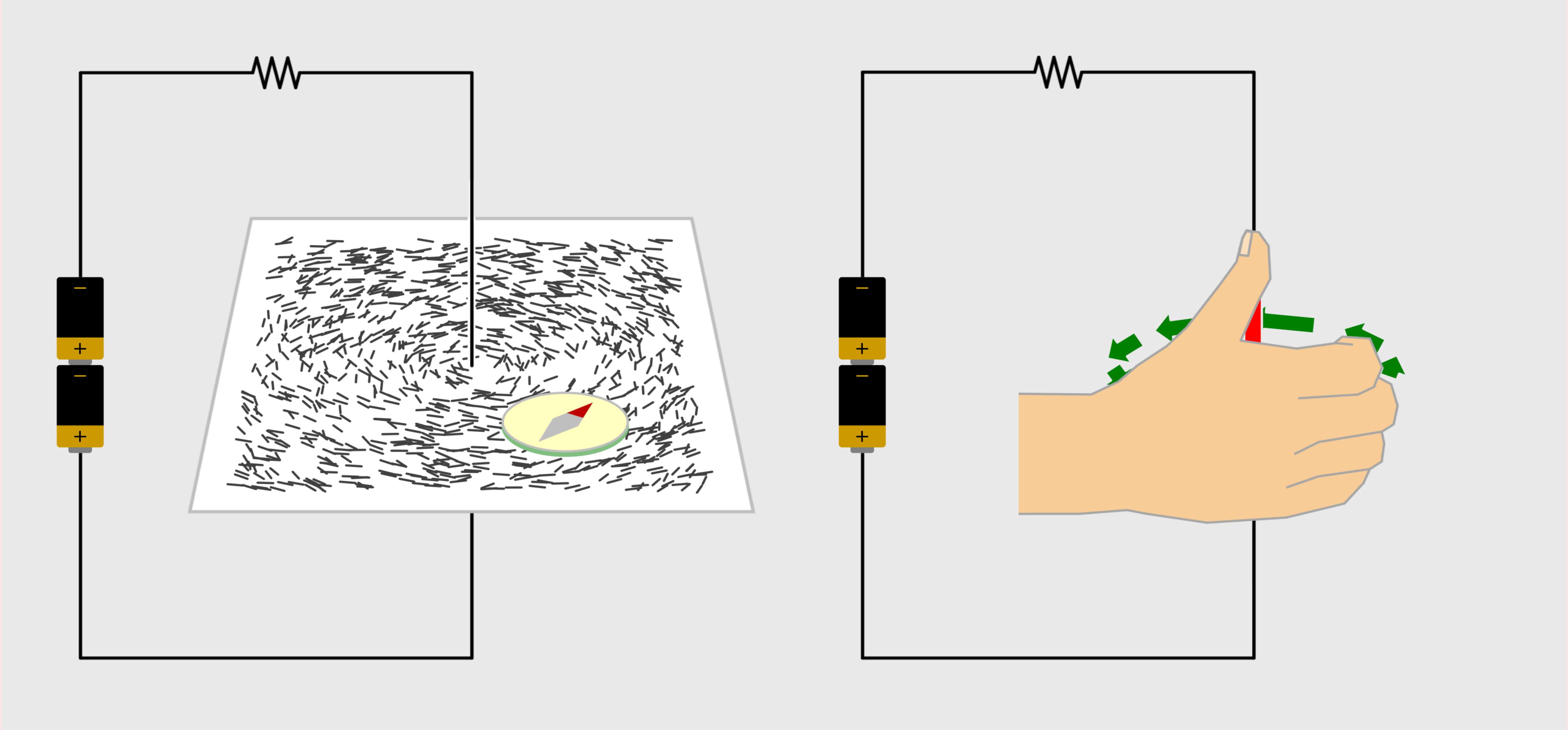

Figure: Magnetic Field around a Wire and its direction is give by right hand thumb rule. Source : Java Lab

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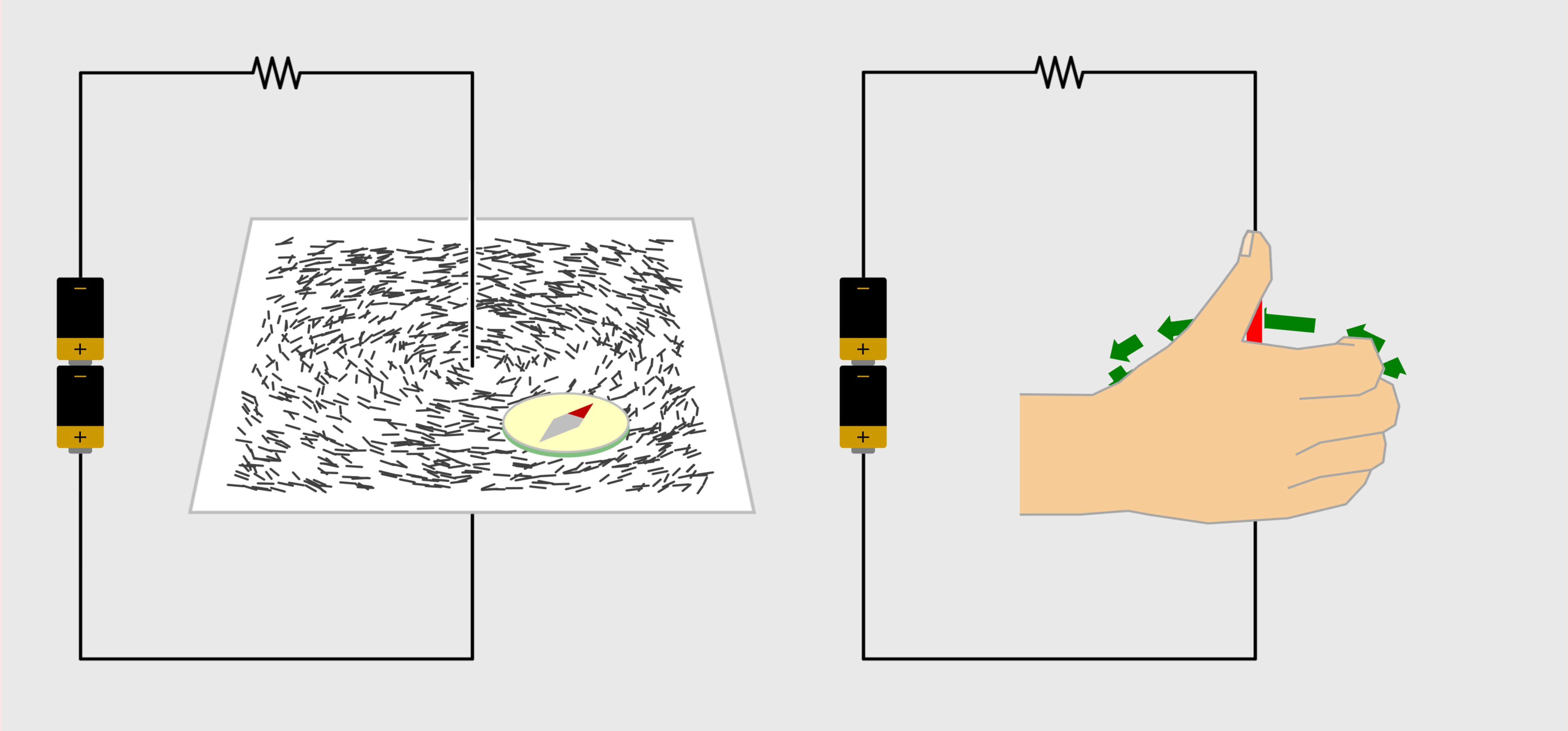

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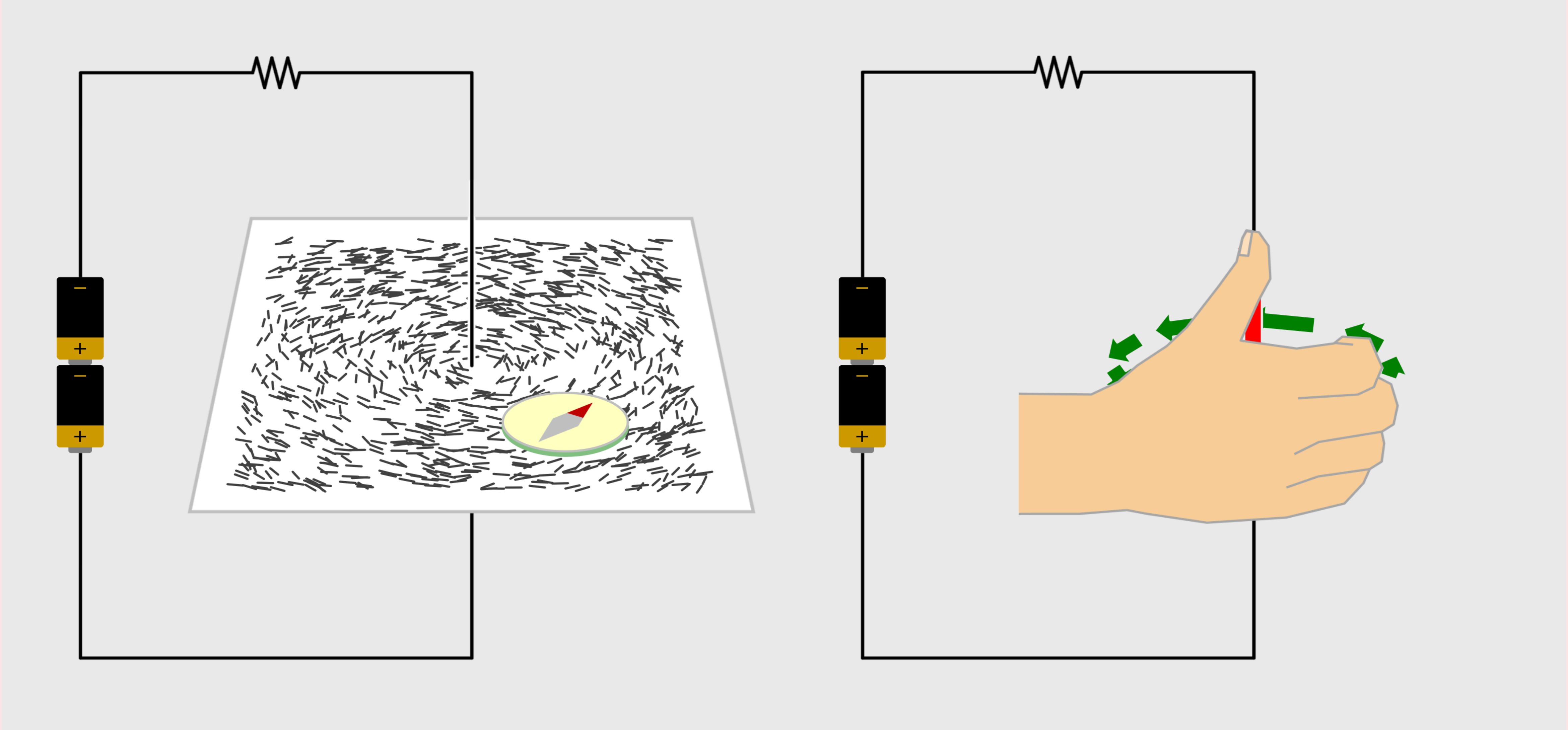

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- **Oersted concluded** that **moving charges or currents produced a magnetic field** in the surrounding space.

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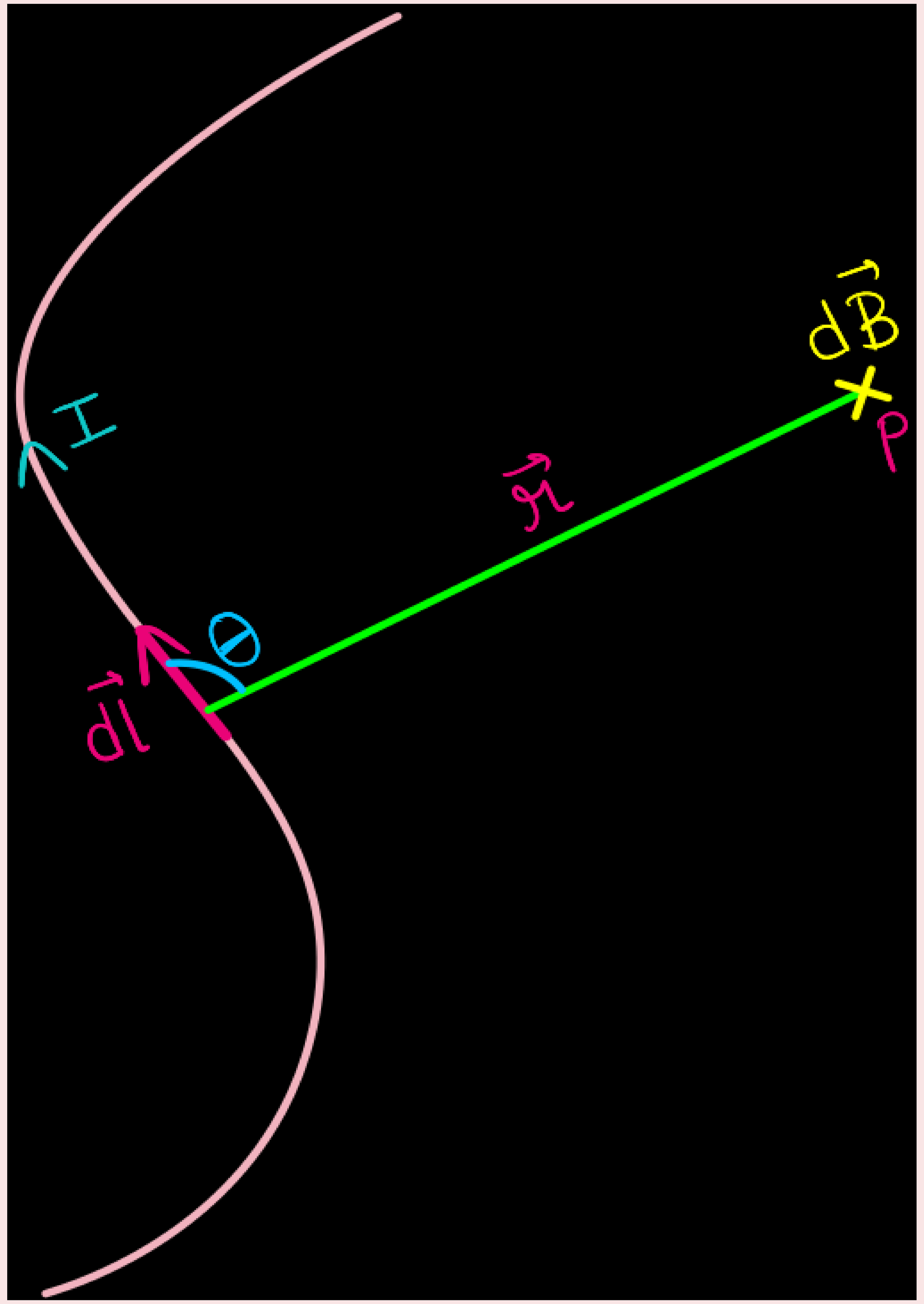

Magnetic Field Due to a Current Element

- Figure shows a finite conductor carrying **current I**.

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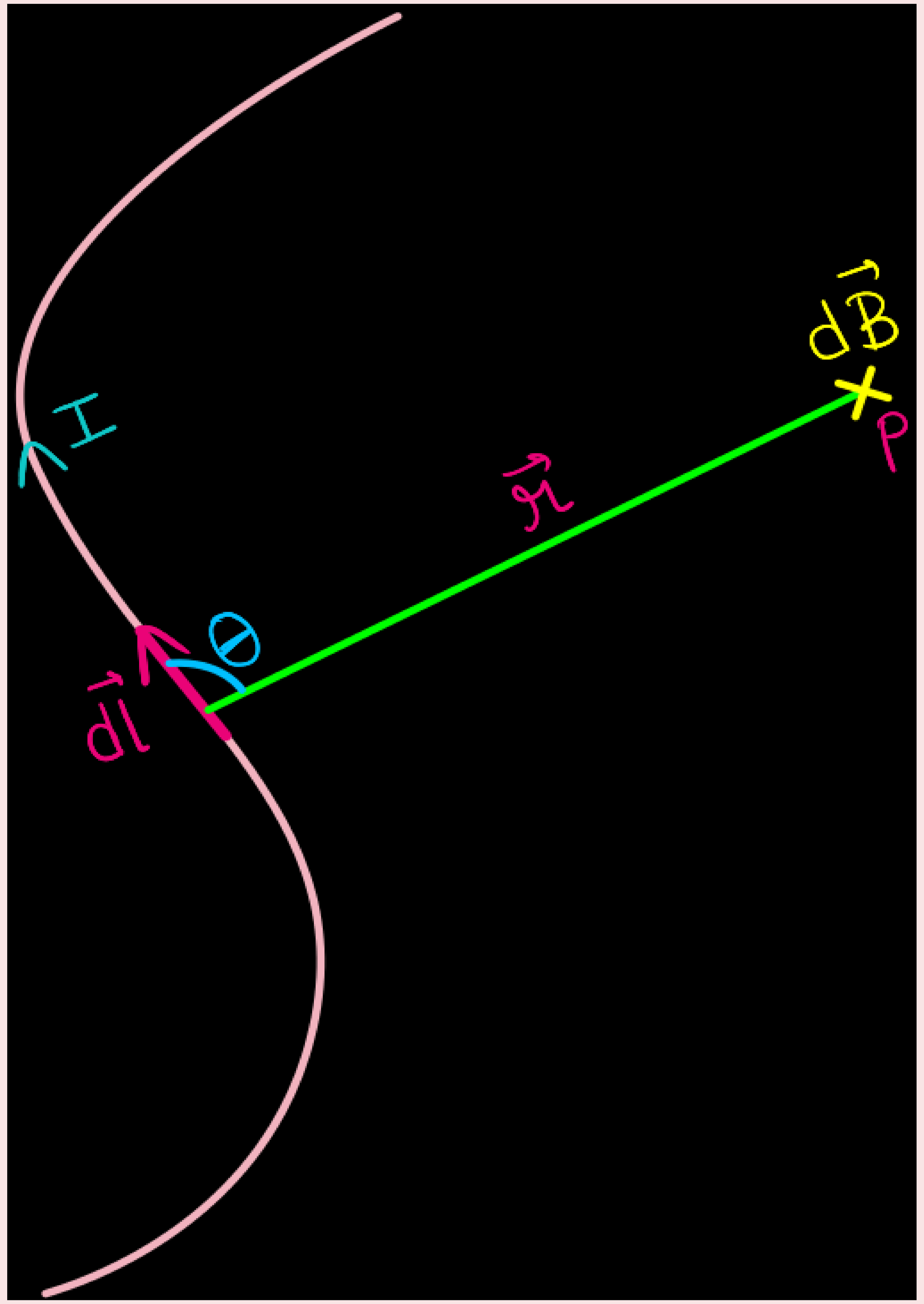

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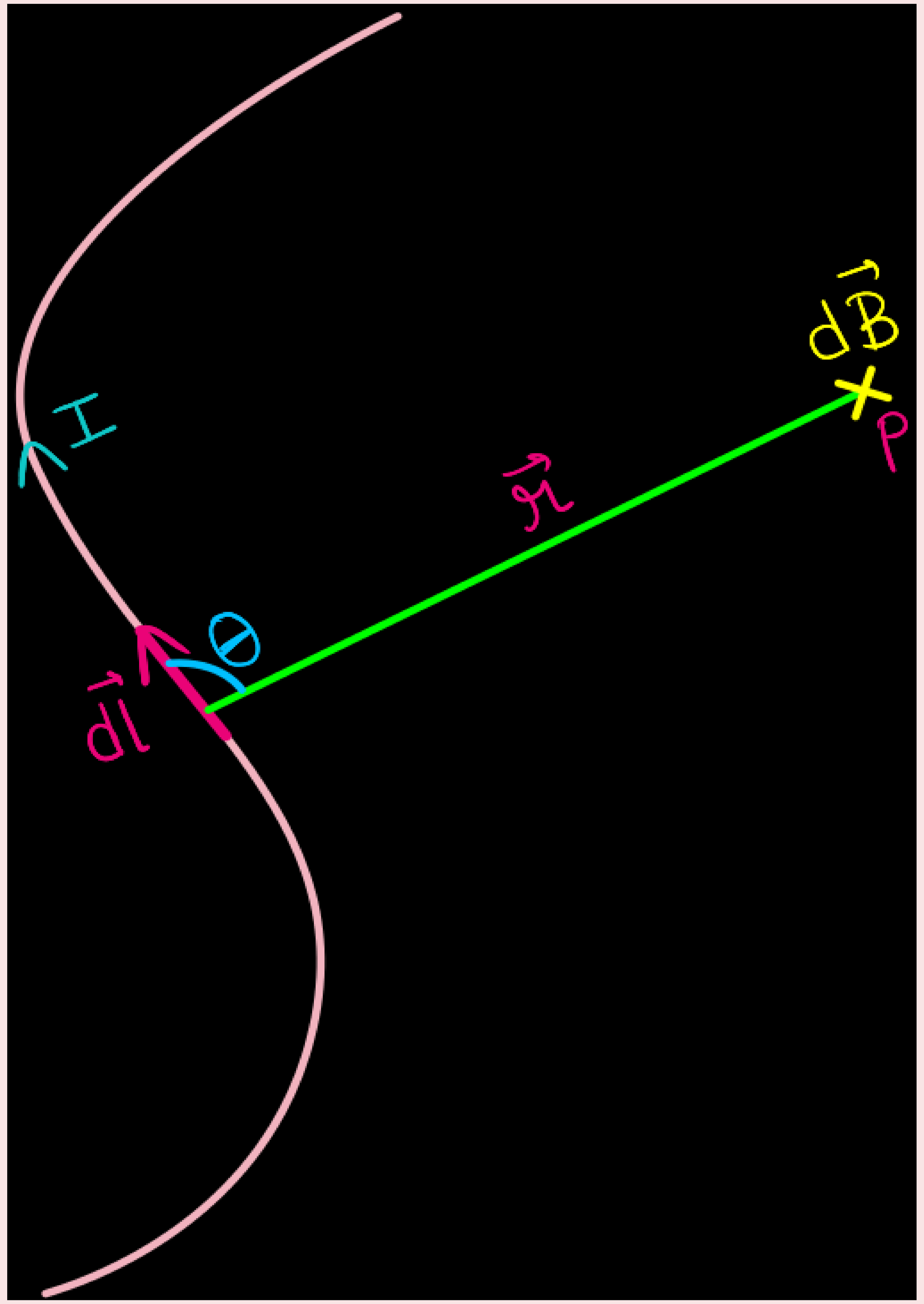

Magnetic Field Due to a Current Element

- Figure shows a finite conductor carrying **current I**.
- Consider an **infinitesimal element** $d\vec{l}$ of the conductor.
- The **magnetic field** $d\vec{B}$ due to this element is to be determined **at a point P** which is at a **distance** $\boldsymbol{r}$ from it. Let $\boldsymbol{\theta}$ be the **angle between** dl **and the displacement vector** $\boldsymbol{r}$.

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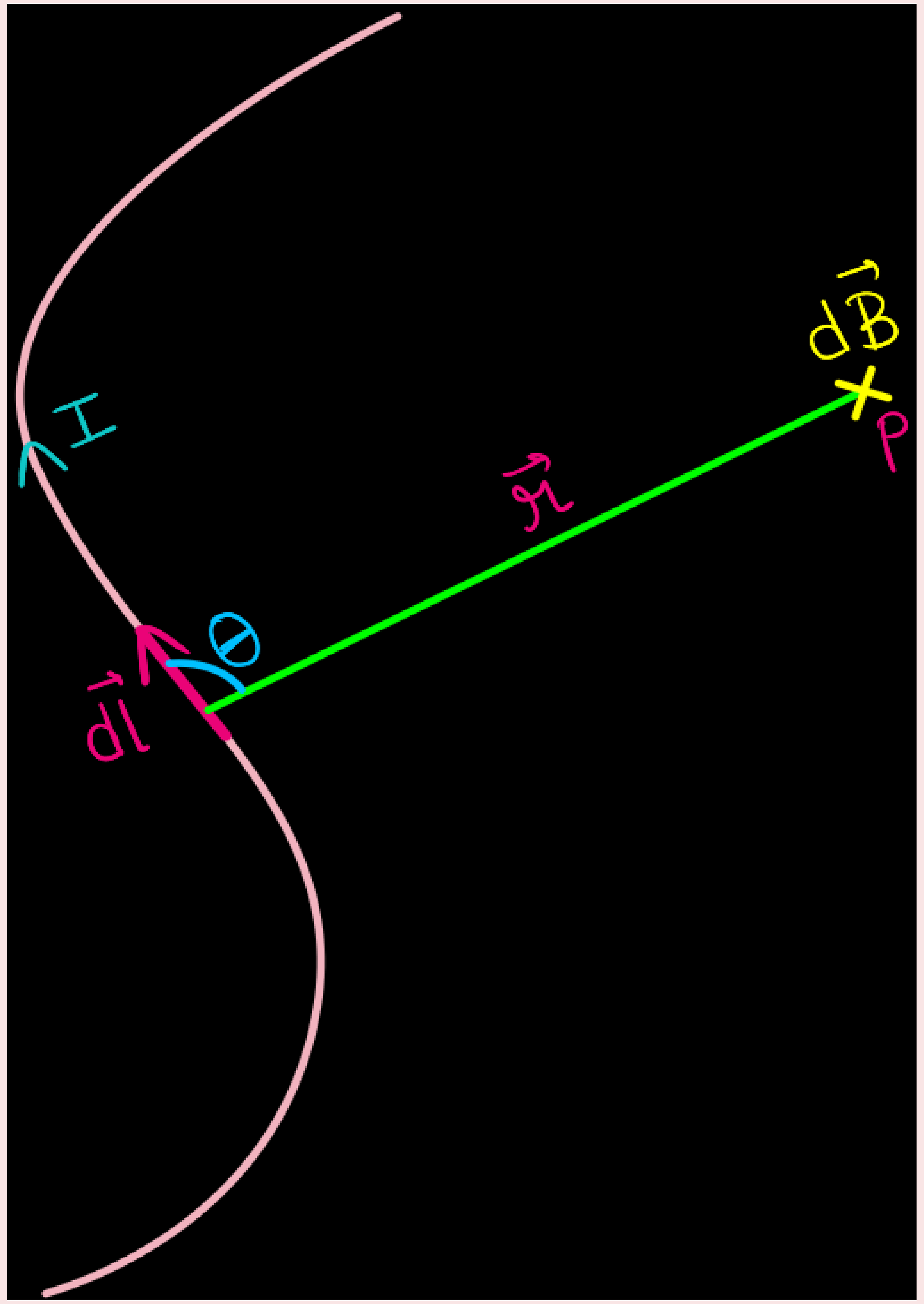

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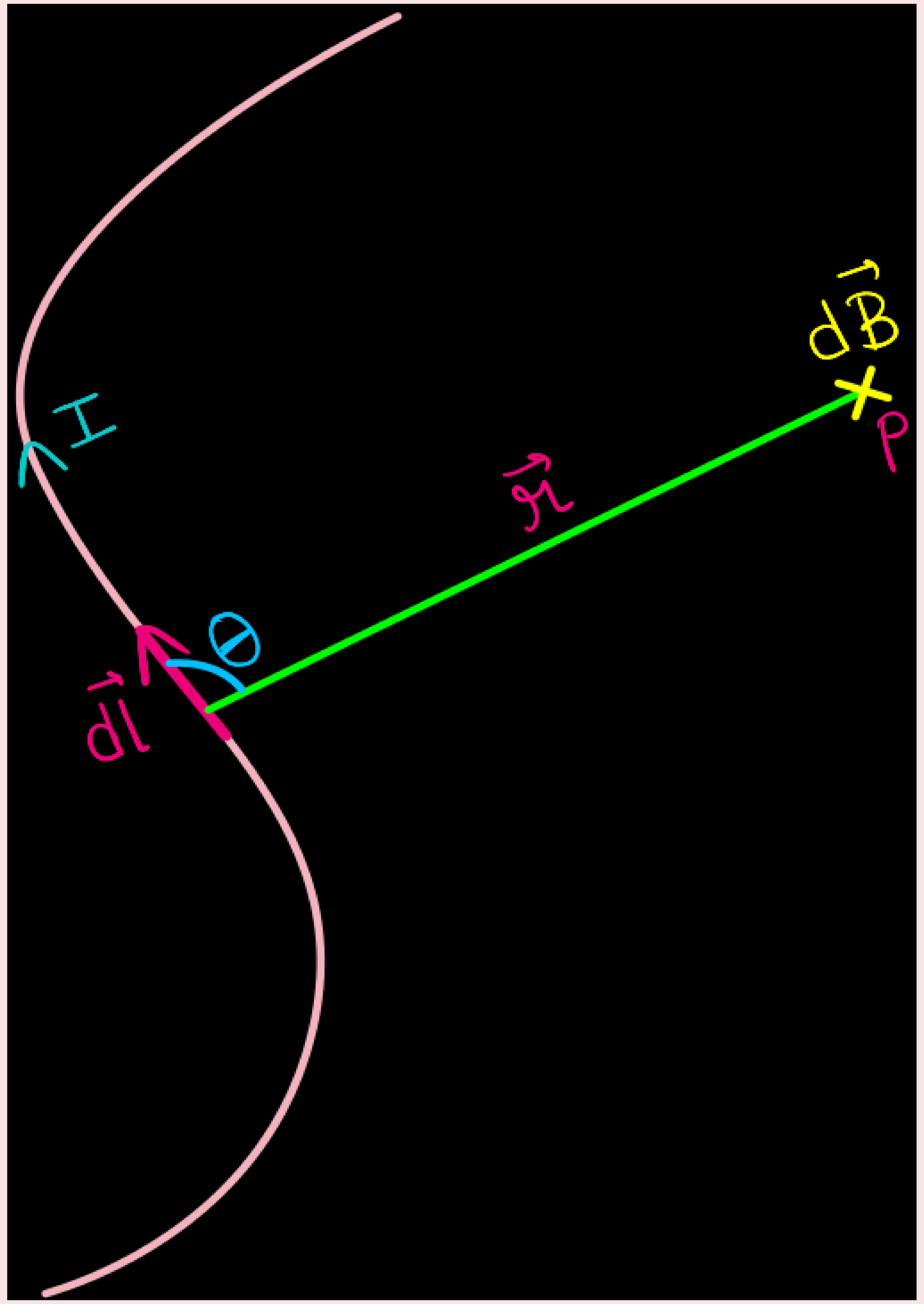

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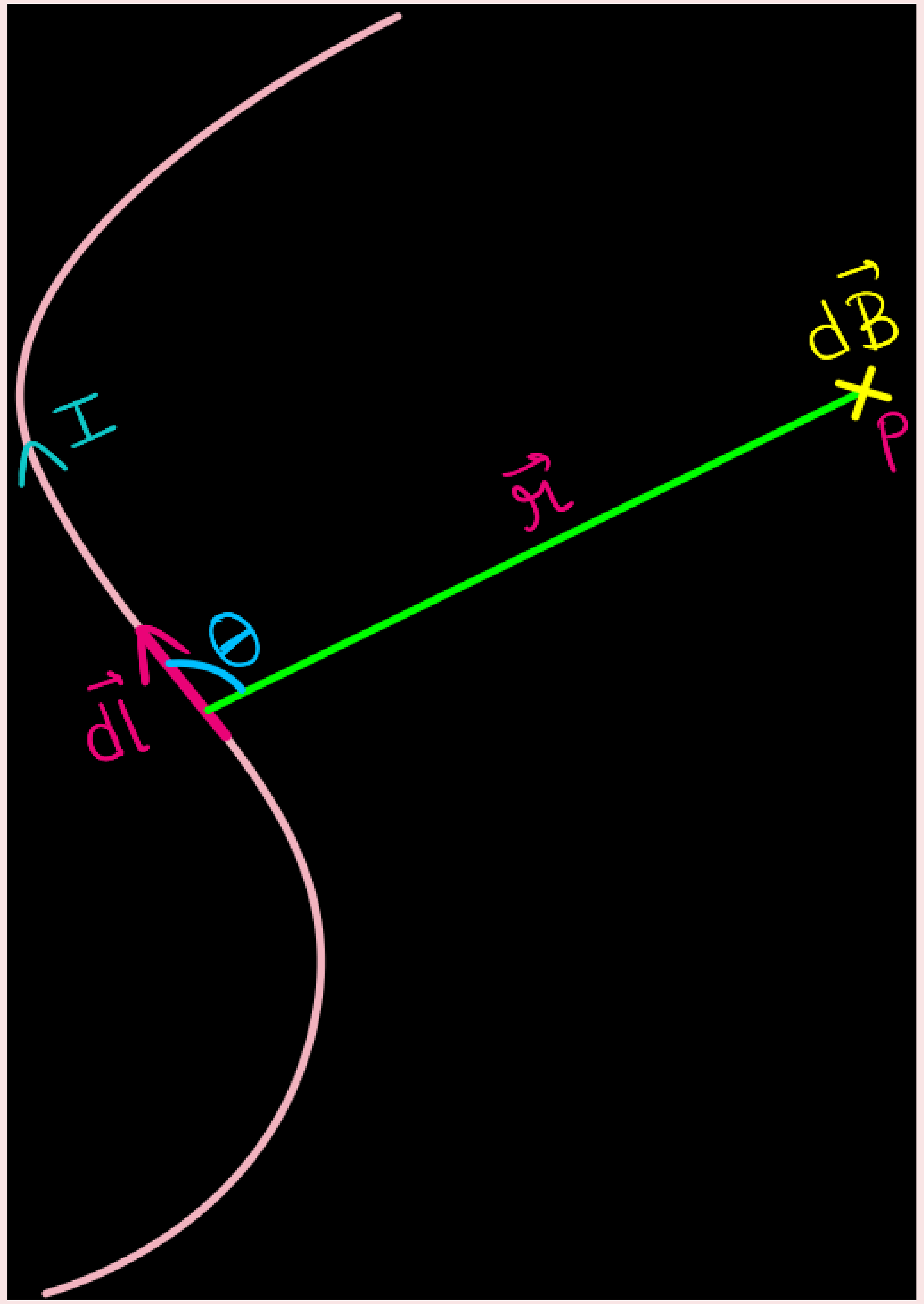

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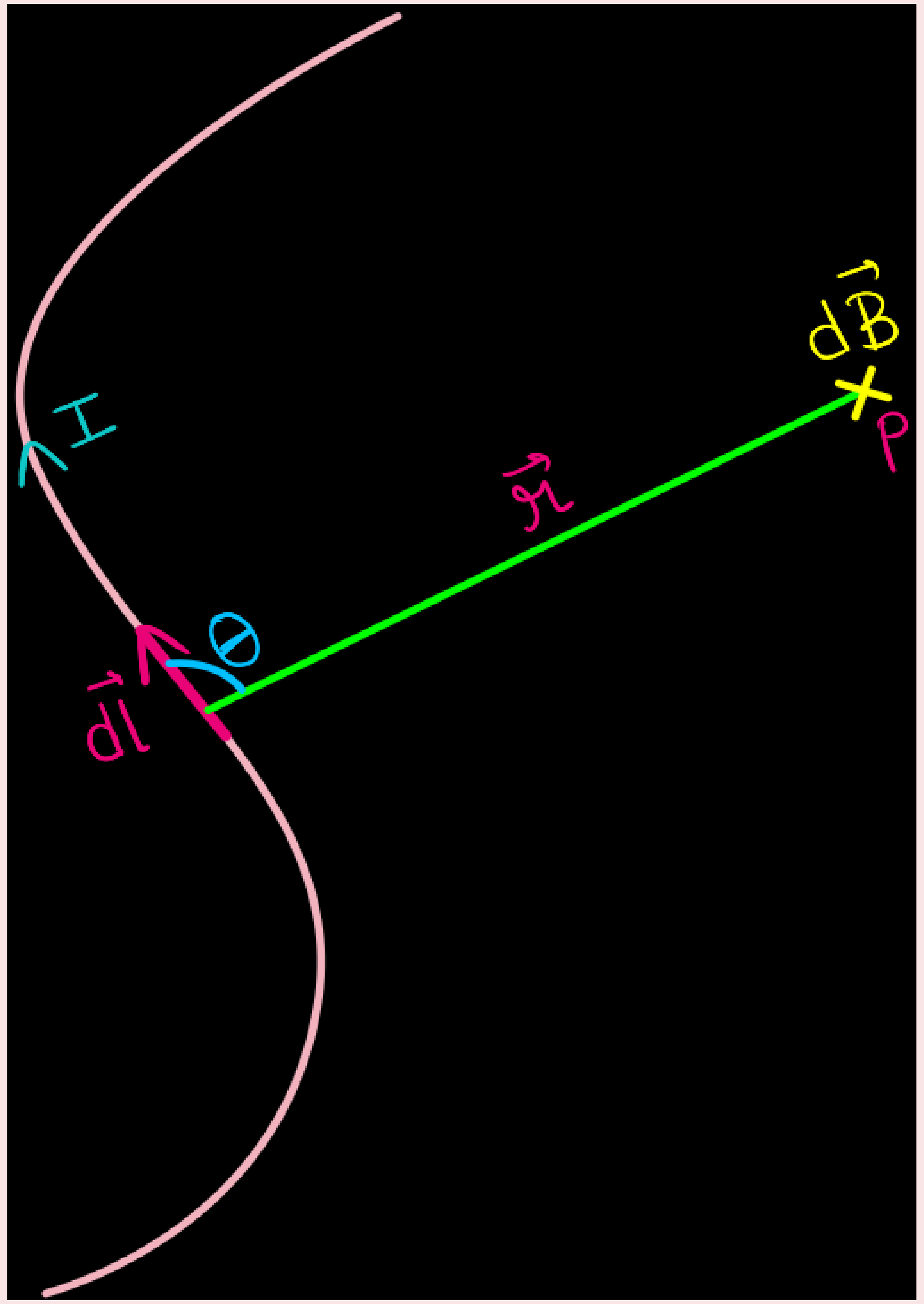

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  3. **Inversely proportional** to the **square of the distance r**
  $$\left(d\vec{B} \propto \frac{1}{r^2}\right).$$

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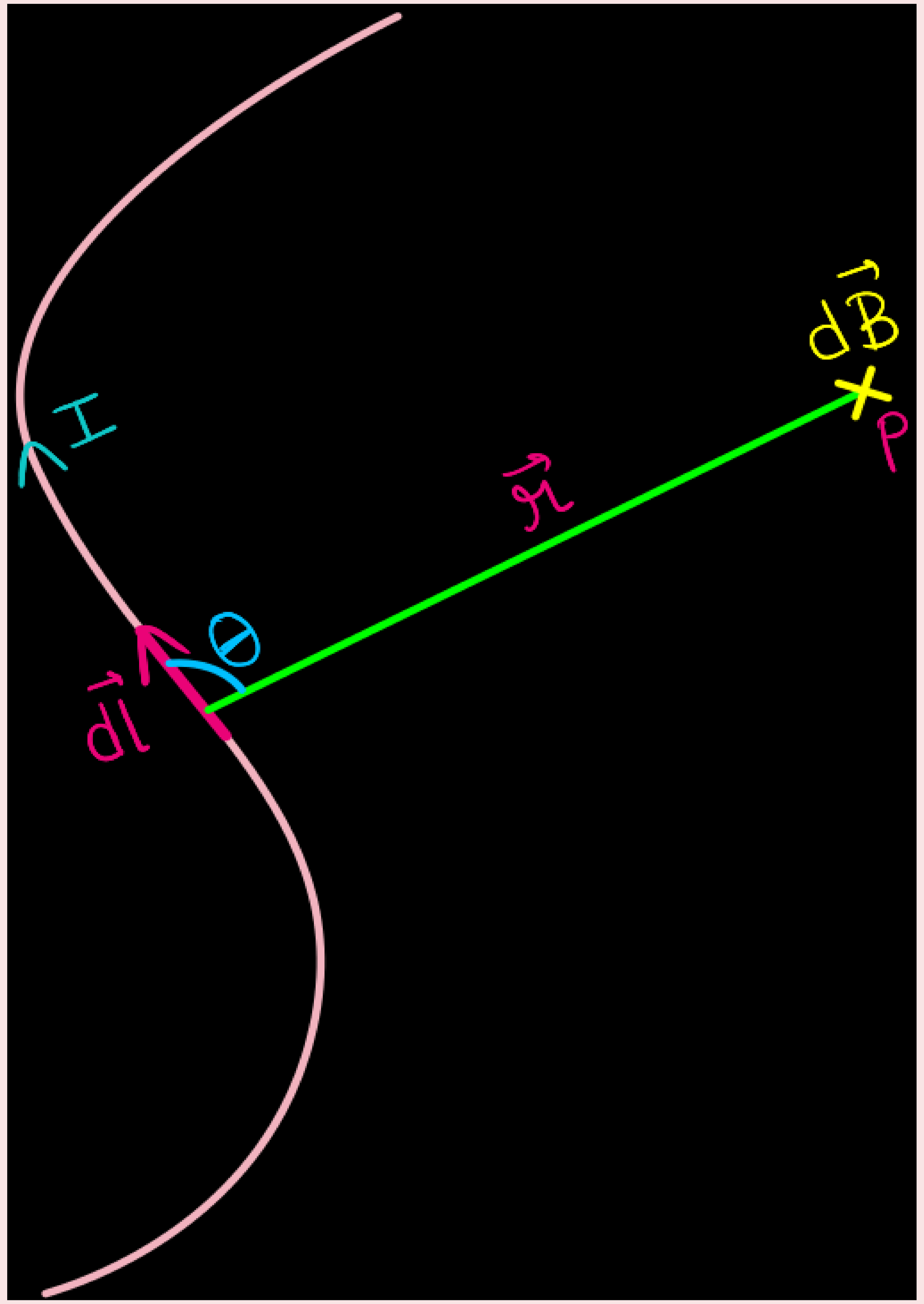

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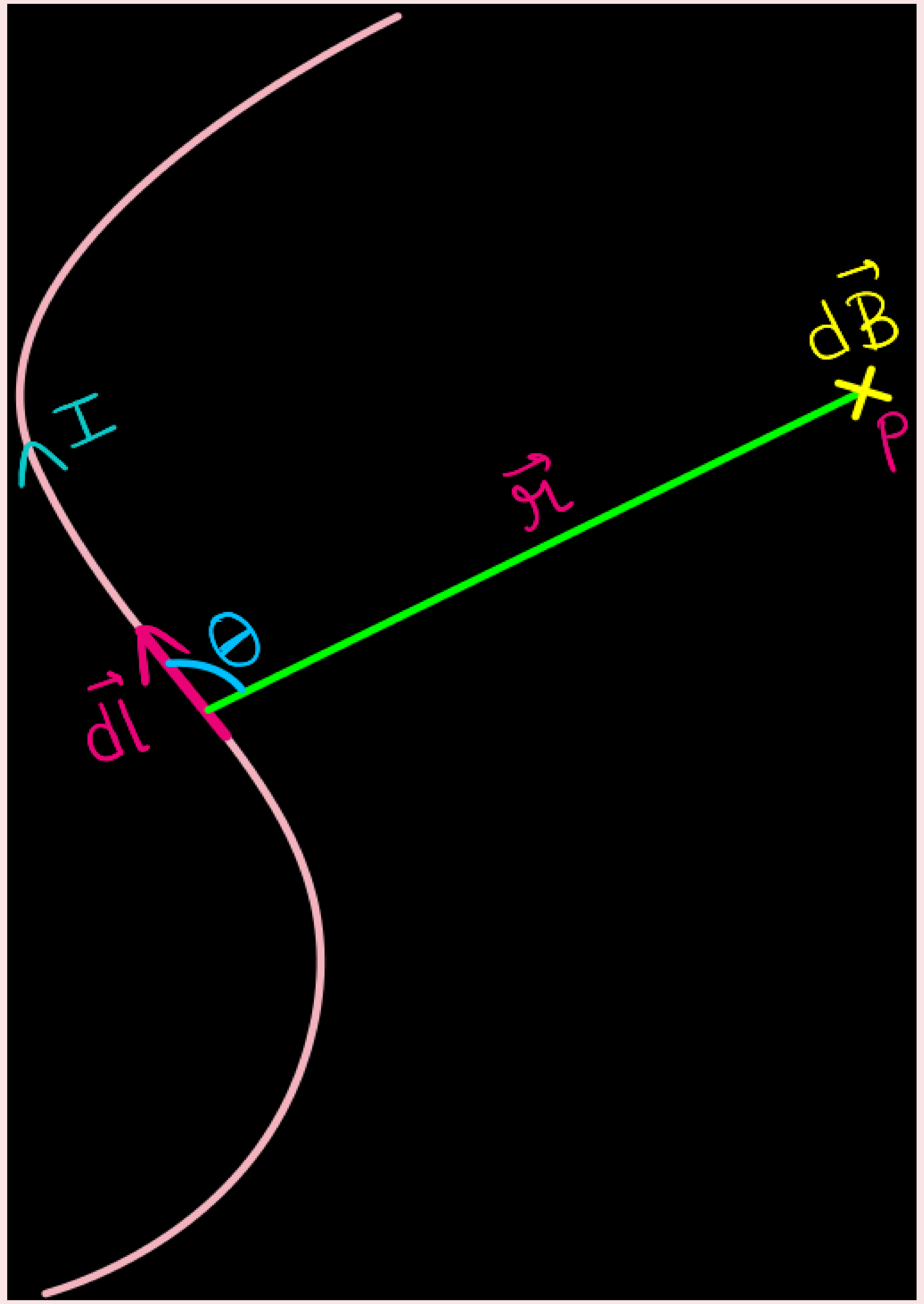

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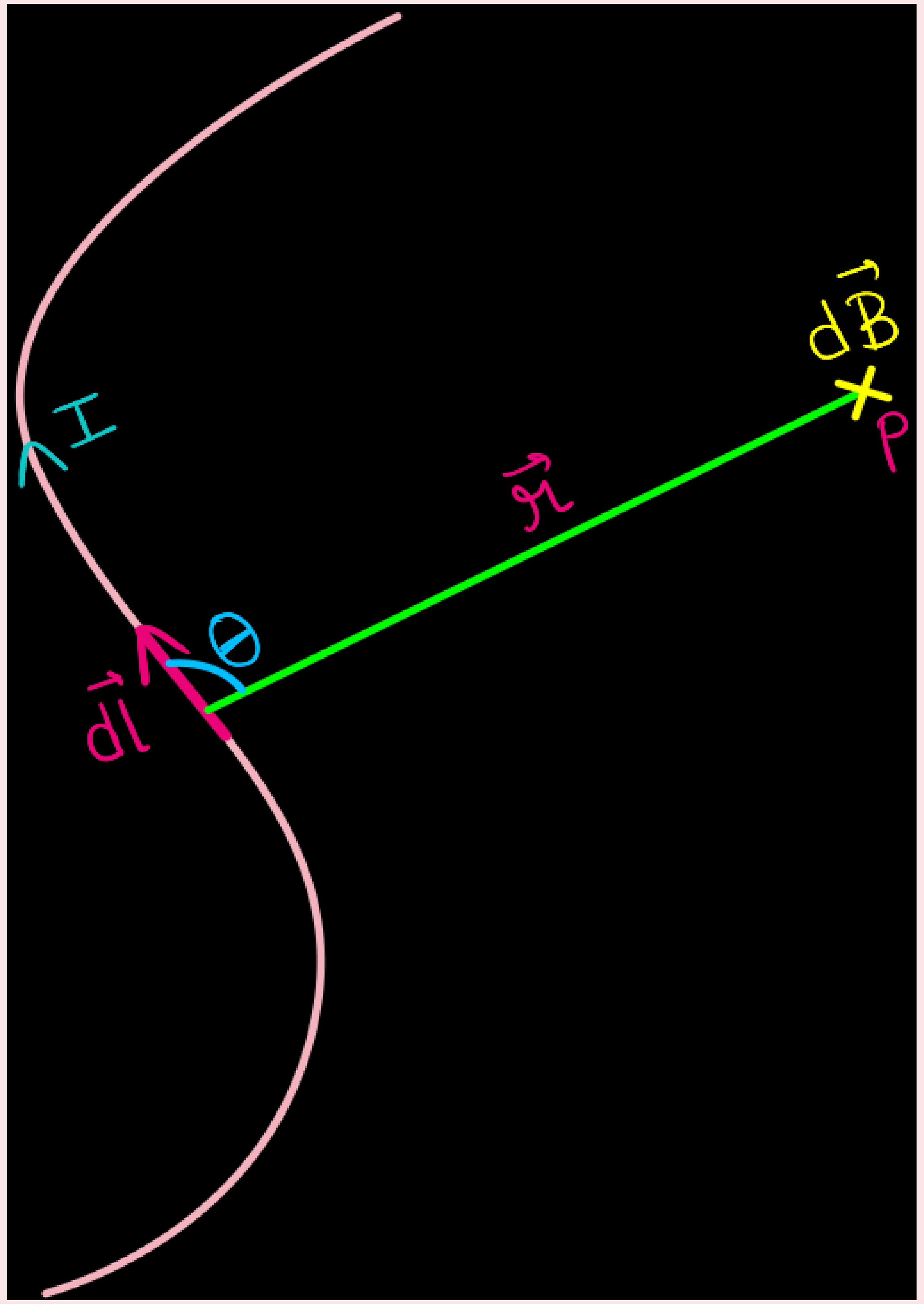

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$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\ d\vec{l} \times \hat{r}}{r^2}}$$

or

$$\boxed{|d\vec{B}| = \frac{\mu_0}{4\pi}\frac{I\ dl\ \sin\theta}{r^2}}$$

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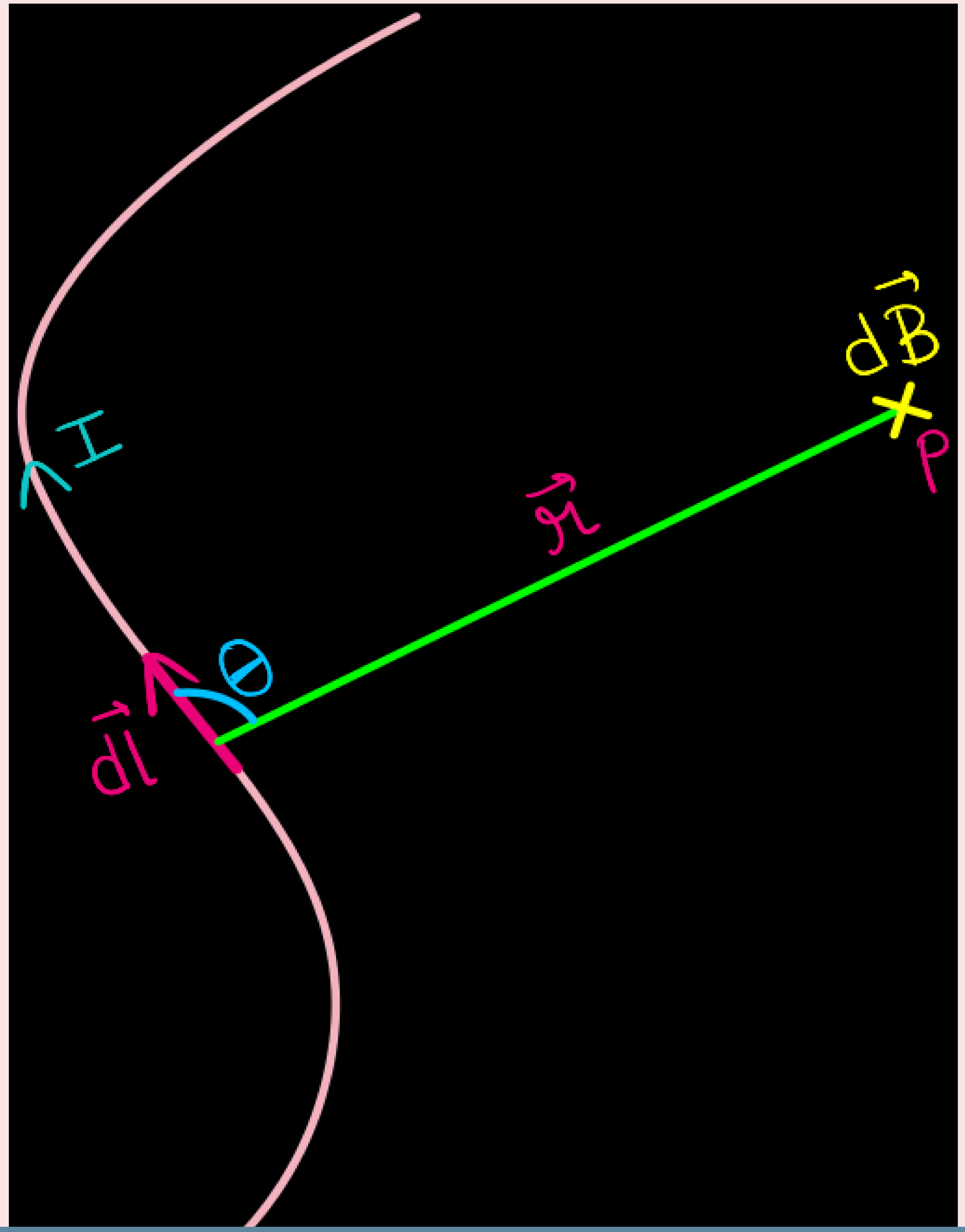

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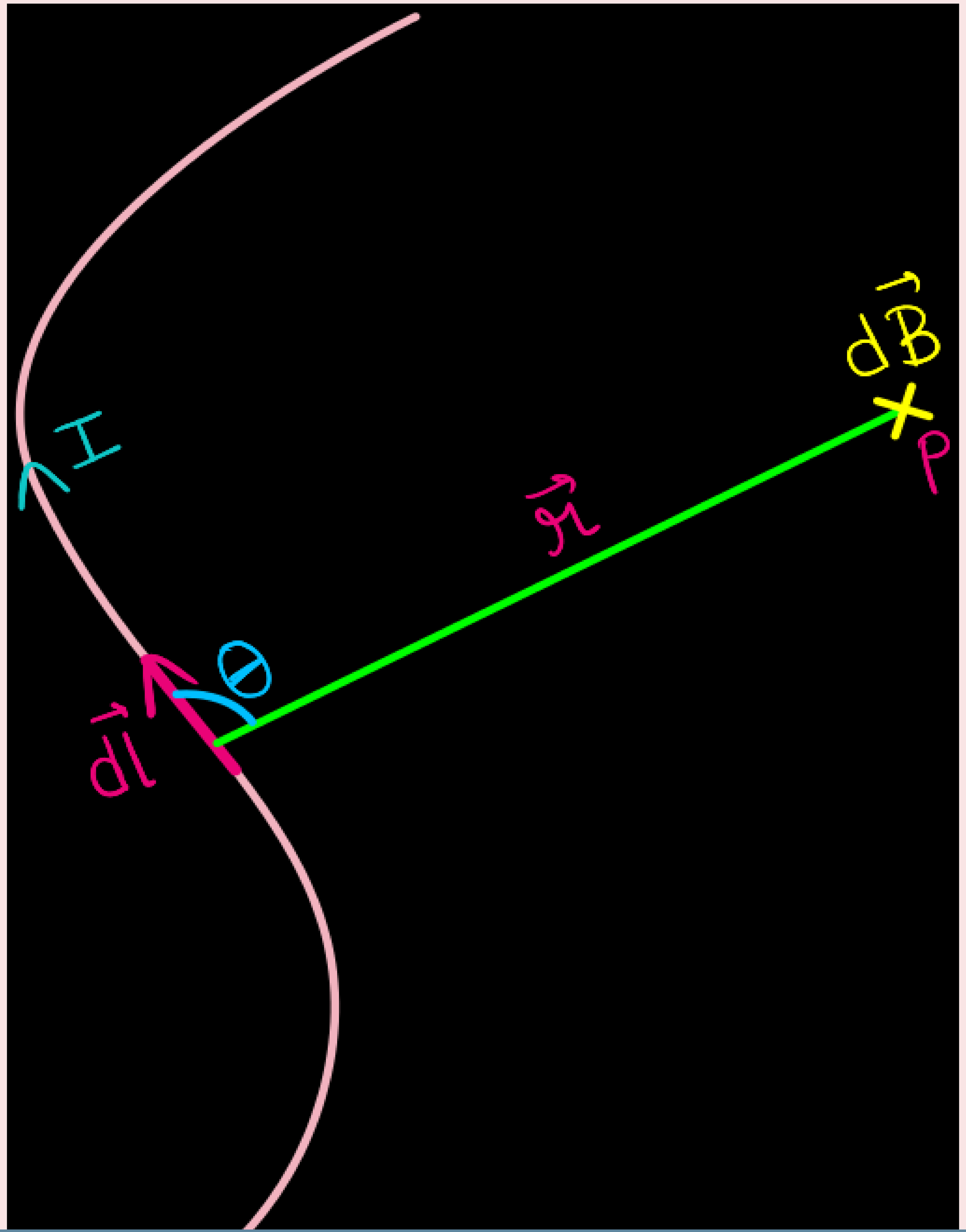

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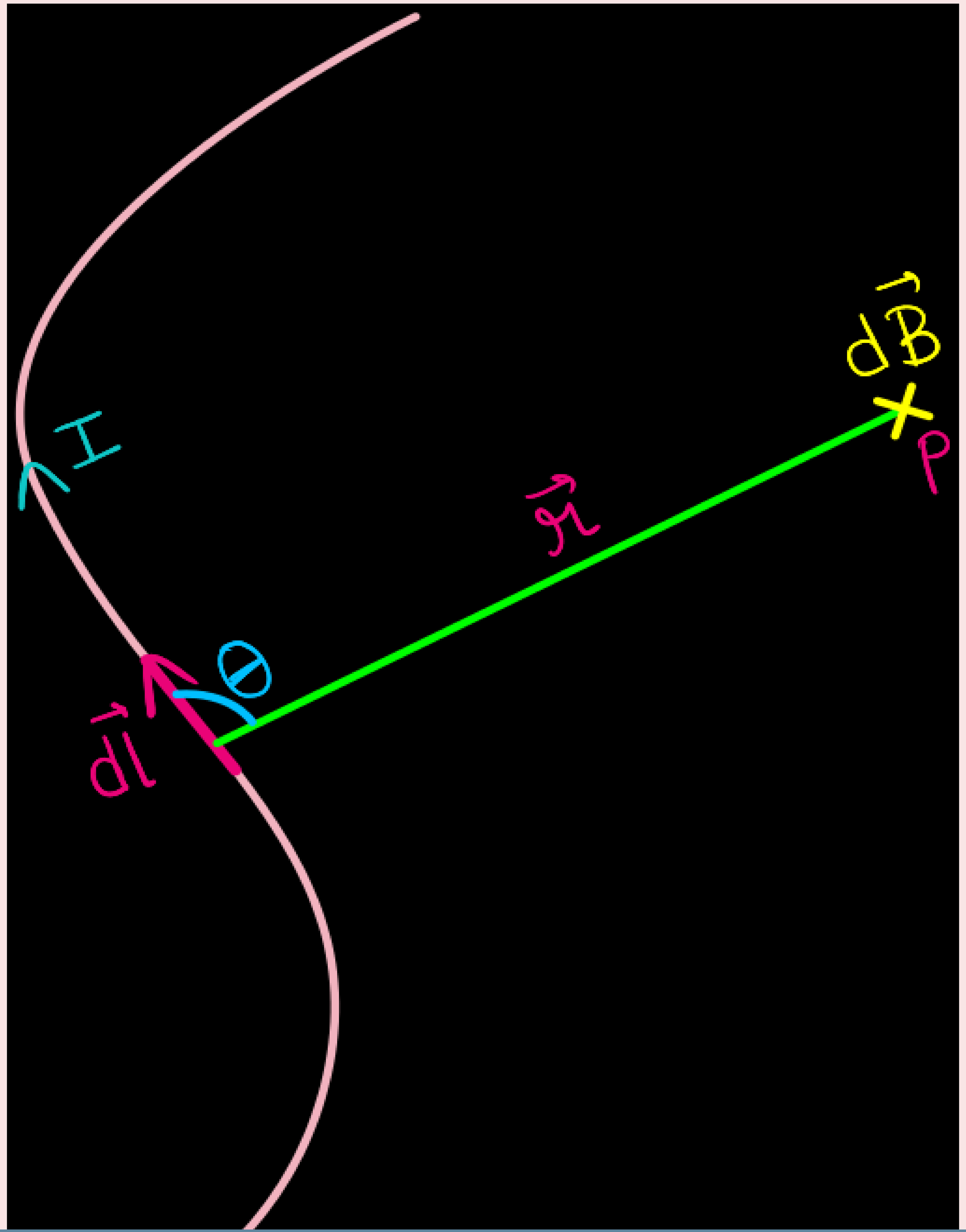

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- We call $\boldsymbol{\mu_0}$ the **permeability of free space (or vacuum)**.

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- There is an **angle dependence in the Biot-Savart law** which is **not present in the electrostatic case**.

Relation between Permittivity of free space (ϵ_0) and the permeability of free space (μ_0)

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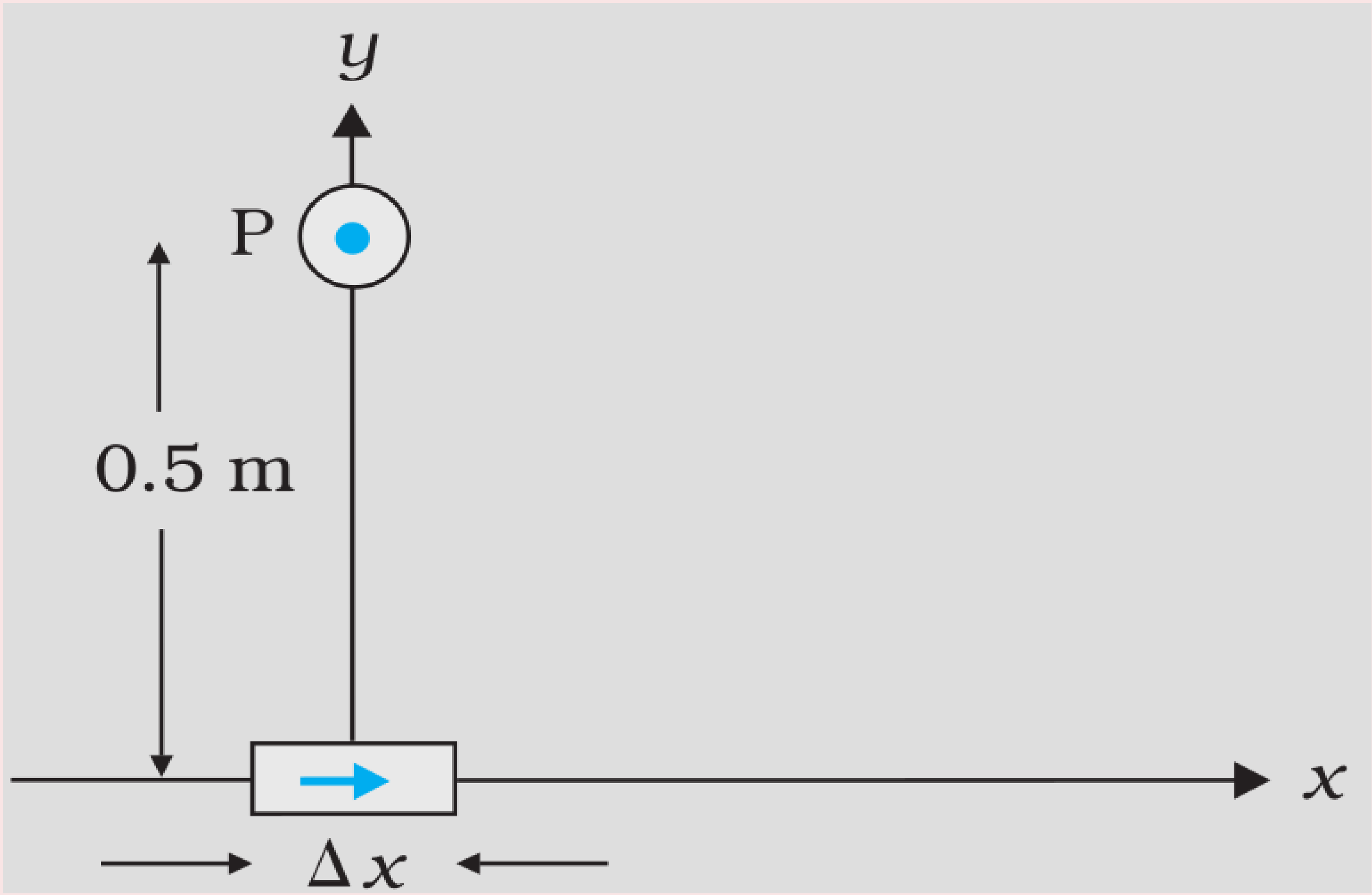
$$\epsilon_0 \times \mu_0 = \frac{1}{c^2} \text{ Or } \boxed{c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}}$$

Example 4.5



An element $\Delta l = \Delta x \hat{\boldsymbol{i}}$ is placed at the origin and carries a large current $I = 10\ A$. What is the magnetic field on the y-axis at a distance of $0.5\ m$. $\Delta x = 1\ cm$.

Applications of Biot-Savart Law

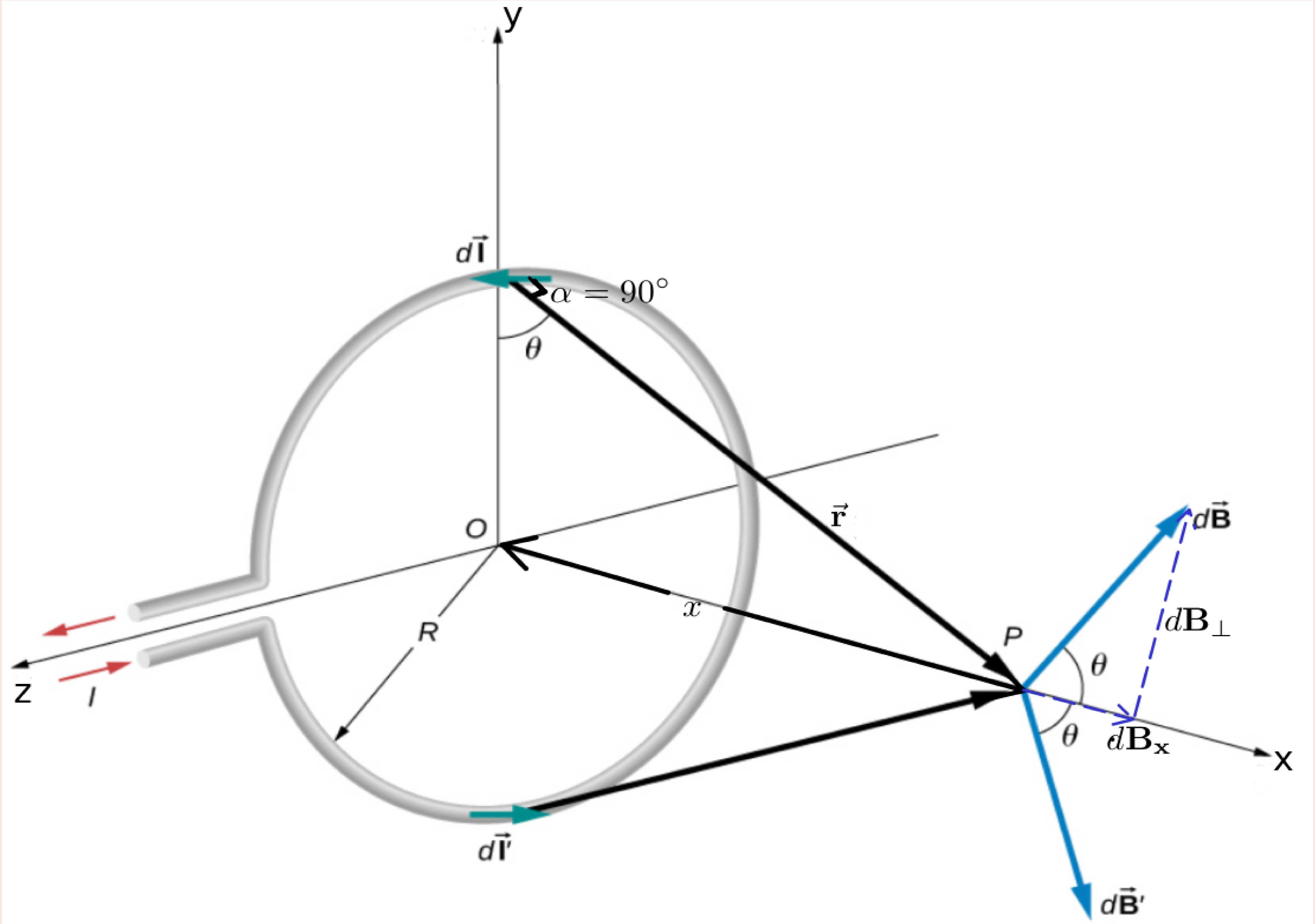


Figure 12.11 Determining the magnetic field at point *P* along the axis of a current-carrying loop of wire.

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- Figure depicts a **circular loop carrying a steady current I**.

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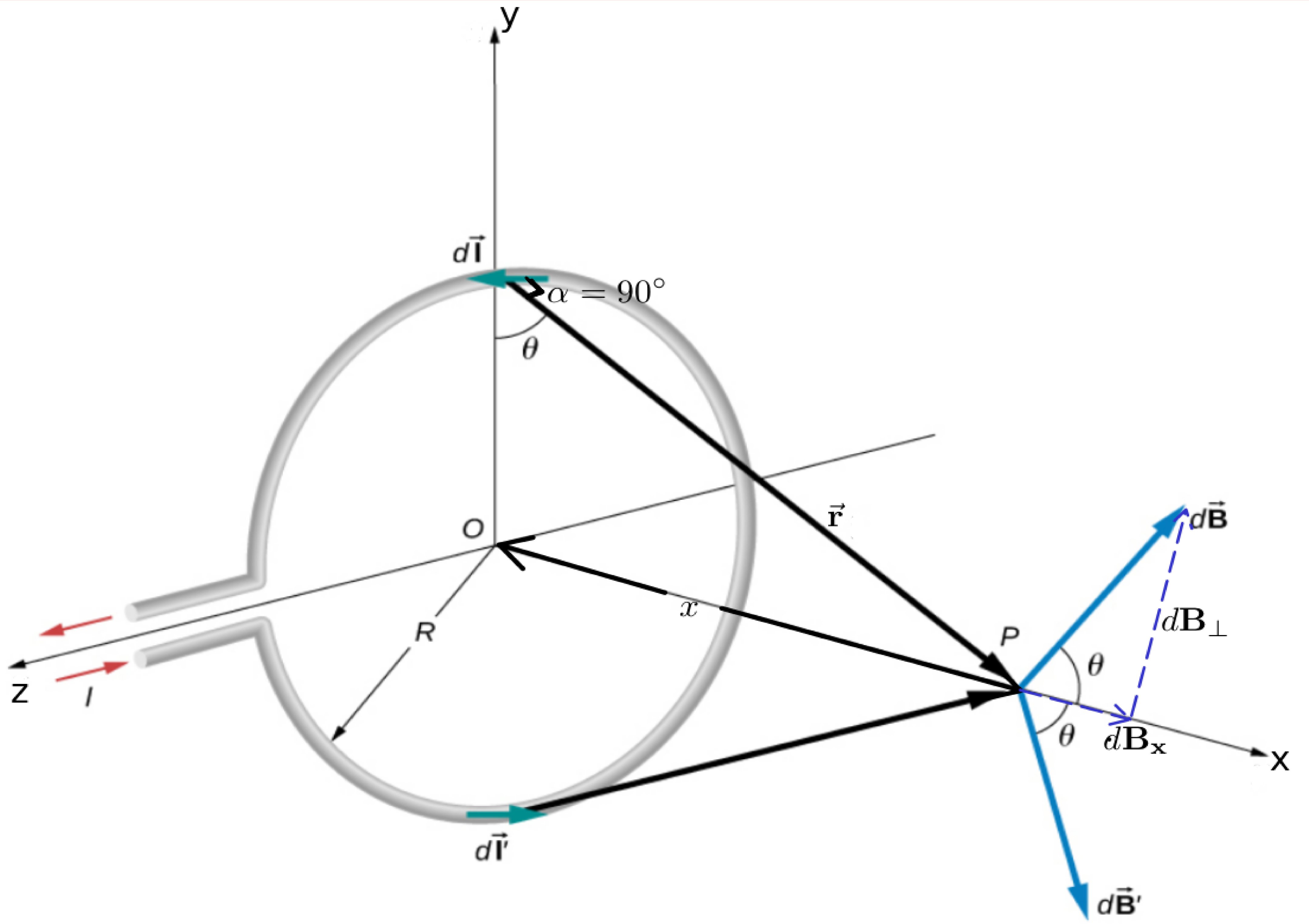

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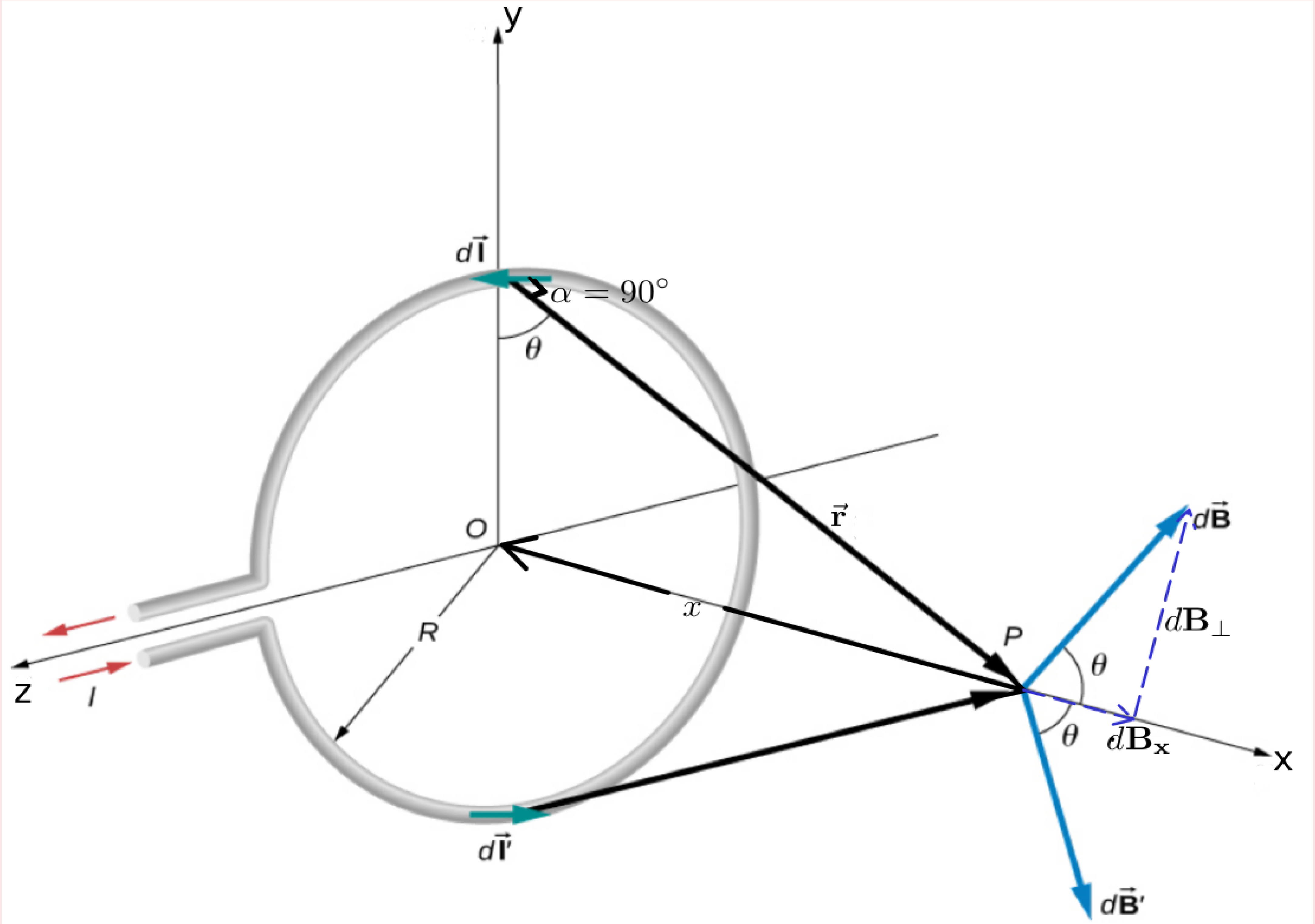

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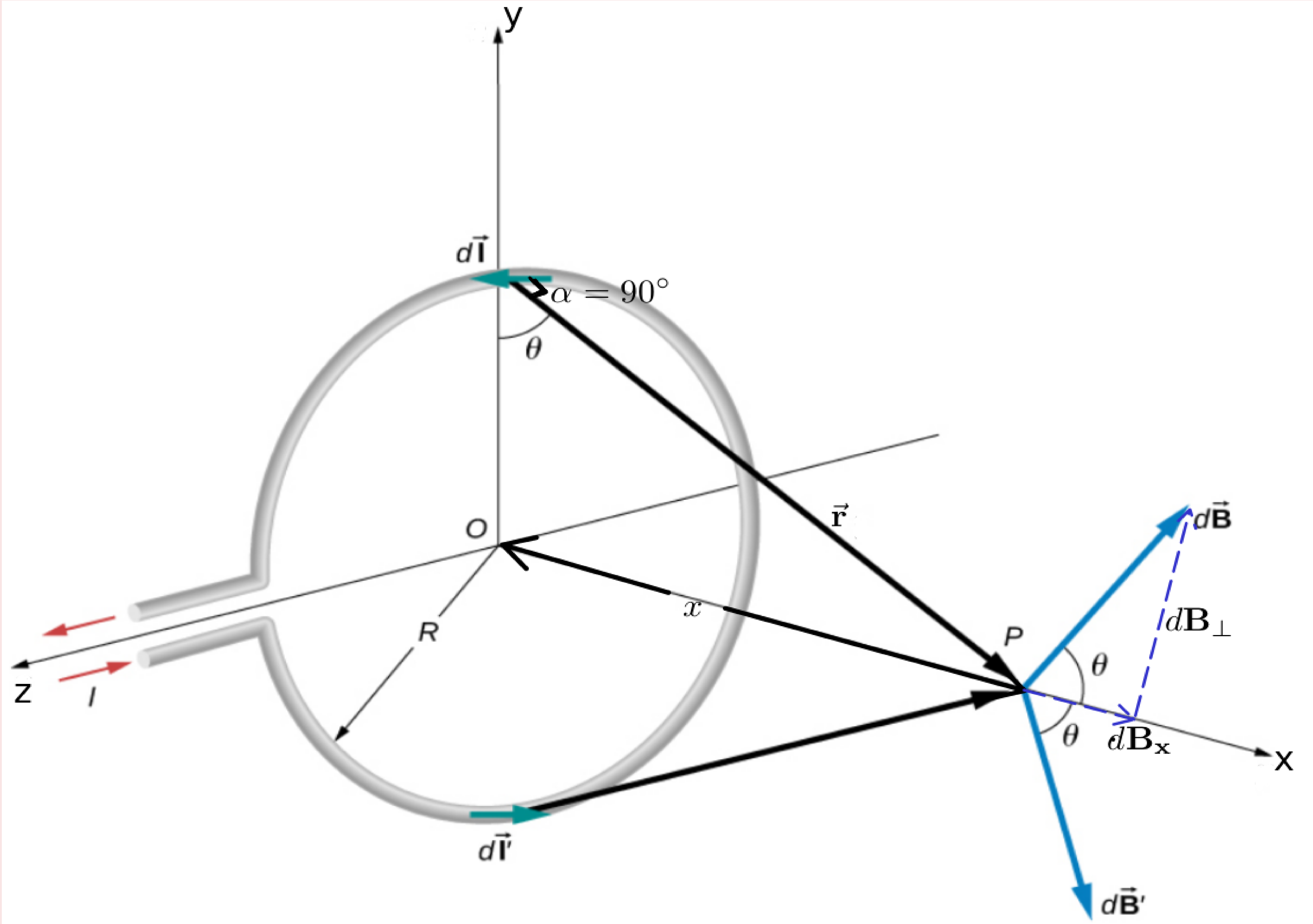

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- Let **x be the distance of P from the center O** of the loop.

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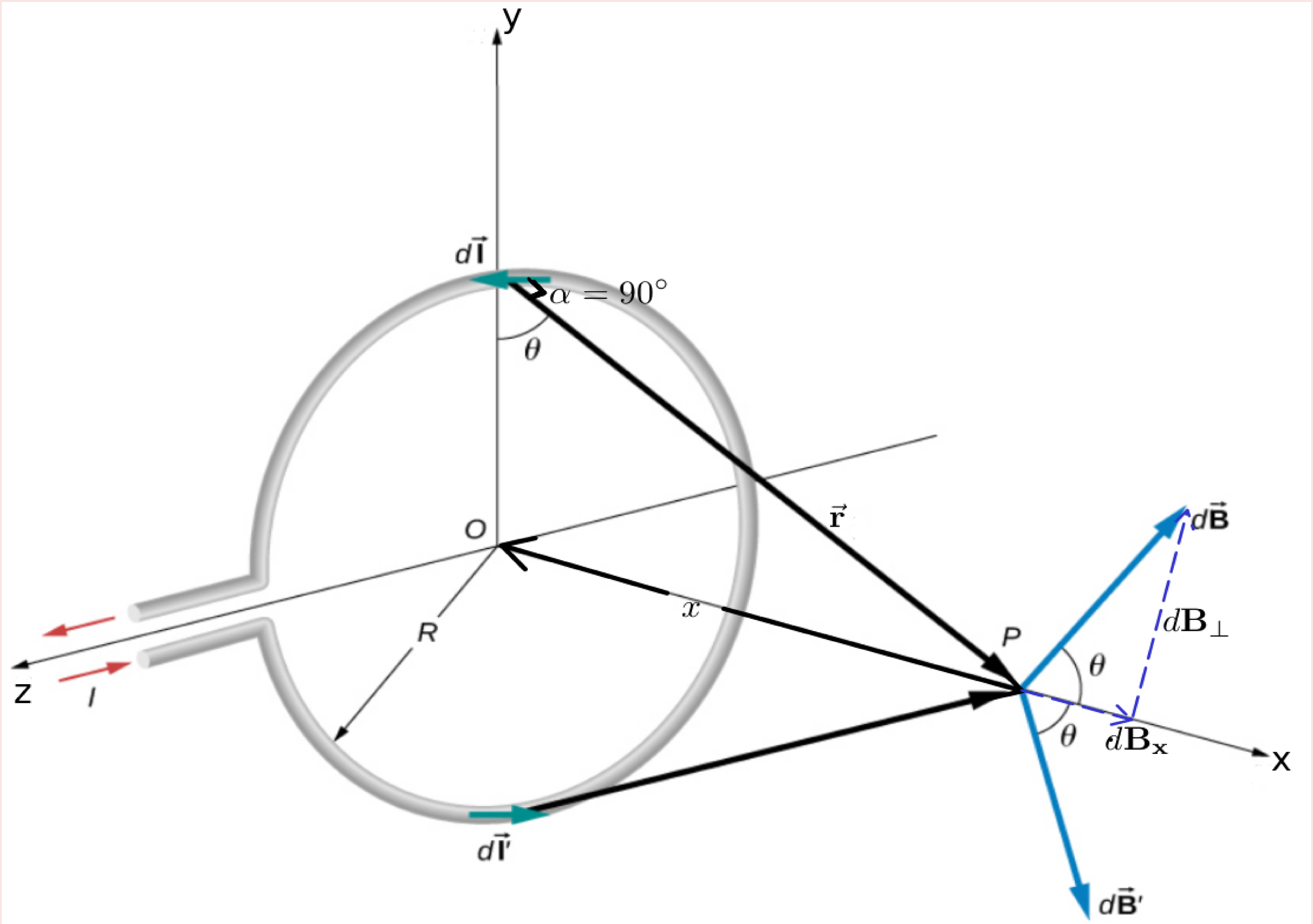

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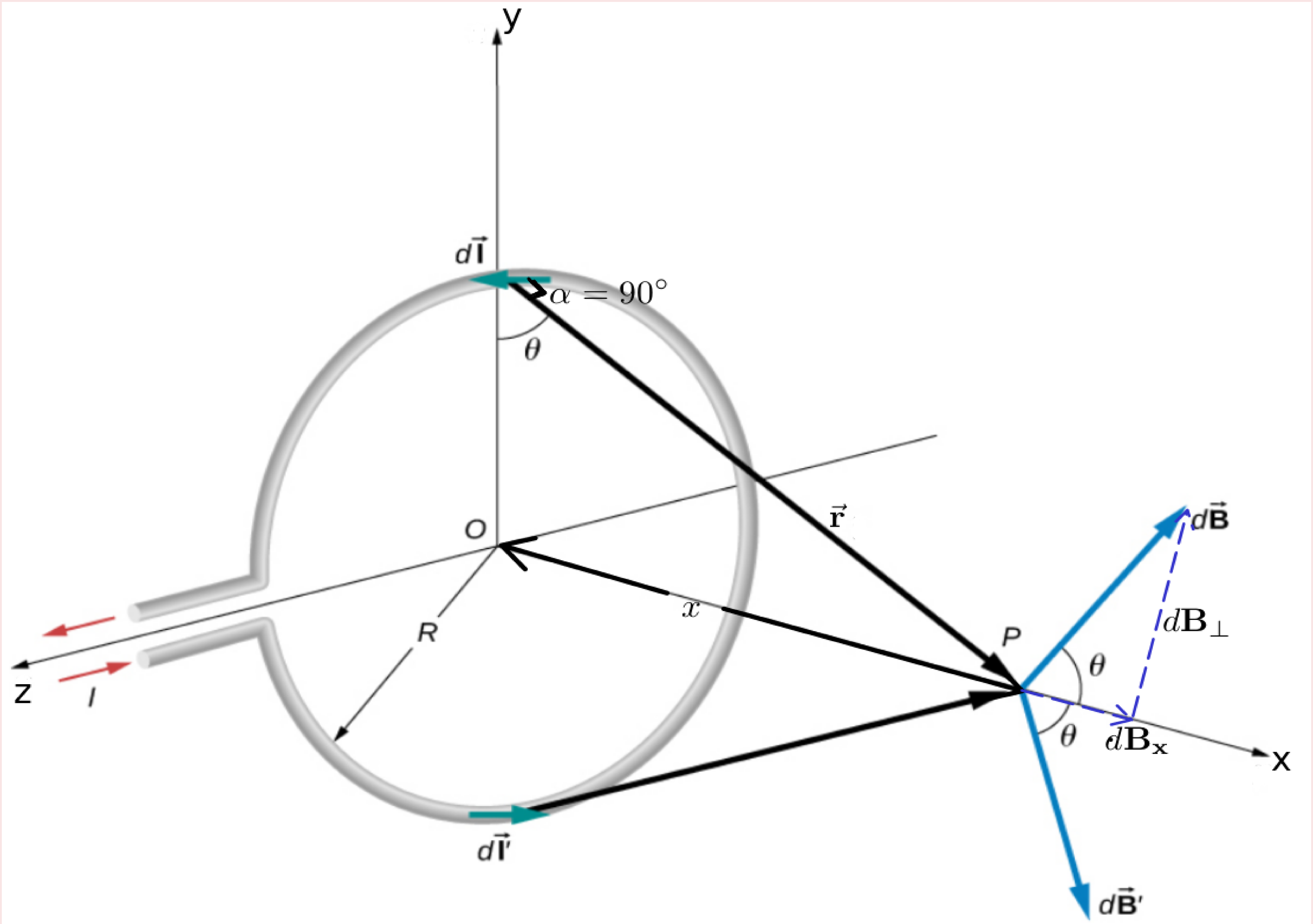

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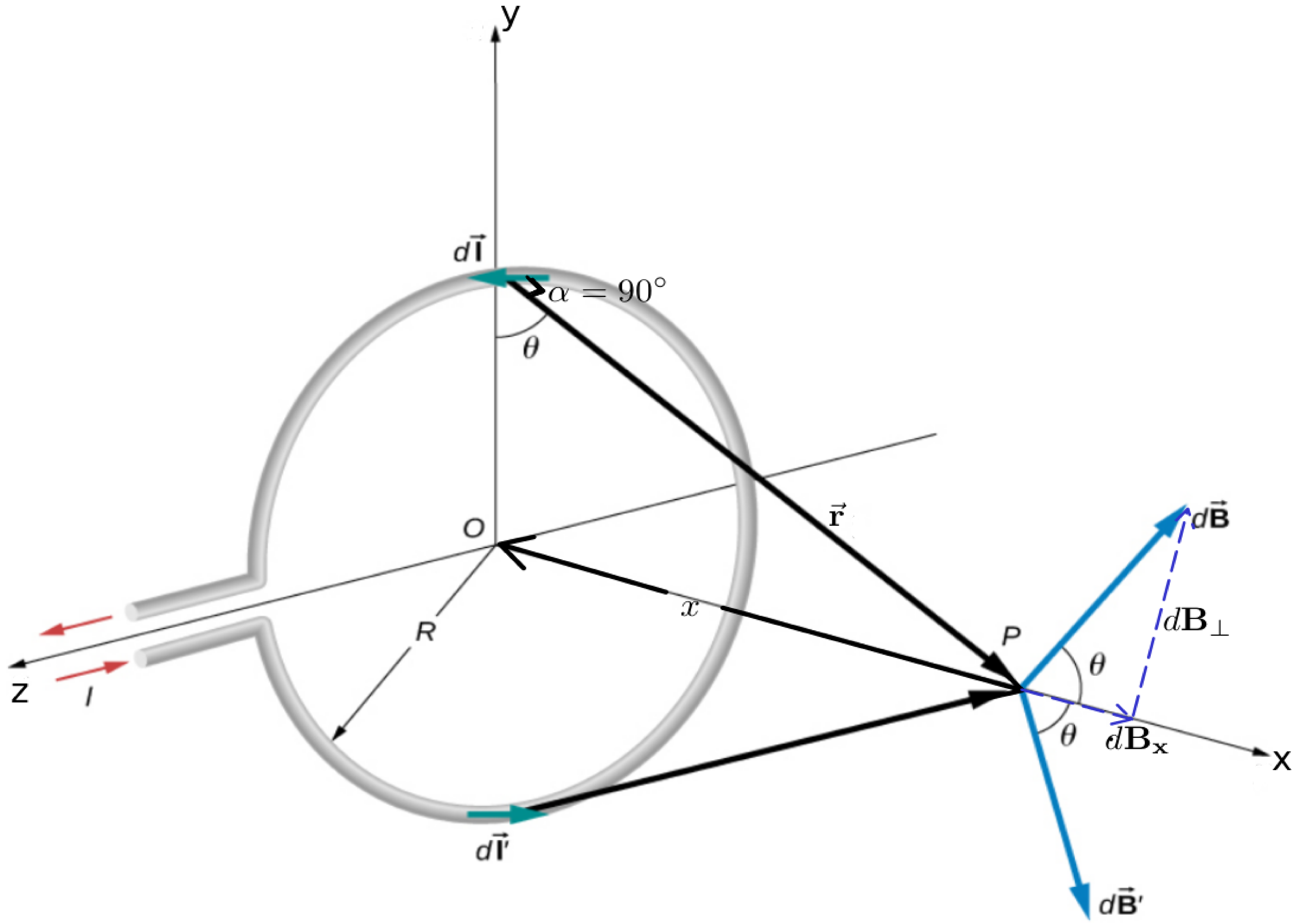

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- The element $d\vec{l}$ will be perpendicular to the displacement vector $\vec{r}$ (Or $\alpha = 90°$)

Applications of Biot-Savart Law

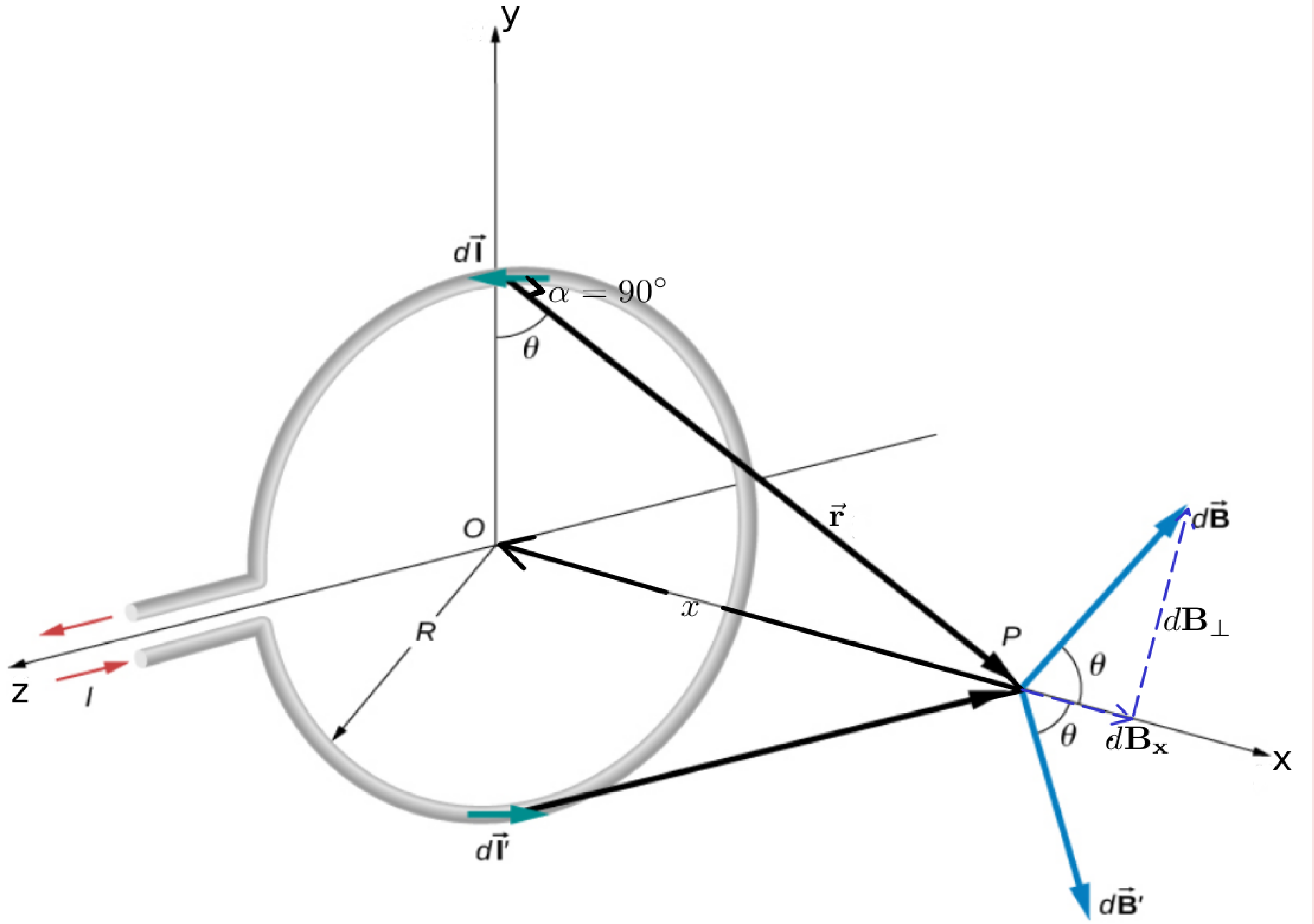


Figure 12.11 Determining the magnetic field at point *P* along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- Thus,

$$\boxed{dB = \frac{\mu_0}{4\pi}\frac{I dl}{(x^2 + R^2)}} \quad .....(1)$$

Applications of Biot-Savart Law

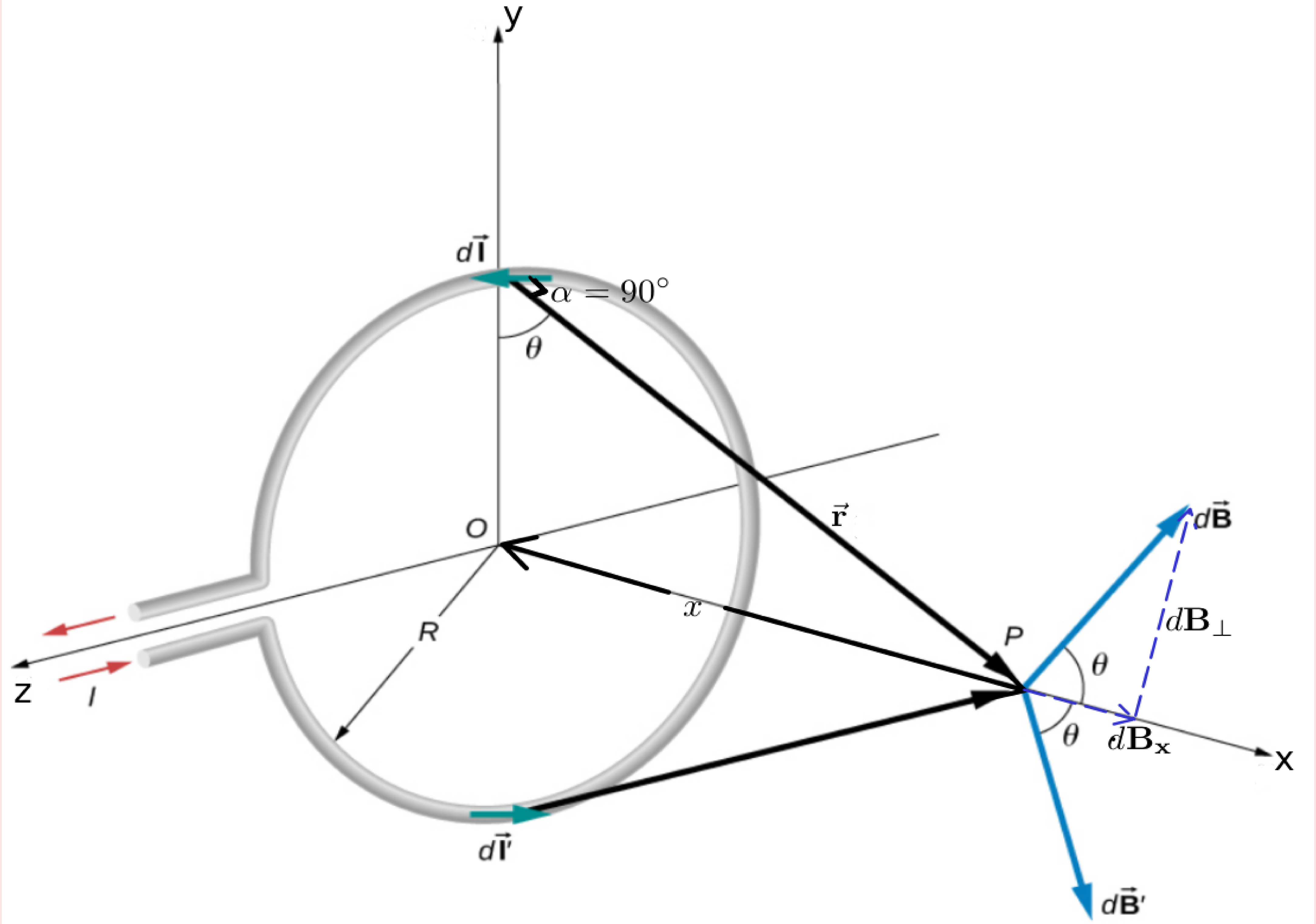


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

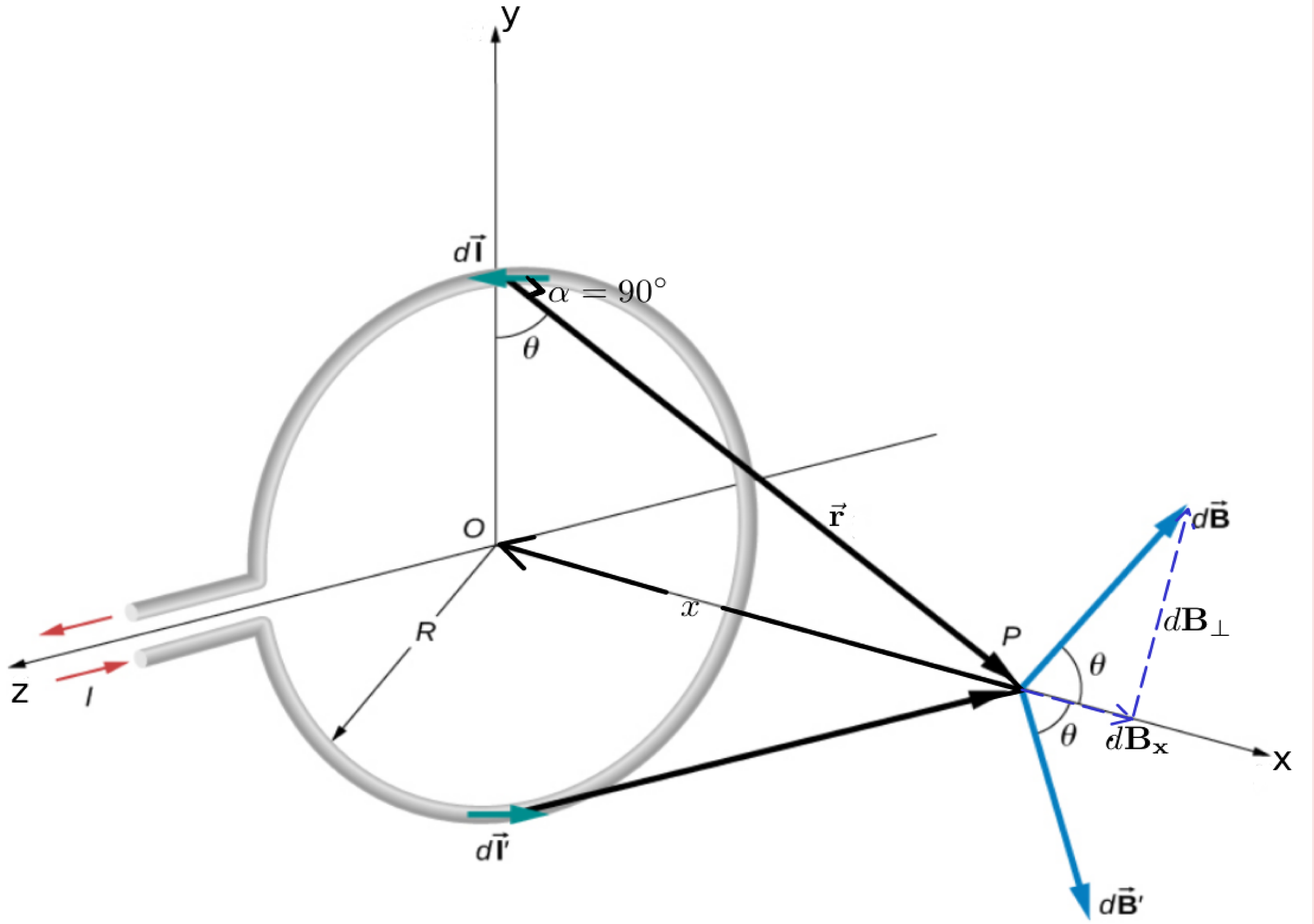
Magnetic Field On The Axis Of a Circular Current Loop

- Thus,

$$\boxed{dB = \frac{\mu_0}{4\pi}\frac{I\,dl}{(x^2+R^2)}} \quad .....(1)$$

- $d\vec{B}$ has an **x-component** $d\boldsymbol{B_x}$ and **a component perpendicular** to x-axis, $d\boldsymbol{B}_\perp$.

Applications of Biot-Savart Law



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Magnetic Field On The Axis Of a Circular Current Loop

- Thus,

$$\boxed{dB = \frac{\mu_0}{4\pi}\frac{Idl}{(x^2+R^2)}} \quad .....(1)$$

- $d\vec{B}$ has an **x-component** dB_x and **a component perpendicular** to x-axis, $dB_\perp$.
- When the **perpendicular components** $(dB_\perp)$ are **summed** over, **they cancel out**.

Applications of Biot-Savart Law

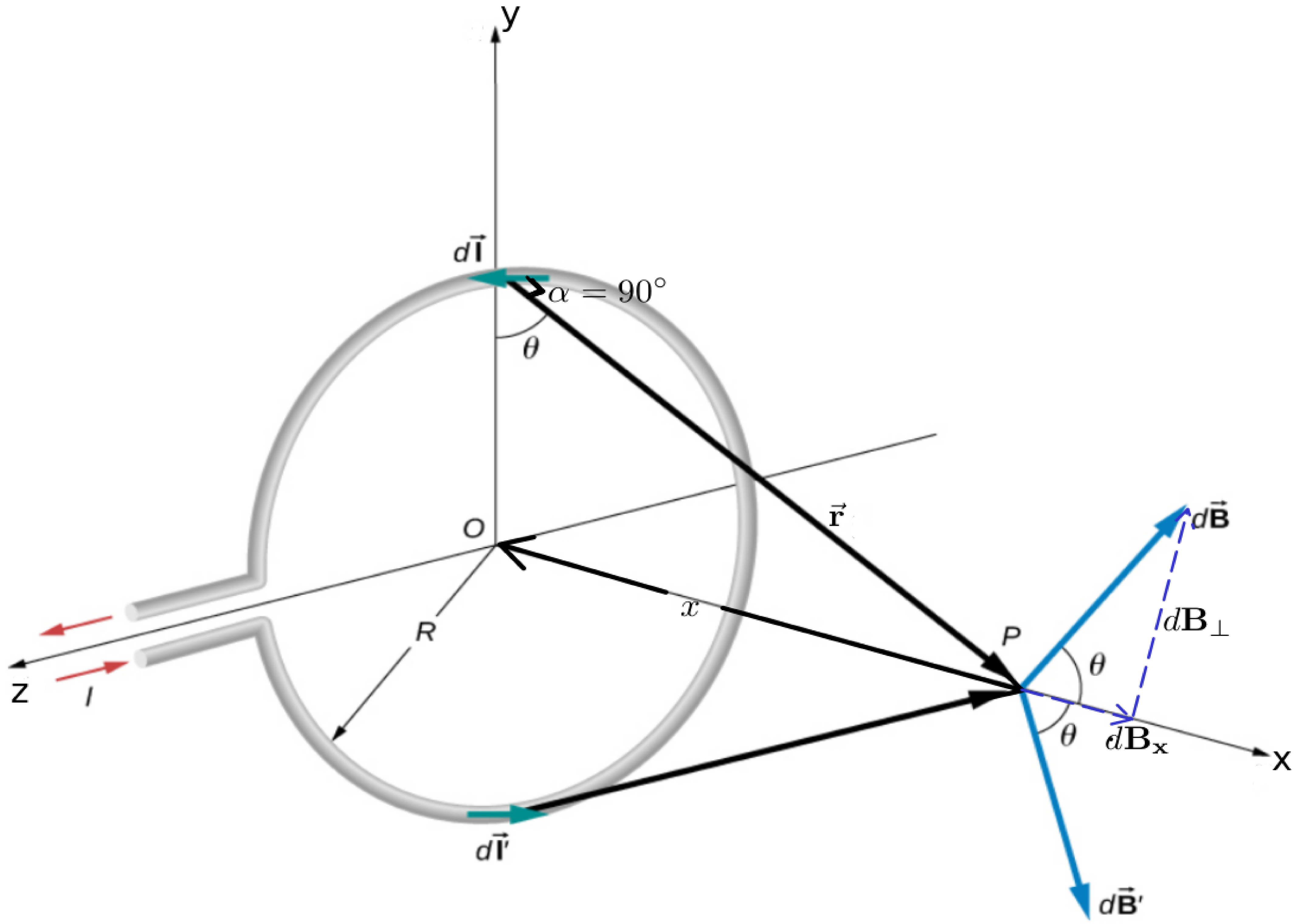

Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- Thus,

$$\boxed{dB = \frac{\mu_0}{4\pi}\frac{I dl}{(x^2 + R^2)}} \quad .....(1)$$

- $d\vec{B}$ has an **x-component** dB_x and **a component perpendicular** to x-axis, $dB_\perp$.
- When the **perpendicular components** $(dB_\perp)$ are **summed** over, **they cancel out**.
- Thus, only the x-component (dB_x) survives

Applications of Biot-Savart Law

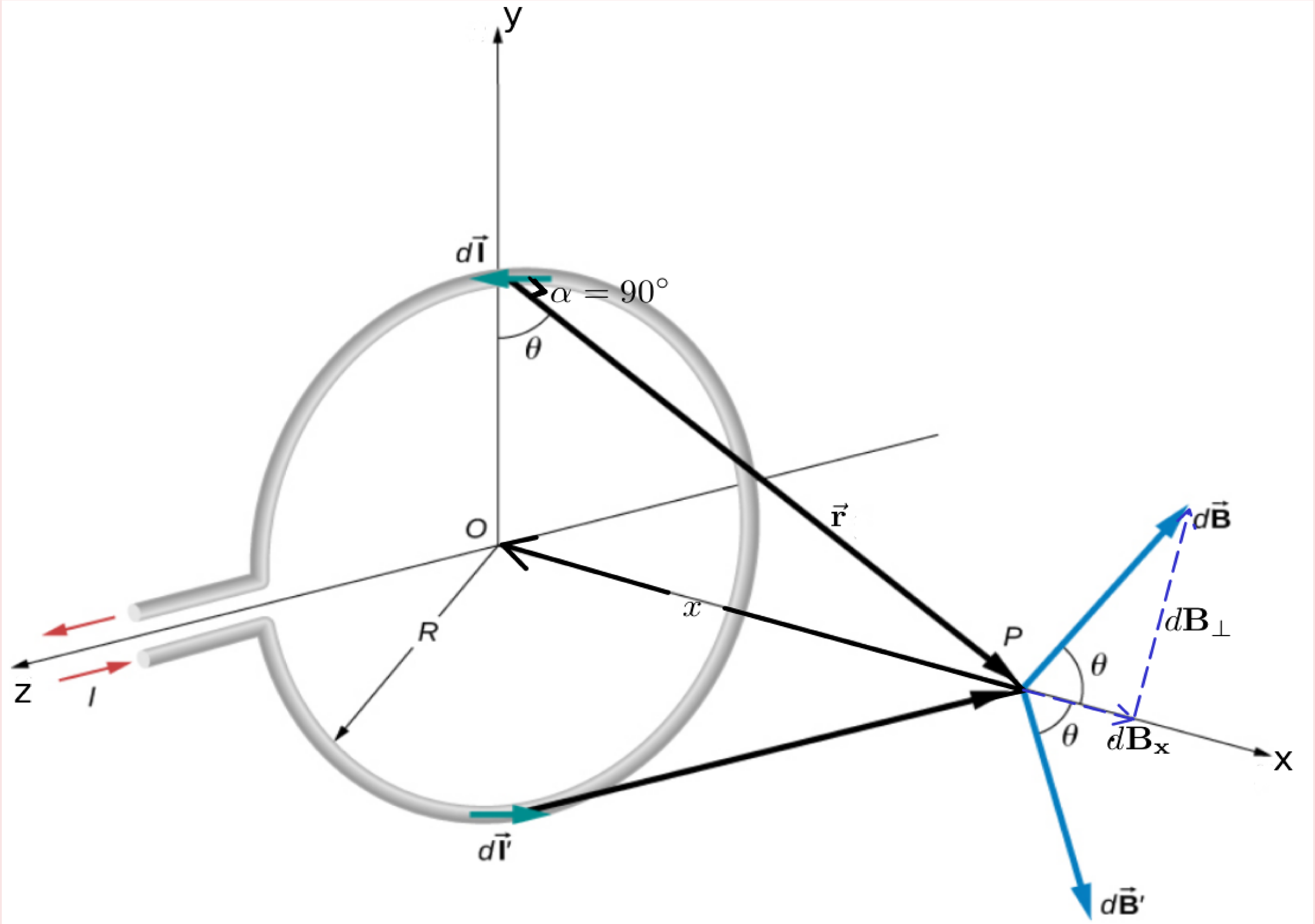

Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- The **net contribution along x-direction** can be **obtained by integrating** $dB_x = dB\cos\theta$ over the loop.

Applications of Biot-Savart Law

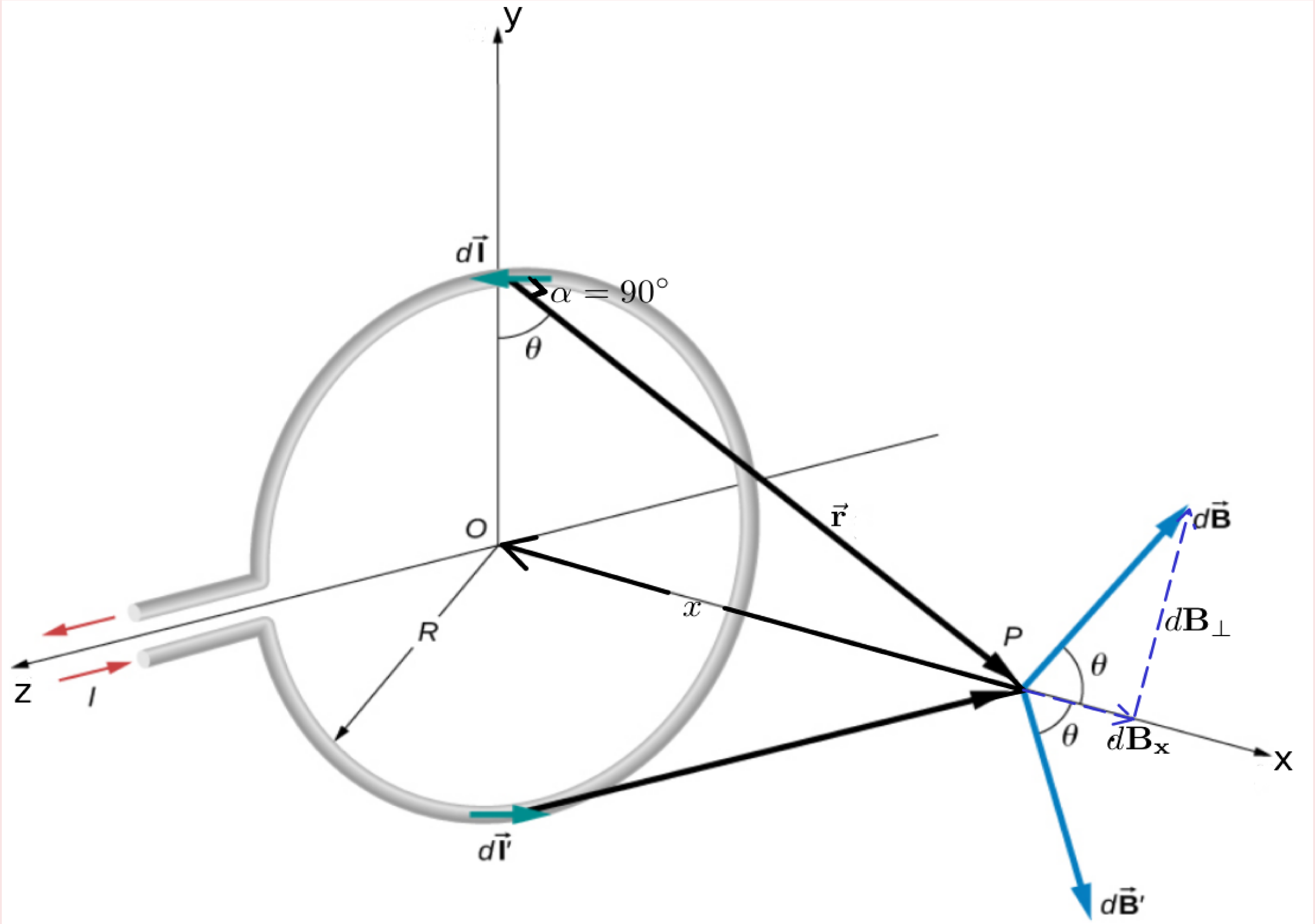

Figure 12.11 Determining the magnetic field at point *P* along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- The **net contribution along x-direction** can be **obtained by integrating** $dB_x = dB\cos\theta$ over the loop.

$$\boxed{\cos\theta = \frac{R}{(x^2+R^2)^{\frac{1}{2}}}} \qquad \ldots\ldots(2)$$

Applications of Biot-Savart Law

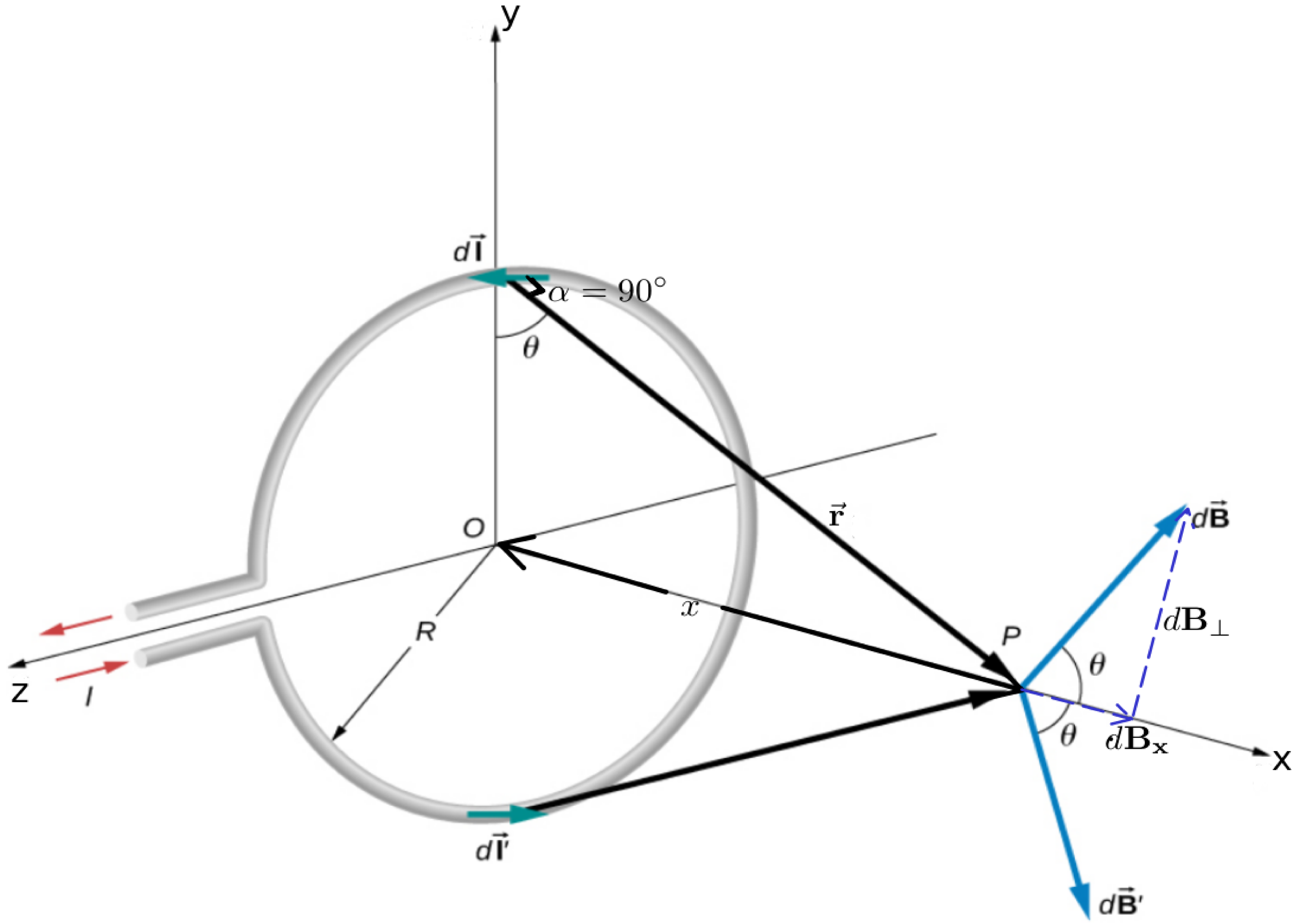

Figure 12.11 Determining the magnetic field at point *P* along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

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$$\boxed{\cos\theta = \frac{R}{(x^2+R^2)^{\frac{1}{2}}}} \qquad \ldots\ldots(2)$$

- From Eqs. (1) and (2),

$$dB_x = \frac{\mu_0}{4\pi}\frac{Idl}{(x^2+R^2)} \times \frac{R}{(x^2+R^2)^{\frac{1}{2}}}$$

Applications of Biot-Savart Law

Magnetic Field On The Axis Of a Circular Current Loop

- $dB_x = \dfrac{\mu_0}{4\pi} \dfrac{IRdl}{(x^2 + R^2)^{\frac{3}{2}}}$

Applications of Biot-Savart Law

Magnetic Field On The Axis Of a Circular Current Loop

- $dB_x = \frac{\mu_0}{4\pi}\frac{IRdl}{(x^2+R^2)^{\frac{3}{2}}}$
- Thus, the **magnetic field at P due to entire circular loop** is

$$\vec{\boldsymbol{B}} = B_x\hat{i} = \int_0^{2\pi R}\frac{\mu_0}{4\pi}\frac{IRdl}{(x^2+R^2)^{\frac{3}{2}}}\hat{i}$$

Applications of Biot-Savart Law

Magnetic Field On The Axis Of a Circular Current Loop

- $$dB_x = \frac{\mu_0}{4\pi} \frac{IRdl}{(x^2 + R^2)^{\frac{3}{2}}}$$

- Thus, the **magnetic field at P due to entire circular loop** is

$$\vec{\boldsymbol{B}} = B_x \hat{i} = \int_0^{2\pi R} \frac{\mu_0}{4\pi} \frac{IRdl}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}$$

$$\vec{\boldsymbol{B}} = \frac{\mu_0}{4\pi} \frac{IR \times 2\pi R}{(x^2 + R^2)^{\frac{3}{2}}} \hat{\boldsymbol{i}} \text{ Or } \boxed{\vec{\boldsymbol{B}} = \frac{\mu_0}{2} \frac{IR^2}{(x^2 + R^2)^{\frac{3}{2}}} \hat{\boldsymbol{i}}}$$

Applications of Biot-Savart Law

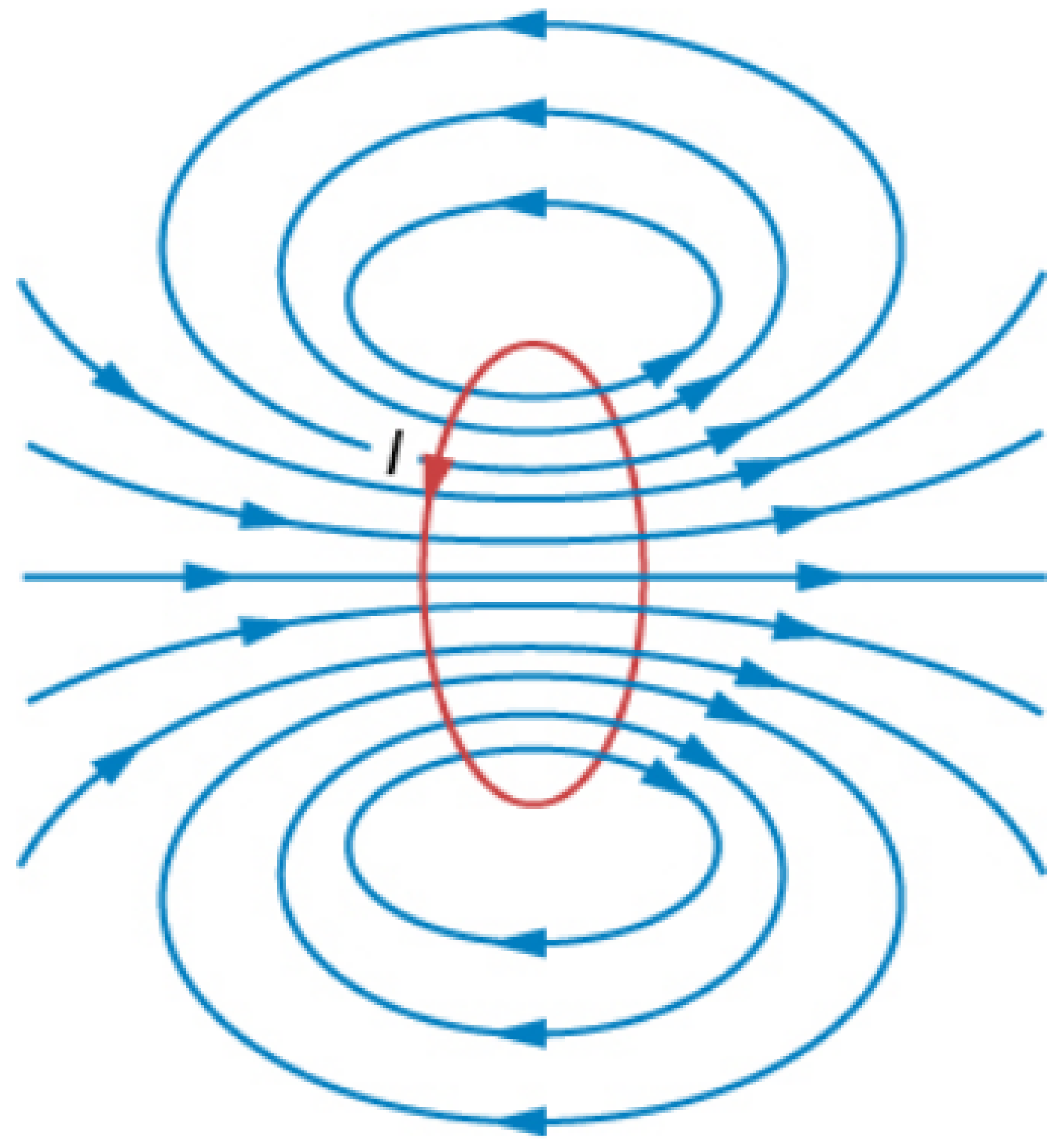
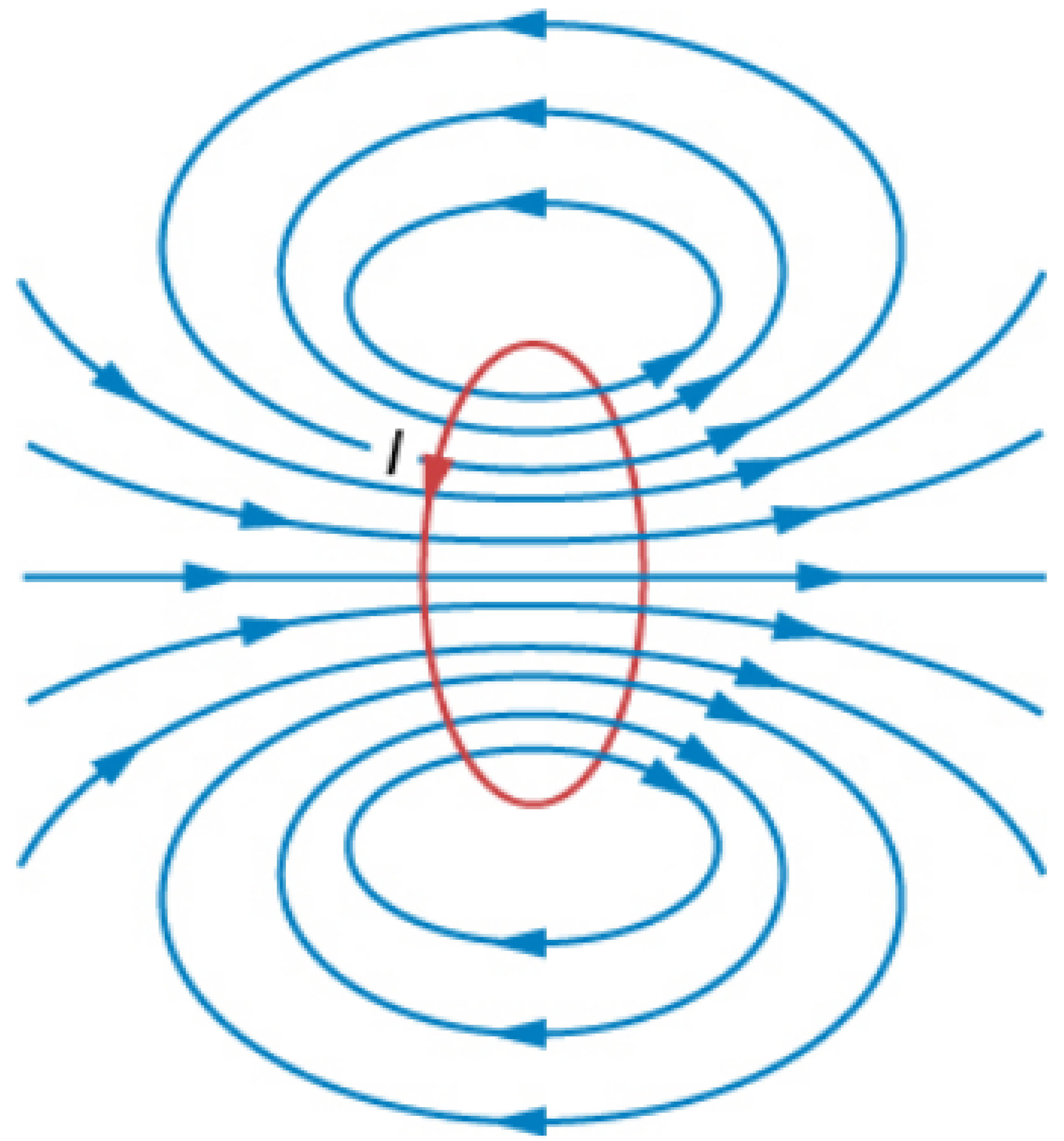


Figure 12.12 Sketch of the magnetic field lines of a circular current loop.

Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The **field at the center of the loop** $(x = 0)$.

Applications of Biot-Savart Law



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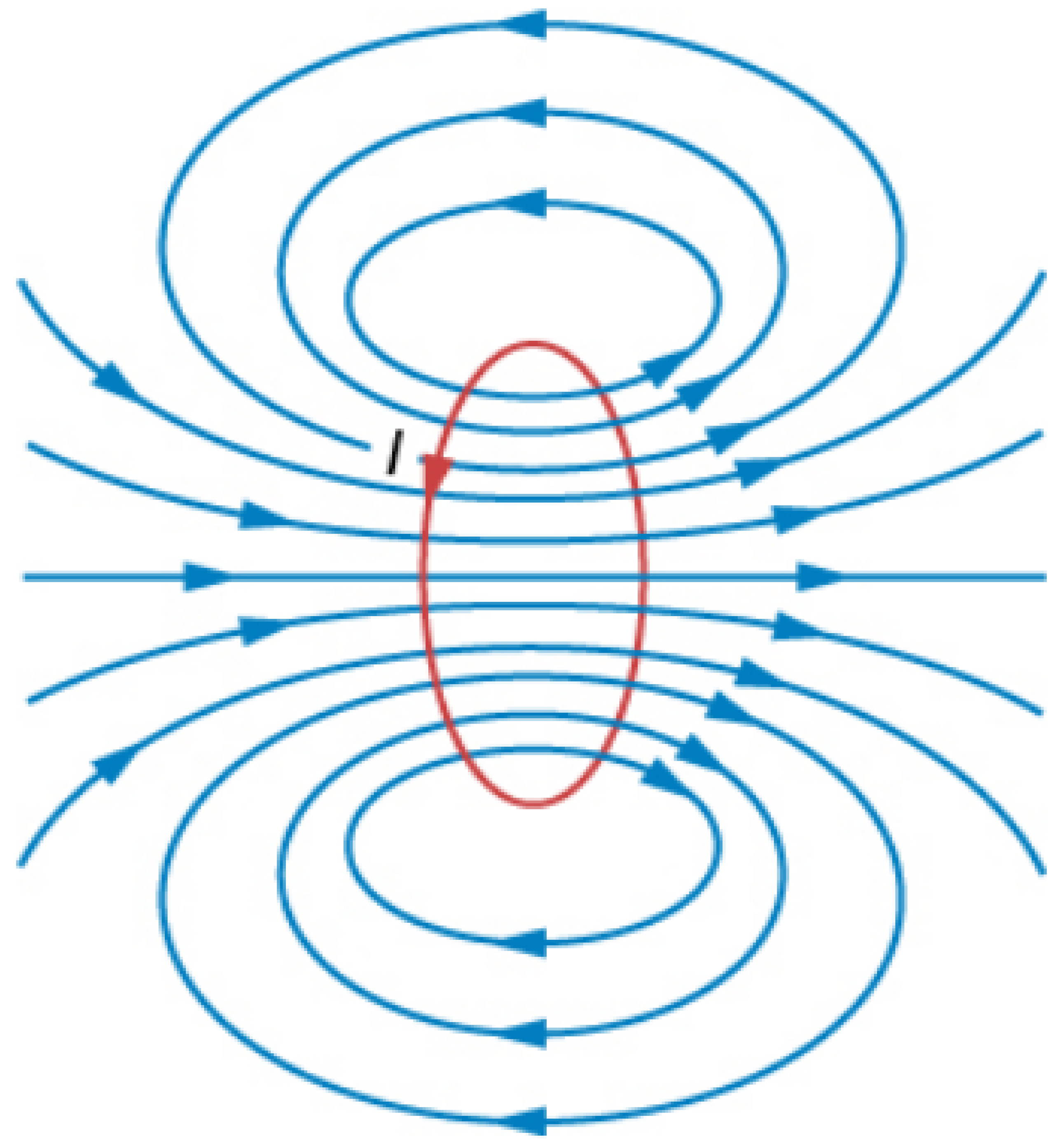
Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The **field at the center of the loop** $(x = 0)$.

$$\vec{\boldsymbol{B}} = \frac{\mu_0}{2} \frac{IR^2}{(0 + R^2)^{\frac{3}{2}}} \hat{\boldsymbol{i}}$$

Applications of Biot-Savart Law



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Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The **field at the center of the loop** $(x = 0)$.

$$\vec{\boldsymbol{B}} = \frac{\mu_0}{2} \frac{IR^2}{(0 + R^2)^{\frac{3}{2}}} \hat{\boldsymbol{i}}$$

$$\boxed{\vec{\boldsymbol{B}} = \frac{\mu_0}{2} \frac{I}{R} \hat{\boldsymbol{i}}}$$

Applications of Biot-Savart Law

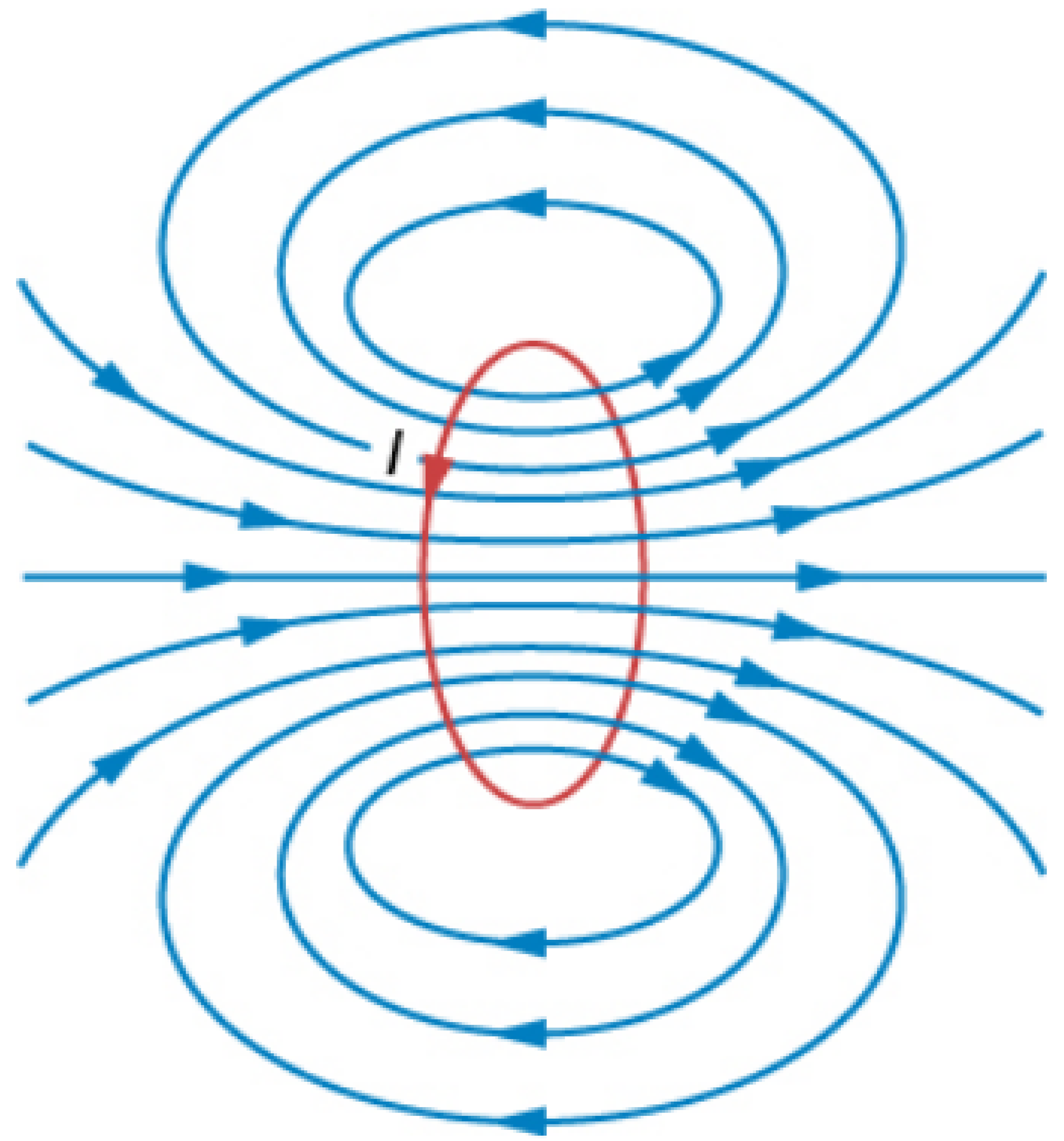


Figure 12.12 Sketch of the magnetic field lines of a circular current loop.

Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The **field at the center of the loop** $(x = 0)$.

$$\vec{\boldsymbol{B}} = \frac{\mu_0}{2} \frac{I R^2}{(0 + R^2)^{\frac{3}{2}}} \hat{\boldsymbol{i}}$$

$$\boxed{\vec{\boldsymbol{B}} = \frac{\mu_0}{2} \frac{I}{R} \hat{\boldsymbol{i}}}$$

- The **magnetic field lines** due to a circular wire **form closed loops**

Example 4.6

A straight wire carrying a **current of 12 A** is bent into a semi-circular arc of **radius 2.0 cm** as shown in Fig. 4.13(a). Consider the **magnetic field B** at the **center** of the arc.
(a) What is the **magnetic field due to the straight segments**?
(b) In what way the **contribution to B** from the **semicircle differs from that of a circular loop** and in **what way does it resemble**?
(c) Would your answer be different **if the wire were bent** into a semi-circular arc of the same radius but **in the opposite way** as shown in Fig. 4.13(b)?

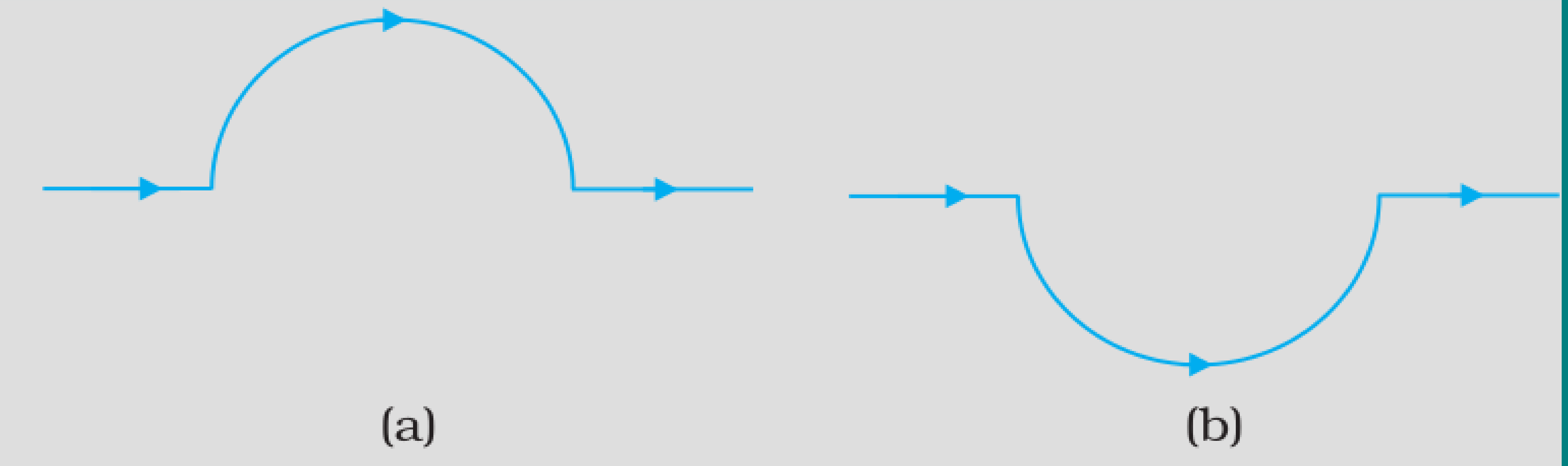


FIGURE 4.13

Example 4.7

Consider a tightly wound **100 turn** coil of **radius 10 cm**, carrying a **current of 1 A**. **What is the magnitude of the magnetic field at the center** of the coil?

6. Two identical coils carrying equal currents are held concentric with their planes perpendicular to each other. The ratio of the magnitude of magnetic field at the centre of one coil to that of the resultant magnetic field at the centre is :

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$

(c) $\frac{1}{2}$ (d) 2

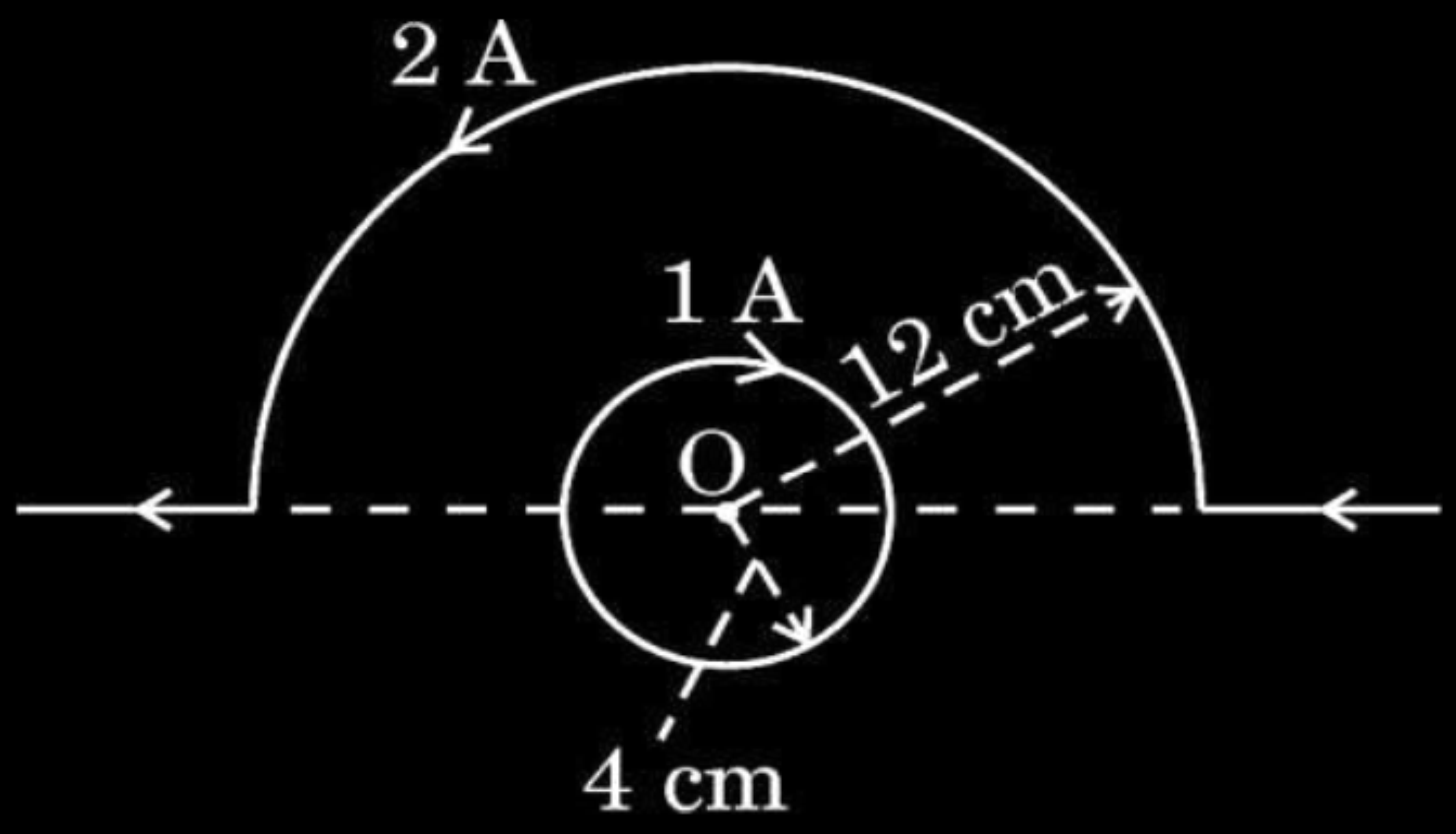
6. Two identical coils carrying equal currents are held concentric with their planes perpendicular to each other. The ratio of the magnitude of magnetic field at the centre of one coil to that of the resultant magnetic field at the centre is :

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$

(c) $\frac{1}{2}$ (d) 2

(a) $\frac{1}{\sqrt{2}}$

27. A current carrying circular loop and a straight wire bent partly in the form of a semicircle are placed as shown in the figure. Find the magnitude and direction of net magnetic field at point O. *3*



Magnetic field due to inner circle at O,

$$B_1 = \frac{\mu_0 I}{2r}$$ ½

$$= \frac{2\pi\times10^{-7}\times1}{4\times10^{-2}}$$

$$= \frac{\pi}{2}\times10^{-5}\ T \qquad \text{into the page}$$ ½

Magnetic field due to semi- circle at O,

$$B_2 = \frac{\mu_0 I}{2r}\cdot\frac{1}{2}$$ ½

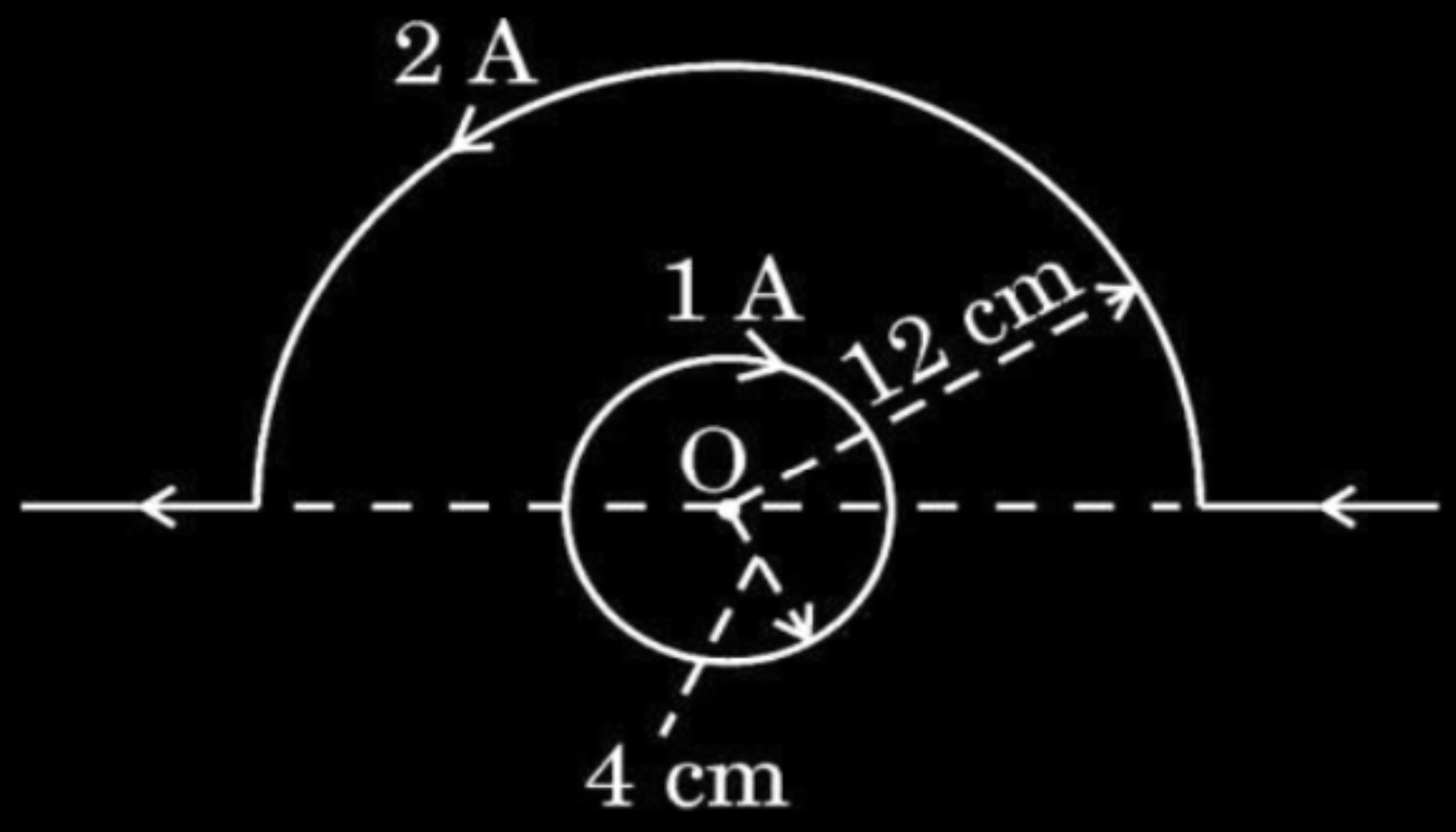
$$= \frac{2\pi\times10^{-7}\times2}{12\times10^{-2}}$$

$$= \frac{\pi}{6}\times10^{-5}\ T \qquad \text{out of the page}$$ ½

$$B_{net} = B_1 - B_2$$

$$= \left(\frac{\pi}{2}\times10^{-5} - \frac{\pi}{6}\times10^{-5}\right)T \approx 1.0\times10^{-5}\ T$$ ½

Direction of the net magnetic field is into the page, since $B_1 > B_2$. ½



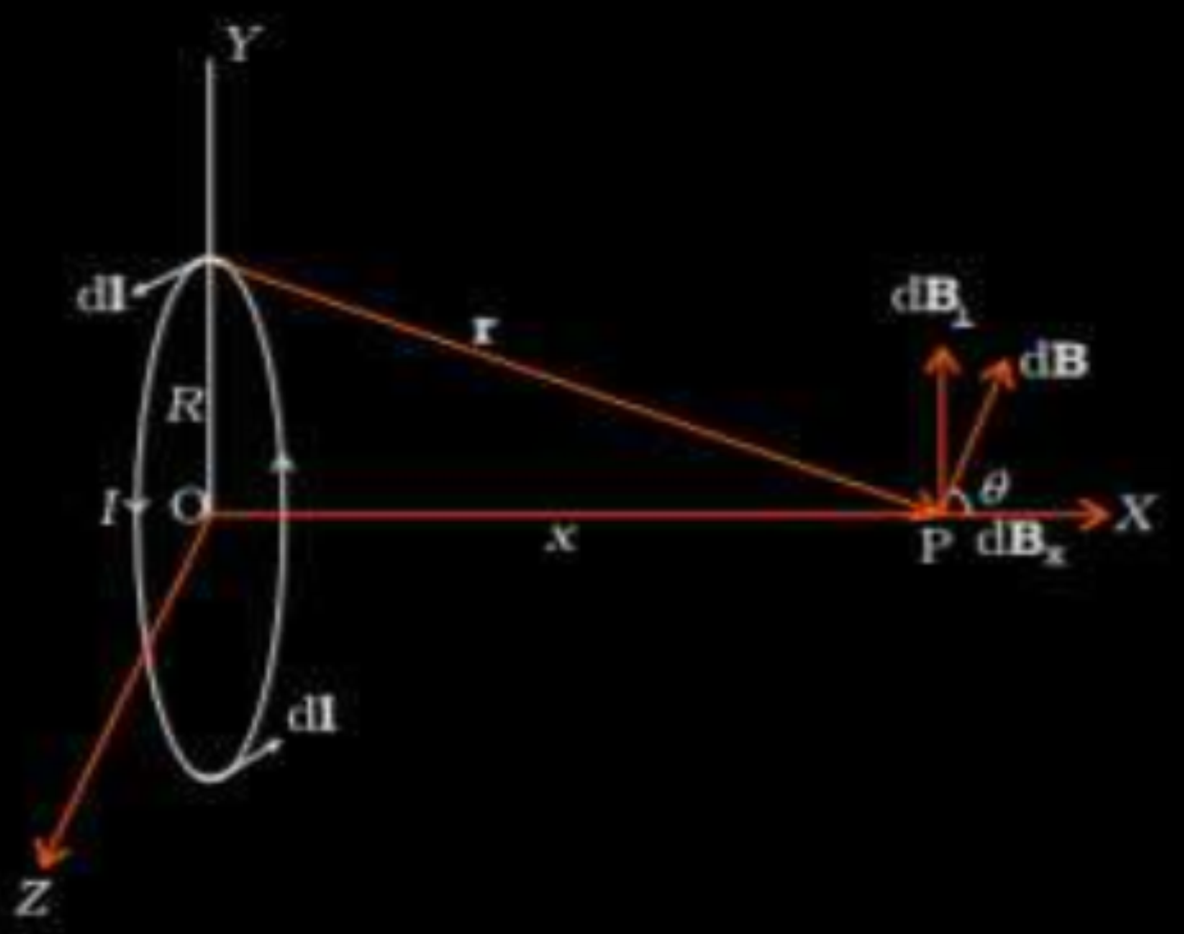
26. Two circular loops A and B, each of radius 3 m, are placed coaxially at a distance of 4 m. They carry currents of 3 A and 2 A in opposite directions respectively. Find the net magnetic field at the centre of loop A. *3*

Calculation of magnetic field due to loop A	1
Calculation of magnetic field due to loop B	1
Net magnetic field	1

$B_1 = \dfrac{\mu_0 I}{2r}$	
$B_1 = \dfrac{4\pi \times 10^{-7} \times 3}{2 \times 3} = 2\pi \times 10^{-7} T = 6.28 \times 10^{-7} T$	1
$B_2 = \dfrac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$	
$= \dfrac{2\pi \times 10^{-7} \times 2 \times 9}{(3^2 + 4^2)^{3/2}}$, opposite to B_1	
$B_2 = \dfrac{36\pi \times 10^{-7}}{125} T = 0.9 \times 10^{-7} T$	1
$B_{net} = B_1 - B_2$	
$= 6.28 \times 10^{-7} - 0.9 \times 10^{-7}$	
$B_{net} = 5.38 \times 10^{-7}\ T$	1

(b) (i) Derive an expression for magnetic field on the axis of a current carrying circular loop.

(i)



½

$$dB = \frac{\mu_0}{4\pi}\frac{I|d\mathbf{l} \times \mathbf{r}|}{r^3}$$

½

$$= \frac{\mu_o i dl \sin 90^o}{4\pi\left(x^2 + R^2\right)}$$

½

$dB_{\perp}$ cancels out.

½

Net B = $\int dB_x = \int dB \cos\theta$

$$= \frac{\mu_o}{4\pi}\int \frac{idl}{\left(x^2+R^2\right)} \times \frac{R}{\left(x^2+R^2\right)^{1/2}}$$

$$= \frac{\mu_o iR}{4\pi\left(x^2+R^2\right)^{3/2}} \int dl$$

½

$$= \frac{\mu_o iR}{4\pi\left(x^2+R^2\right)^{3/2}} \; (2\pi R)$$

½

$$\mathbf{B} = B_x\hat{\mathbf{i}} = \frac{\mu_0 I R^2}{2\left(x^2+R^2\right)^{3/2}}\hat{\mathbf{i}}$$

19. A wire of length l is in the form of a circular loop A of one turn. This loop is reshaped into loop B of three turns. Find the ratio of the magnetic fields at the centres of loop A and loop B for the same current through them. *2*

$B = \dfrac{\mu_o NI}{2r}$	½
$l = 2\pi r$; $2\pi r_A = 3(2\pi r_B)$	
$r_B = \dfrac{r_A}{3}$	½
$\dfrac{B_A}{B_B} = \dfrac{\mu_o N_A I}{2r_A} \text{X} \dfrac{2r_B}{\mu_0 N_B I}$	½
$\dfrac{B_A}{B_B} = \dfrac{1}{9}$	½

33. (a) (i) State Biot-Savart's law for the magnetic field due to a current carrying element. Use this law to obtain an expression for the magnetic field at the centre of a circular loop of radius 'a' and carrying a current 'I'. Draw the magnetic field lines for a current loop indicating the direction of magnetic field.

(ii) An electron is revolving around the nucleus in a circular orbit with a speed of 10^7 m s^{-1}. If the radius of the orbit is 10^{-10} m, find the current constituted by the revolving electron in the orbit. *5*

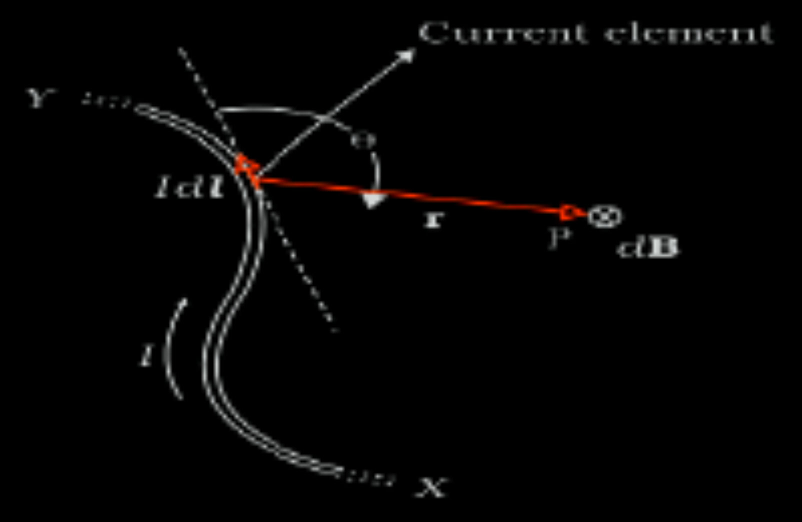
i) Statement of Biot-Savart's law	1
Expression for magnetic field	2
Diagram for magnetic field lines	½
ii) Finding current by revolving electron	1 ½

(i)

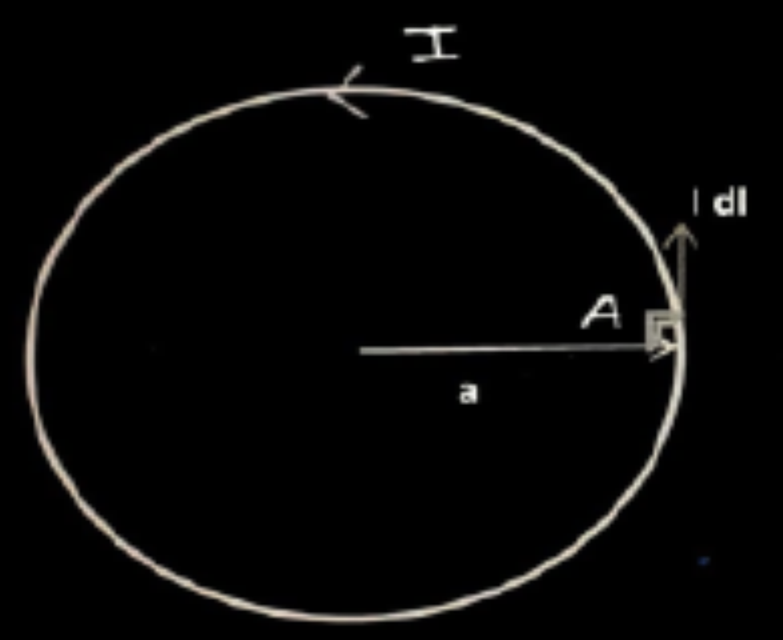
The magnetic field at a point due to a current carrying element is proportional to magnitude of current, element length and inversely proportional to the square of the distance from the element. 1

$$\vec{dB} = \frac{\mu_o}{4\pi} I \frac{\vec{dl} \times \vec{r}}{r^3}$$

$$|dB| = \frac{\mu_o}{4\pi} \frac{Idl\sin\theta}{r^2}$$



Consider a circular coil of radius a carrying current I.



½

According to Biot-Savart's law

$$|\vec{dB}| = \frac{\mu_o}{4\pi} \frac{Idl\sin\theta}{r^2}$$

At point A $\quad I\,\vec{dl} \perp \vec{a}$ ½

$\therefore \theta = 90^o, \sin 90^o = 1$

Hence Db= $\dfrac{\mu_o}{4\pi} \dfrac{Idl}{a^2}$

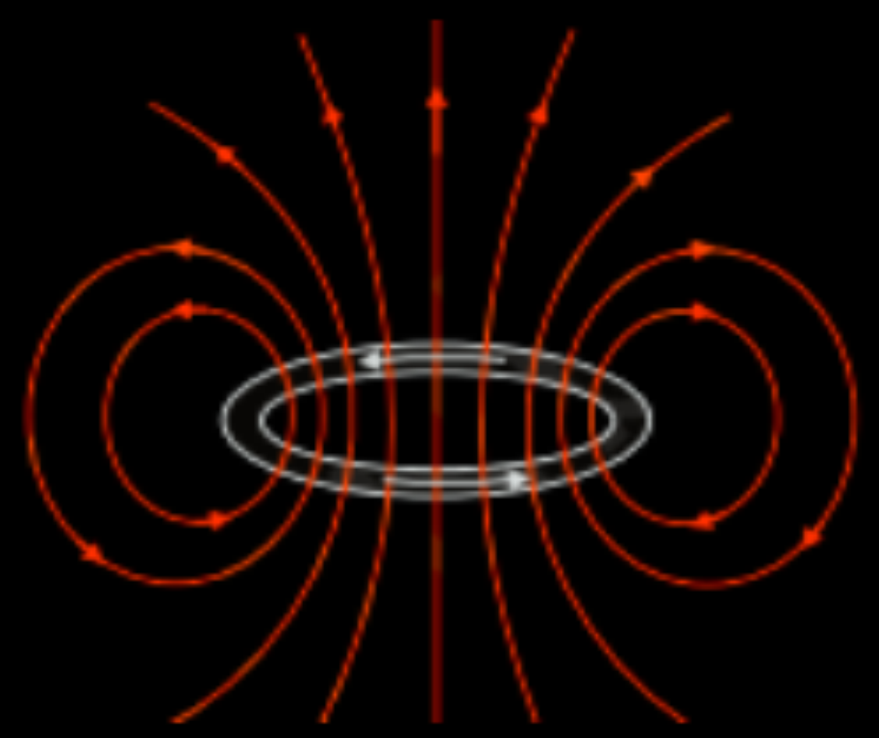
Magnetic field at centre

B= $\displaystyle\int_o^{2\pi a} dB = \int_0^{2\pi a} \frac{\mu_o}{4\pi} \frac{Idl}{a^2}$ ½

B= $\dfrac{\mu_o}{4\pi} \times \dfrac{I}{a^2} \times 2\pi a$

B = $\dfrac{\mu_o I}{2a}$ ½

Note: Give full credit of 2 marks if a student derives the expression for magnetic field at the axis of the loop and then puts distance of point as 0 from the centre.



½

ii) $\quad q=e\ ,\quad v=10^7 ms^{-1}, r=10^{-10}m$

i= $\dfrac{q}{T}$ ½

$= \dfrac{qv}{2\pi r}$

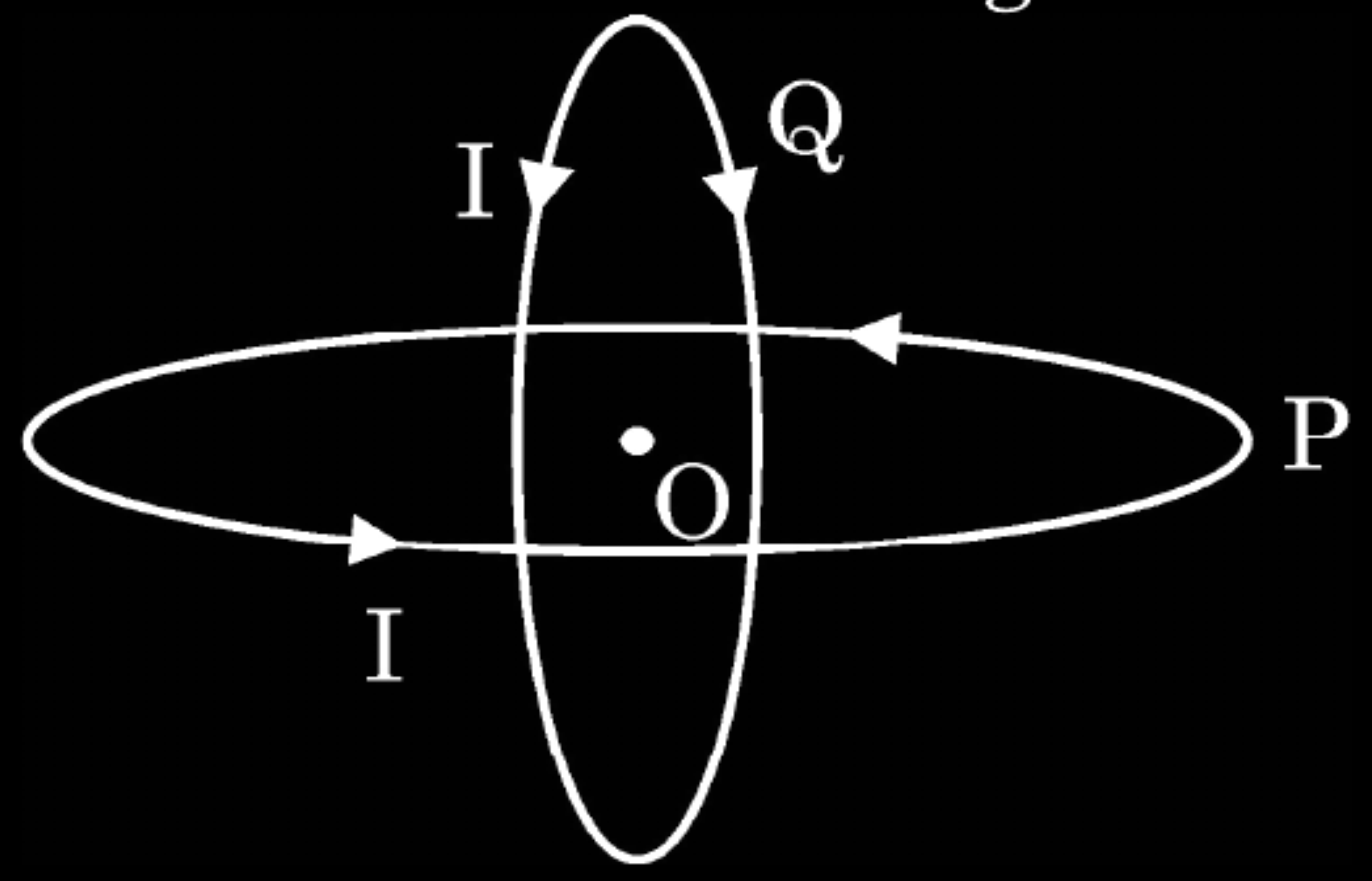
$= \dfrac{ev}{2\pi r}$

$= \dfrac{1.6\times10^{-19}\times10^7}{2\times\pi\times10^{-10}}$ ½

$= \dfrac{0.8}{\pi}\times10^{-2} A$ ½

$= 0.255 \times10^{-2} A \quad = 2.55$ Ma

23. (a) Two identical circular loops P and Q, each of radius R carrying current I are kept in perpendicular planes such that they have a common centre O as shown in the figure.



Find the magnitude and direction of the net magnetic field at point O. **2**

(a) Calculation of magnitude of field	(2)

$B_1 = \frac{\mu_0 I}{2R}$	½
$B_2 = \frac{\mu_0 I}{2R}$	½
$B = \sqrt{B_1^2 + B_2^2}$	½
$B = \sqrt{2\left(\frac{\mu_0 I}{2R}\right)^2}$	
$B = \sqrt{2}\,\frac{\mu_0 I}{2R}$	½
Note: Since direction of current is shown clockwise as well as anti-clock wise in the vertical loop. So no marks is assigned for direction of magnetic field.	

2. Two insulated concentric coils, each of radius R, placed at right angles to each other, carry currents I and $\sqrt{3}$ I respectively. The magnitude of the net magnetic field at their common centre will be

(A) $\frac{\mu_o I}{R}$

(B) $\frac{\mu_o I}{2R}$

(C) $\frac{\mu_o I}{4R}$

(D) $\frac{2\mu_o I}{R}$

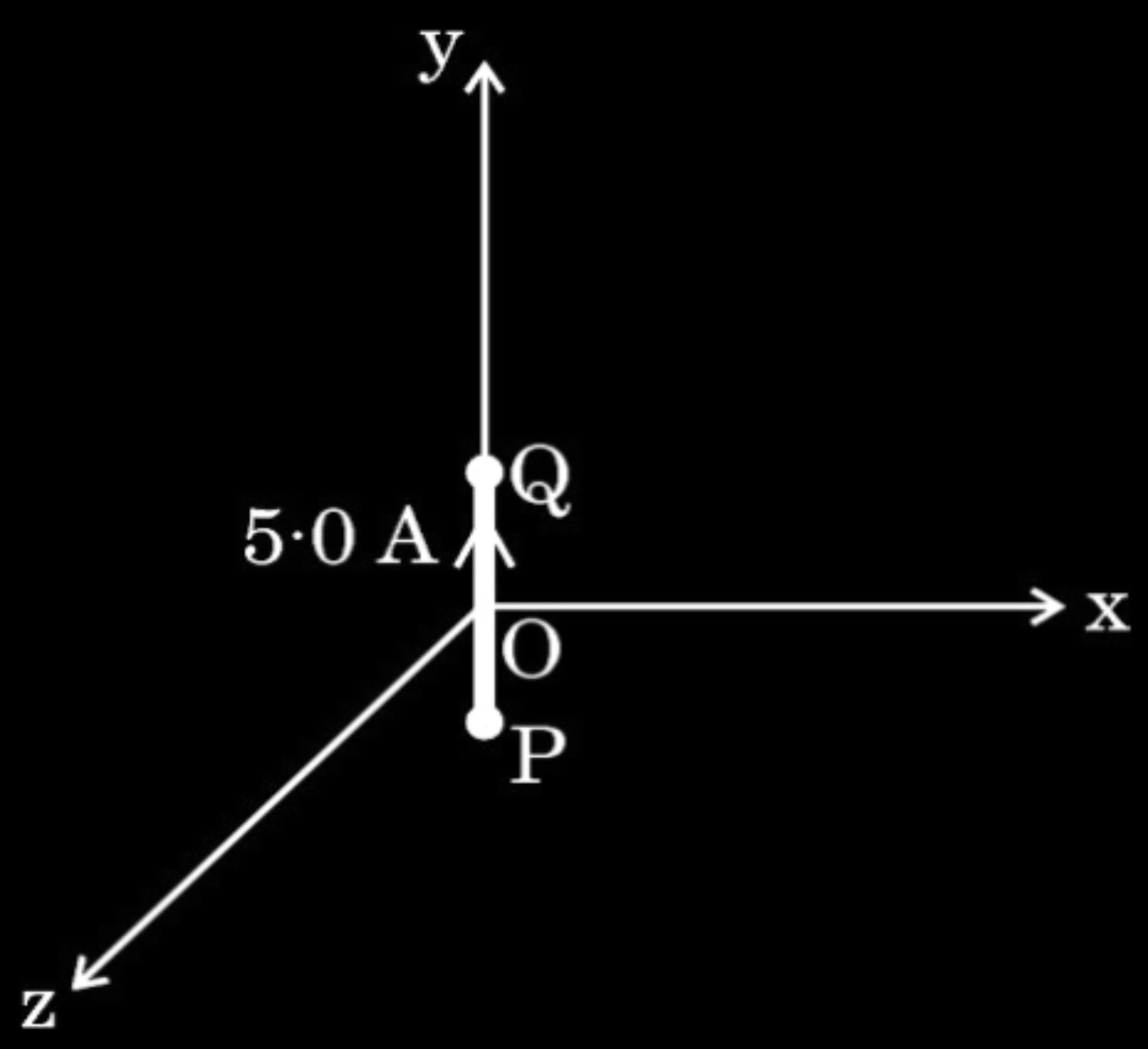
2. Two insulated concentric coils, each of radius R, placed at right angles to each other, carry currents I and $\sqrt{3}$ I respectively. The magnitude of the net magnetic field at their common centre will be

(A) $\frac{\mu_o I}{R}$ (B) $\frac{\mu_o I}{2R}$

(C) $\frac{\mu_o I}{4R}$ (D) $\frac{2\mu_o I}{R}$

(A) $\frac{\mu_0 I}{R}$

3. A 2·0 cm segment of wire, carrying 5·0 A current in positive y-direction lies along y-axis, as shown in the figure. The magnetic field at a point (3 m, 4 m, 0) due to this segment (part of a circuit) is :



(A) $(0{\cdot}12\text{ nT})\ \hat{j}$

(B) $-(0{\cdot}10\text{ nT})\ \hat{j}$

(C) $-(0{\cdot}24\text{ nT})\ \hat{k}$

(D) $(0{\cdot}24\text{ nT})\ \hat{k}$

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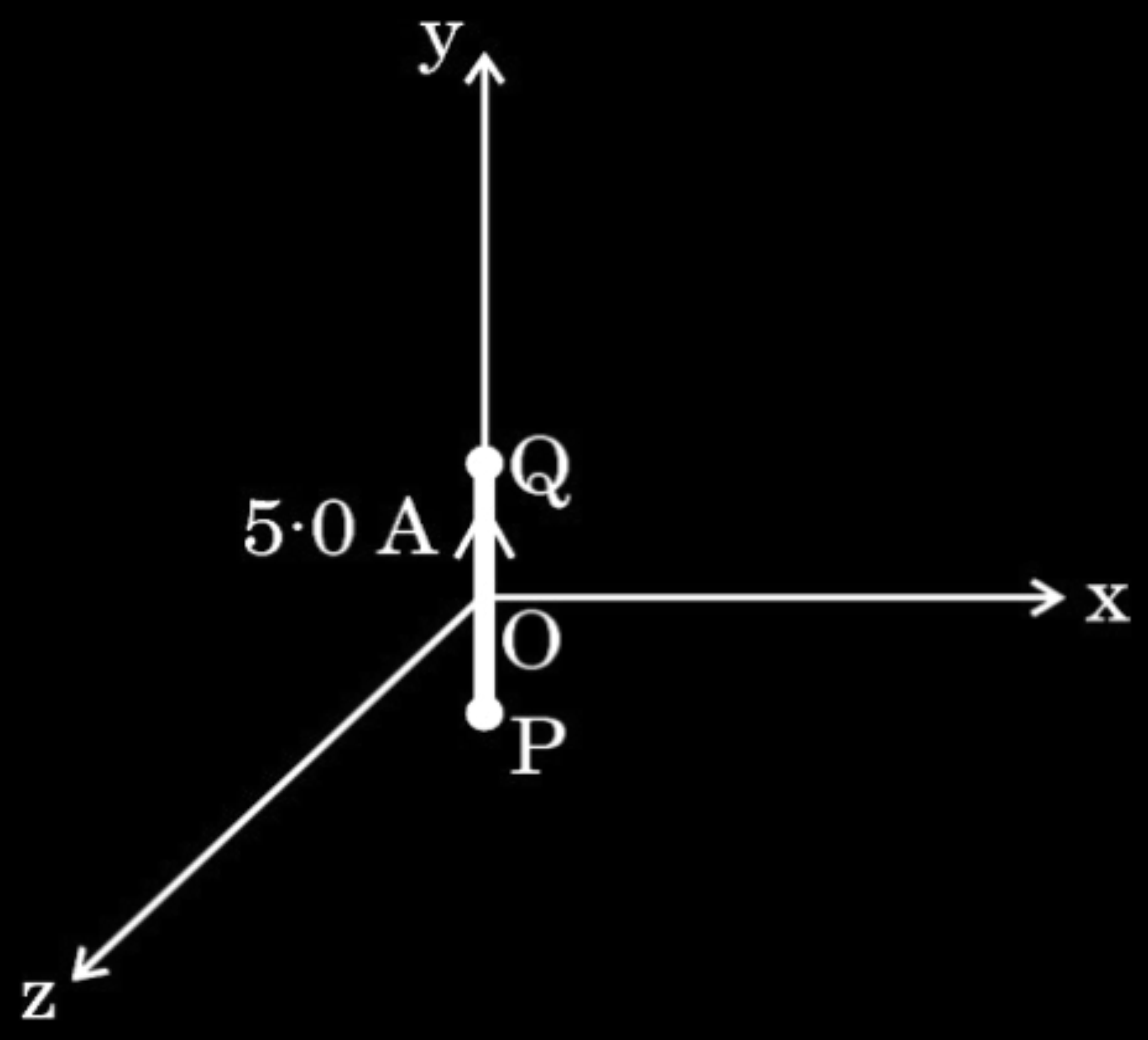

(A) $(0{\cdot}12\ \text{nT})\ \hat{j}$

(B) $-(0{\cdot}10\ \text{nT})\ \hat{j}$

(C) $-(0{\cdot}24\ \text{nT})\ \hat{k}$

(D) $(0{\cdot}24\ \text{nT})\ \hat{k}$

(C) $-(0.24\ \text{nT})\ \hat{k}$

4. A piece of wire bent in the form of a circular loop A carries a current I. The wire is then bent into a circular loop B of two turns and carries the same current. The ratio of magnetic fields at the centre of loop A to that of loop B will be :

(A) $\frac{1}{16}$

(B) 16

(C) 4

(D) $\frac{1}{4}$

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(A) $\frac{1}{16}$

(B) 16

(C) 4

(D) $\frac{1}{4}$

(D) $\frac{1}{4}$

3. The magnetic field due to a current carrying circular coil at a point far off on its axis at a distance ‘r’ from the centre of the coil is proportional to :

(A) $\frac{1}{r}$

(B) $\frac{1}{r^2}$

(C) $\frac{1}{r^{3/2}}$

(D) $\frac{1}{r^3}$

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(A) $\frac{1}{r}$ (B) $\frac{1}{r^2}$

(C) $\frac{1}{r^{3/2}}$ (D) $\frac{1}{r^3}$

(D) $\frac{1}{r^3}$

2. Two circular loops of areas A and 4A carry currents 2I and I respectively. The magnetic fields at their centres will be in the ratio of :

(A) 3 : 1 (B) 4 : 1 (C) 1 : 1 (D) 1 : 2

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(A) 3 : 1 (B) 4 : 1 (C) 1 : 1 (D) 1 : 2

(B) 4:1

27. Two circular coils of radius R each and having equal number of turns N, carry equal currents I in the same direction. They are placed coaxially at a distance $2\sqrt{3}$ R. Find the magnitude and direction of the net magnetic field produced at the midpoint of the line joining their centres. *3*

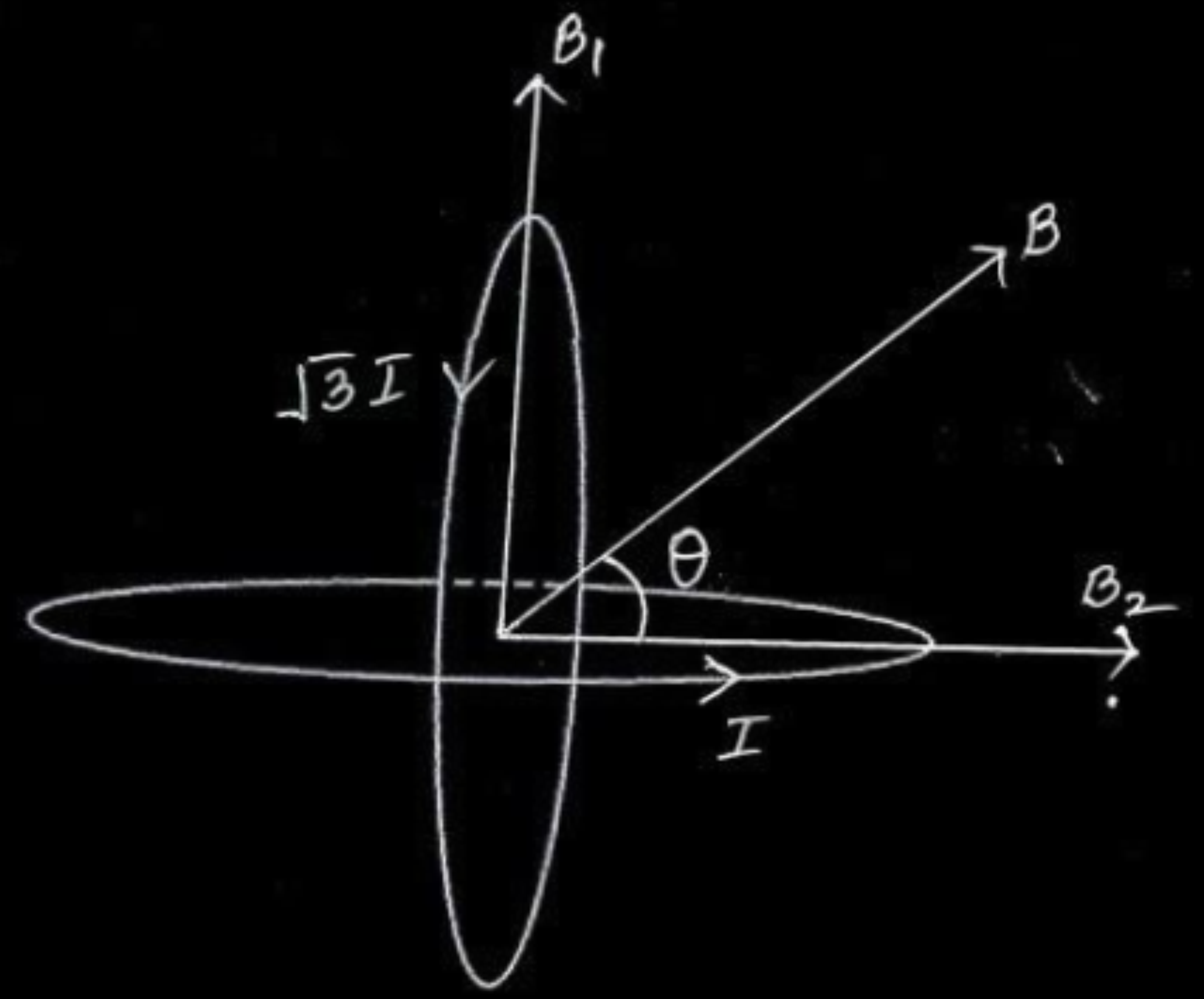
Finding the magnitude of net magnetic field	2
Direction of net magnetic field	1

Magnetic field due to a circular coil having N number of turns at point on the axis of the coil	
$B = \frac{N\mu_0 IR^2}{2(a^2+R^2)^{3/2}}$	½
Net field at mid point $= B_1 + B_2$	½
$= \frac{N\mu_0 IR^2}{(a^2+R^2)^{3/2}}$	
$\left(As \;\; a = \sqrt{3}R\right)$	½
Net magnetic field $B = \frac{N\mu_0 IR^2}{(3R^2+R^2)^{3/2}}$	
$B = \frac{N\mu_0 I}{8R}$	½
Direction of the net field will be towards left or right along the axis depending on direction of current in the two loops.	1

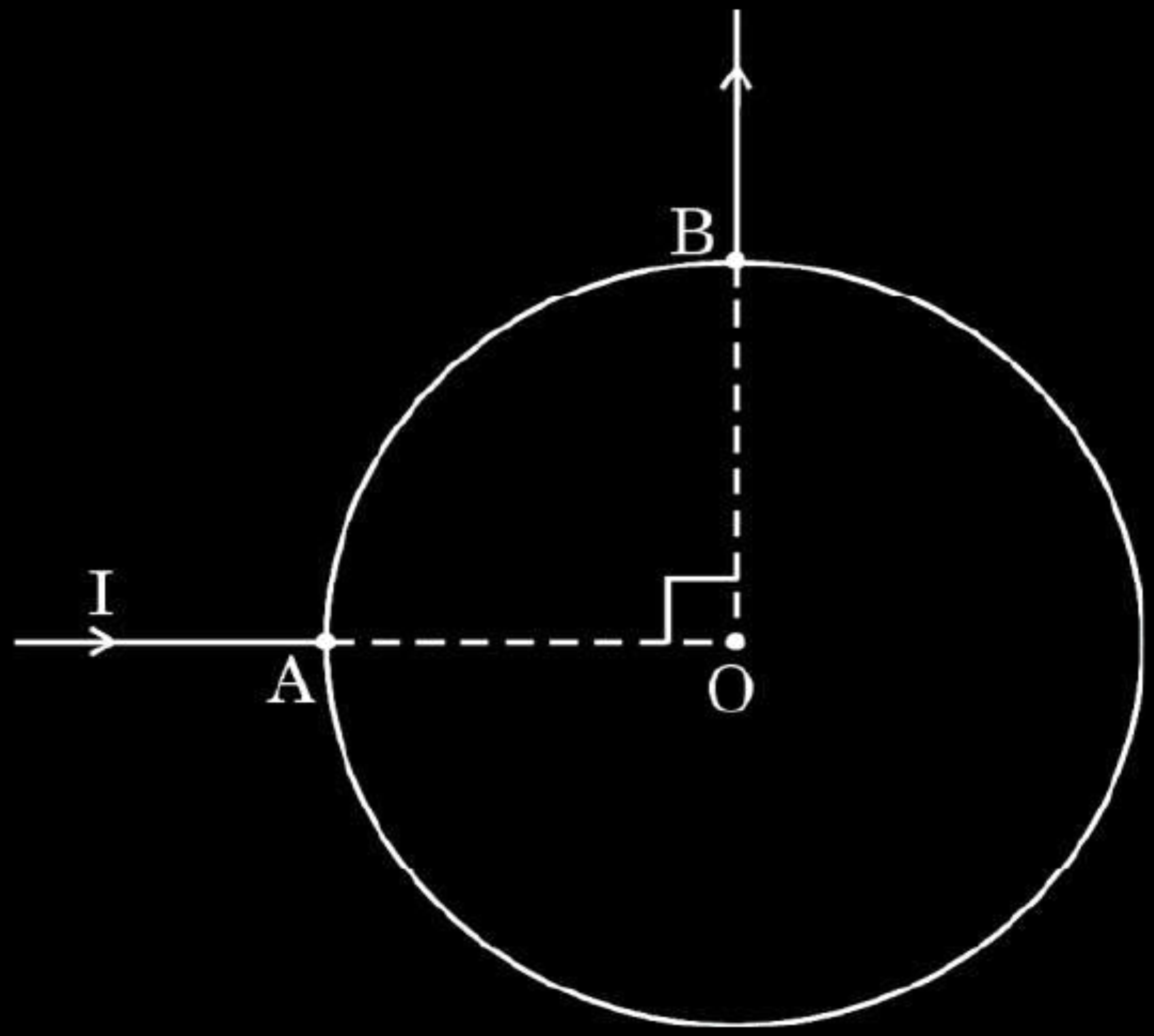
25. (a) (i) Write Biot-Savart's law in vector form.

(ii) Two identical circular coils A and B, each of radius R, carrying currents I and $\sqrt{3}$ I respectively, are placed concentrically in XY and YZ planes respectively. Find the magnitude and direction of the net magnetic field at their common centre. *3*

(i)Writing Biot-Savart's Law in vector form	1
(ii)Finding magnitude & direction of net magnetic field at centre of two current carrying coils	2

(i) $\overrightarrow{dB} = \dfrac{\mu_0}{4\pi}\dfrac{I(\overrightarrow{dl}\times\vec{r})}{r^3}$	1
(ii) $B_1 = \dfrac{\mu_0 I}{2R}$	½
$B_2 = \dfrac{\mu_0 \sqrt{3} I}{2R}$	
$B = \sqrt{B_1^2 + B_2^2}$	½
$\therefore B = \dfrac{\mu_0 I}{2R}\sqrt{1+3}$	
	
$B = \dfrac{\mu_0 I}{R}$	½
$\tan\theta = \dfrac{B_1}{B_2} = \dfrac{1}{\sqrt{3}}$	½
Direction of net magnetic field is 30° with direction of B_2/ 60° with the direction of B_1.	

3. In a circular loop of radius R, current I enters at point A and exits at point B, as shown in the figure. The value of the magnetic field at the centre O of the loop is :



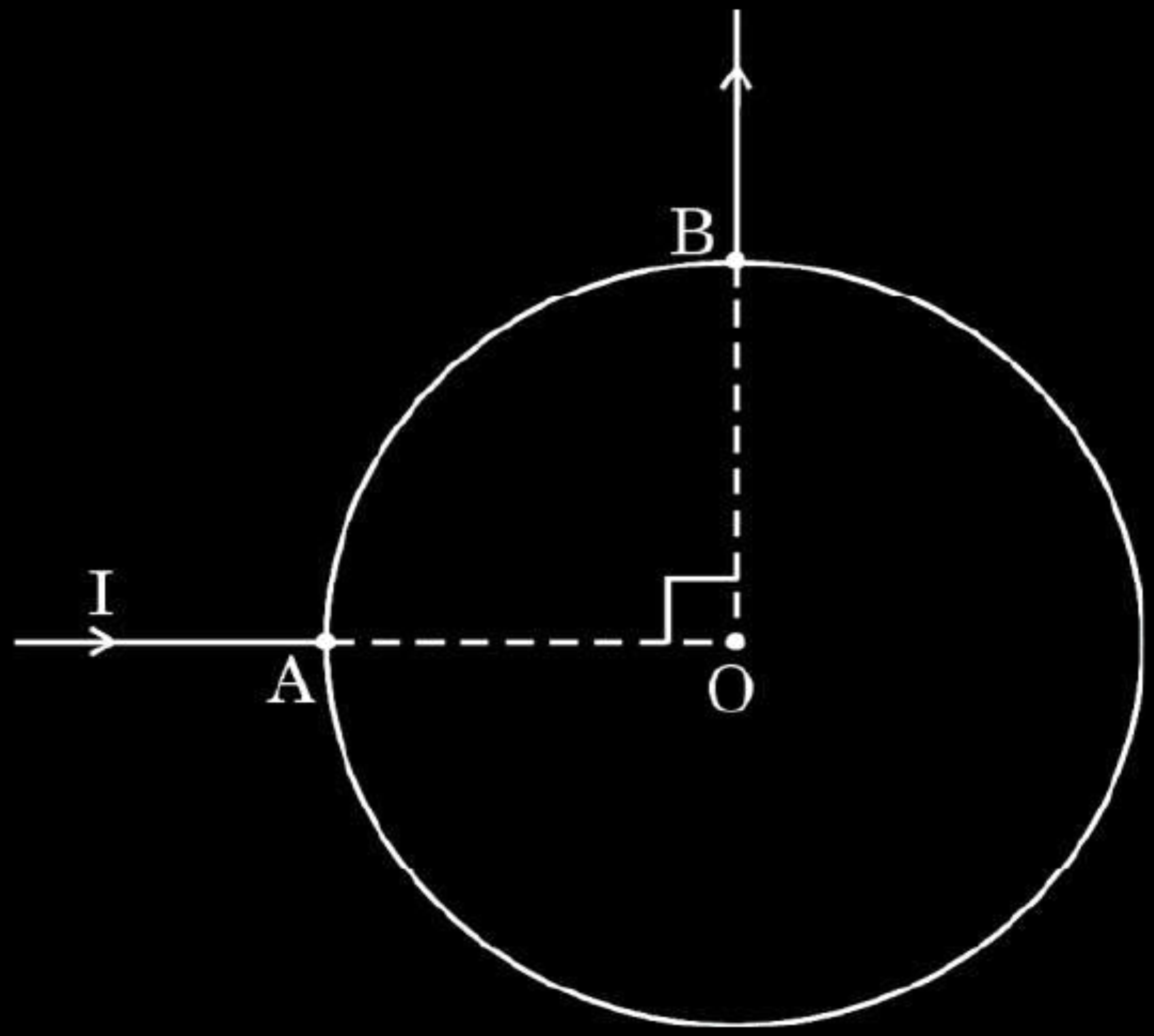
(A) $\frac{\mu_0 I}{R}$

(B) zero

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{4R}$

3. In a circular loop of radius R, current I enters at point A and exits at point B, as shown in the figure. The value of the magnetic field at the centre O of the loop is :



(A) $\frac{\mu_0 I}{R}$

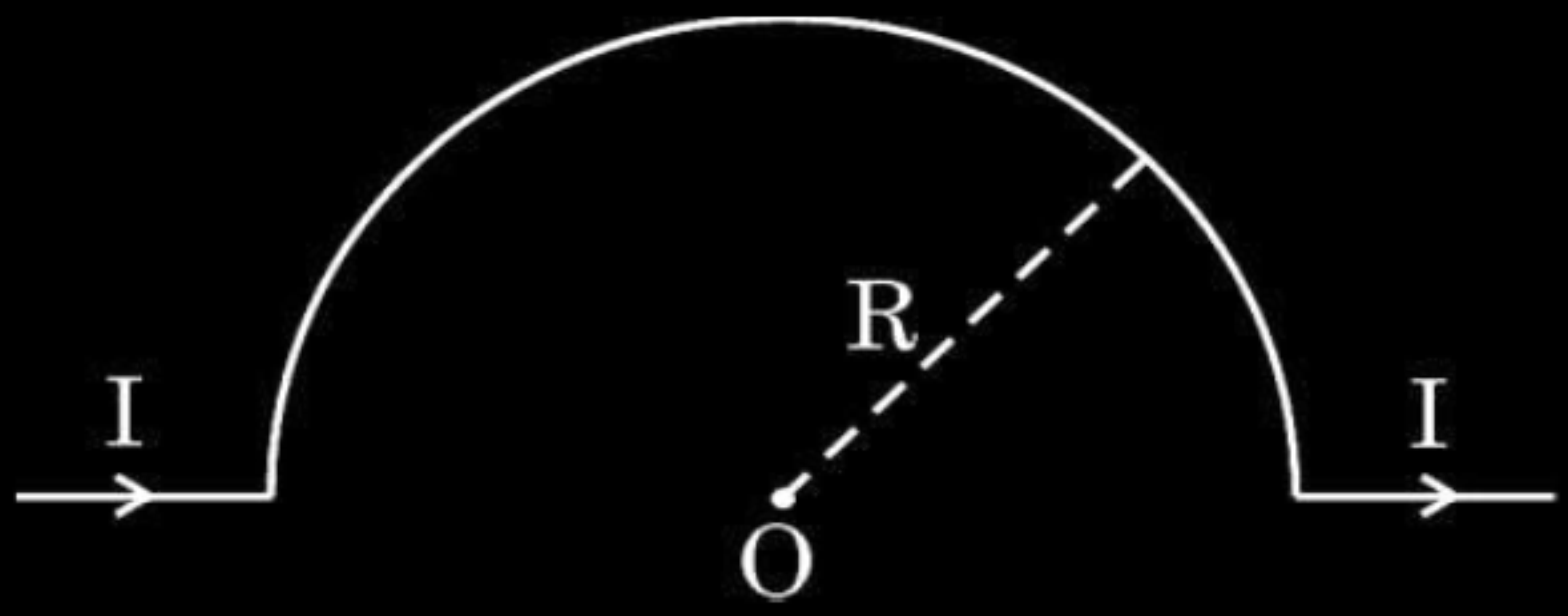
(B) zero

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{4R}$

(B) zero

3. The value of magnetic field at point O in the given figure is :



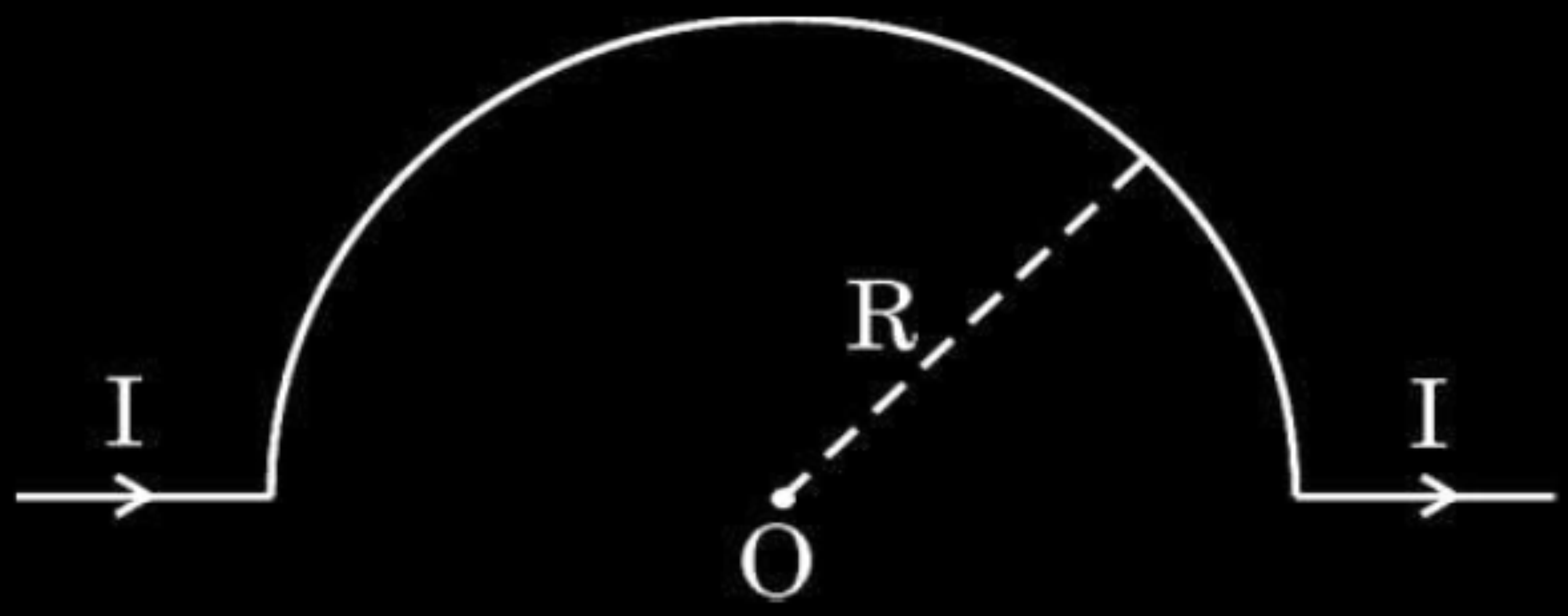
(A) $\frac{\mu_0 I}{2\pi R}$

(B) $\frac{\mu_0 I}{\pi R}$

(C) $\frac{\mu_0 I}{4R}$

(D) $\frac{\mu_0 I}{R}$

3. The value of magnetic field at point O in the given figure is :



(A) $\frac{\mu_0 I}{2\pi R}$

(B) $\frac{\mu_0 I}{\pi R}$

(C) $\frac{\mu_0 I}{4R}$

(D) $\frac{\mu_0 I}{R}$

(C) $\frac{\mu_0 I}{4R}$

23. (a) Write one point of similarity and one point of difference between magnetic field and the electrostatic field. **3**

(b) A wire of length l is first bent into a circular loop and then into a circular coil of two turns. For the same current passing through them, find the ratio of the magnetic fields at their centres.

(a) Writing one	
• Similarity	½
• Difference	½
(b) Finding ratio of magnetic field in two cases	2

(a) **Point of similarity(Any one)** (i) Both obey inverse square law. (ii) Both are long range. (iii) Superposition principle is applicable on both.	½
Point of difference (Any one) (i) The electric field is produced by scalar source , the magnetic field is produced by vector source. (ii) Electric field is radially out ward or inward towards the source charge, the magnetic field is perpendicular to plane. (iii) electric field is due to static charge , magnetic field is due to moving charge.	½
b) Magnetic field at the center of circular coil $B = \frac{N\mu_0 I}{2r}$	½
for coil 1, $l = 2\pi r_1 \Rightarrow r_1 = \frac{l}{2\pi}$	
$B_1 = \frac{1\times\mu_0 I}{2\left(\frac{l}{2\pi}\right)} = \frac{\mu_0 \pi I}{l}$	½
for coil 2 , $l = 2\times 2\pi r_2 \Rightarrow r_2 = \frac{l}{4\pi}$	
$B_2 = \frac{2\times\mu_0 I}{2\times\left(\frac{l}{4\pi}\right)} = \frac{\mu_0 4\pi I}{l}$	½
$\frac{B_1}{B_2} = \frac{1}{4}$	½

18. A current of 5 A is passing along +X direction through a wire lying along X-axis. Find the magnetic field $\vec{B}$ at a point $\vec{r} = (3\hat{i} + 4\hat{j})$ m due to 1 cm element of the wire, centered at origin. *2*

18. A current of 5 A is passing along +X direction through a wire lying along X-axis. Find the magnetic field $\vec{B}$ at a point $\vec{r} = (3\hat{i} + 4\hat{j})$ m due to 1 cm element of the wire, centered at origin. *2*

Finding magnetic field 2 Using Biot – Savart's law $\vec{B} = \frac{\mu_0}{4\pi}\frac{I\vec{dl}\times\vec{r}}{r^3}$ $= \frac{(10^{-7})[5(10^{-2})\hat{i}\times(3\hat{i}+4\hat{j})]}{5^3}$ $= 1.6\times10^{-10}\,\hat{k}\,T$	½ 1 ½

19. A circular coil of wire having 200 turns, each of radius 4·0 cm is placed in a horizontal plane. It carries a current of 0·40 A in clockwise direction. Find the magnitude and direction of the magnetic field at the centre of the coil. *2*

19. A circular coil of wire having 200 turns, each of radius 4·0 cm is placed in a horizontal plane. It carries a current of 0·40 A in clockwise direction. Find the magnitude and direction of the magnetic field at the centre of the coil. *2*

$B = \dfrac{\mu_0 NI}{2r}$	½
$= \dfrac{4\pi \times 10^{-7} \times 200 \times 0.40}{2 \times 4 \times 10^{-2}}$	½
$= 1.256 \times 10^{-3}\ T$	½
Direction is perpendicularly inward into the horizontal plane.	½

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is **1**

(A) $1.0 \times 10^{-9}\,\hat{k}$ T

(B) $-1.0 \times 10^{-9}\,\hat{k}$ T

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10}\,\hat{k}$ T

(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10}\,\hat{k}$ T

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is 1

(A) $1.0 \times 10^{-9}\,\hat{k}$ T

(B) $-1.0 \times 10^{-9}\,\hat{k}$ T

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10}\,\hat{k}$ T

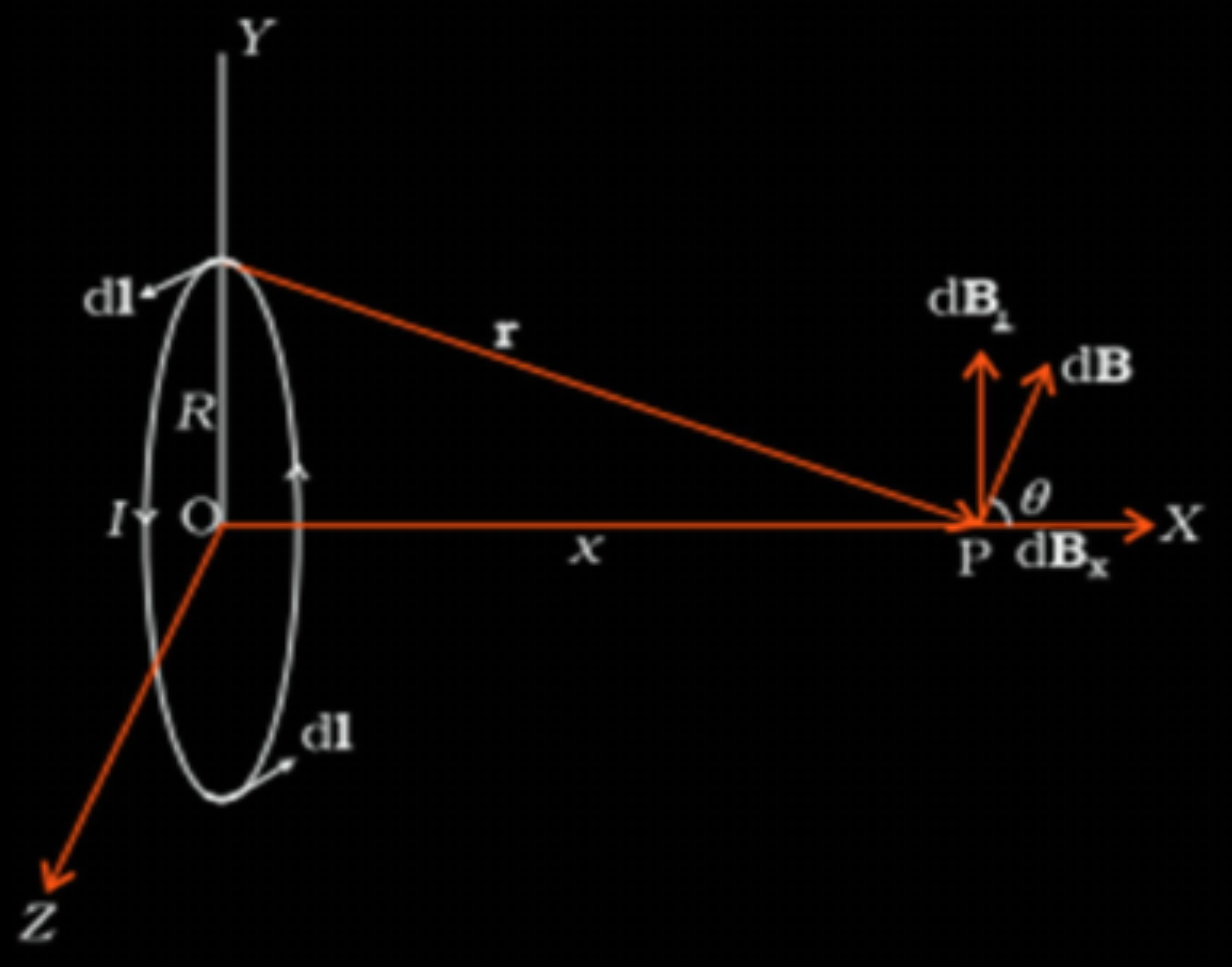
(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10}\,\hat{k}$ T

(C) $\frac{5.0}{\sqrt{2}}$ x $10^{-10}\hat{k}$ T

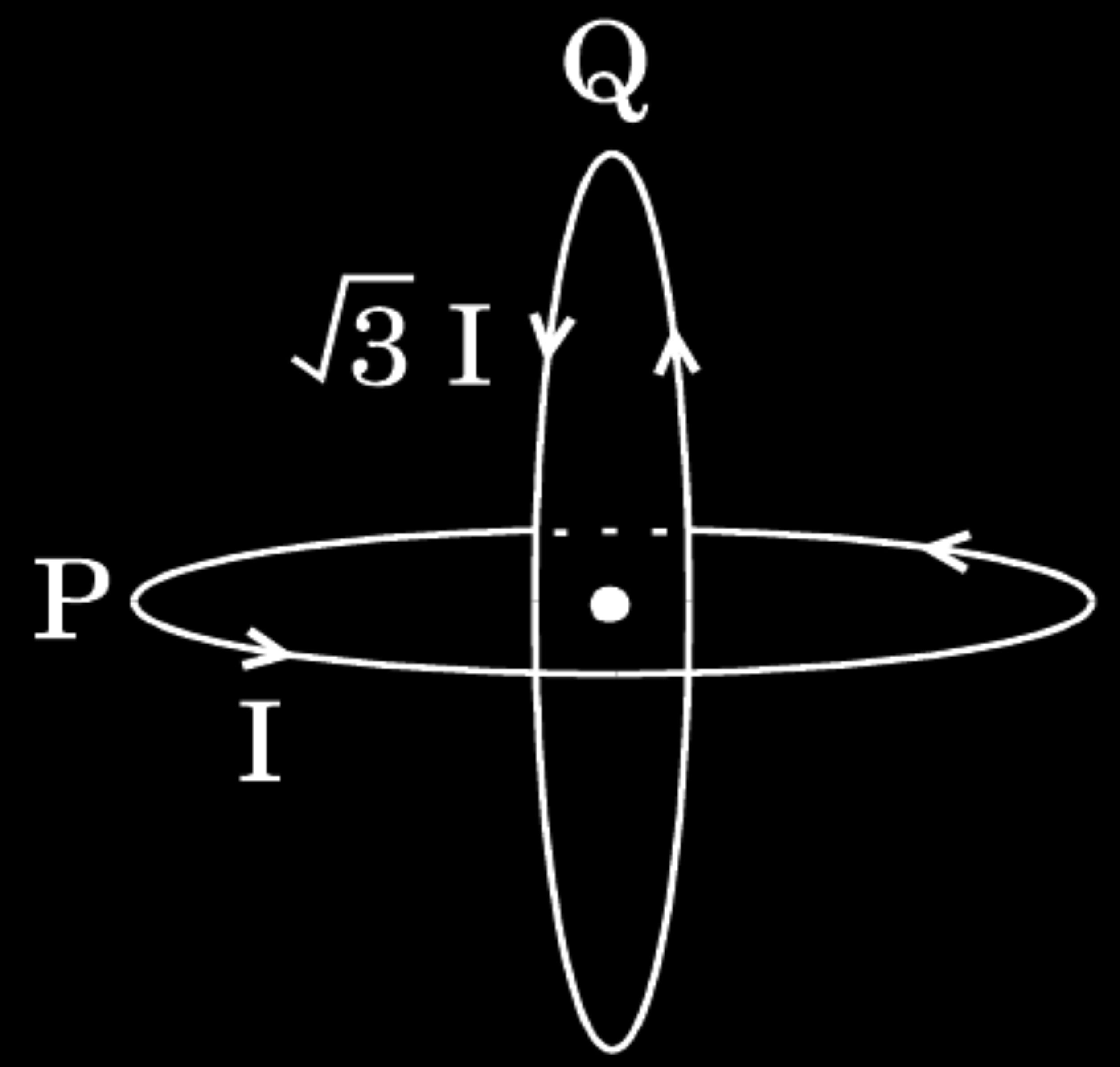
23. Using Biot-Savart law, derive expression for the magnetic field ($\vec{B}$) due to a circular current carrying loop at a point on its axis and hence at its centre. **3**

Derivation for	
Magnetic field on the axis	2 ½
Magnetic field at the centre	½

From Biot Savart' s Law $$\left\|d\vec{B}\right\| = \frac{\mu_\circ}{4\pi}\frac{I\left\|d\vec{l}\times\vec{r}\right\|}{r^3}$$	½
Now $r^2 = x^2 + R^2$ Because $\left\|d\vec{l}\times\vec{r}\right\| = rdl$ $$\therefore dB = \frac{\mu_\circ}{4\pi}\frac{idl}{(x^2+R^2)}$$	½
$d\vec{B}$ has two components. All the components perpendicular to x- axis are summed over and we obtain a null result. Only x- components contribute. The net contribution along x- direction $dB_x = dB\cos\theta$	½
$$\cos\theta = \frac{R}{\left(R^2+x^2\right)^{\frac{1}{2}}}$$ Thus : $$dB_x = \frac{\mu_\circ i}{4\pi}dl\frac{R}{\left(R^2+x^2\right)^{\frac{3}{2}}}$$ Summing dB_x over the entire loop $$\oint dl = 2\pi R$$	
$$\vec{B} = B_x\hat{i} = \frac{\mu_\circ iR^2}{2(x^2+R^2)^{\frac{3}{2}}}\hat{i}$$	½
Magnetic field at the centre of the loop- Here x= 0 $$\therefore \vec{B} = \frac{\mu_\circ i}{2R}\hat{i}$$	½



Two identical coils P and Q each of radius R are lying in perpendicular planes such that they have a common centre. Find the magnitude and direction of the magnetic field at the common centre when they carry currents equal to I and $\sqrt{3}$ I respectively.



Expression for resultant magnetic field -	1½
Direction of resultant field -	½

$$B_1 = \frac{\mu o I}{2R}$$ ½

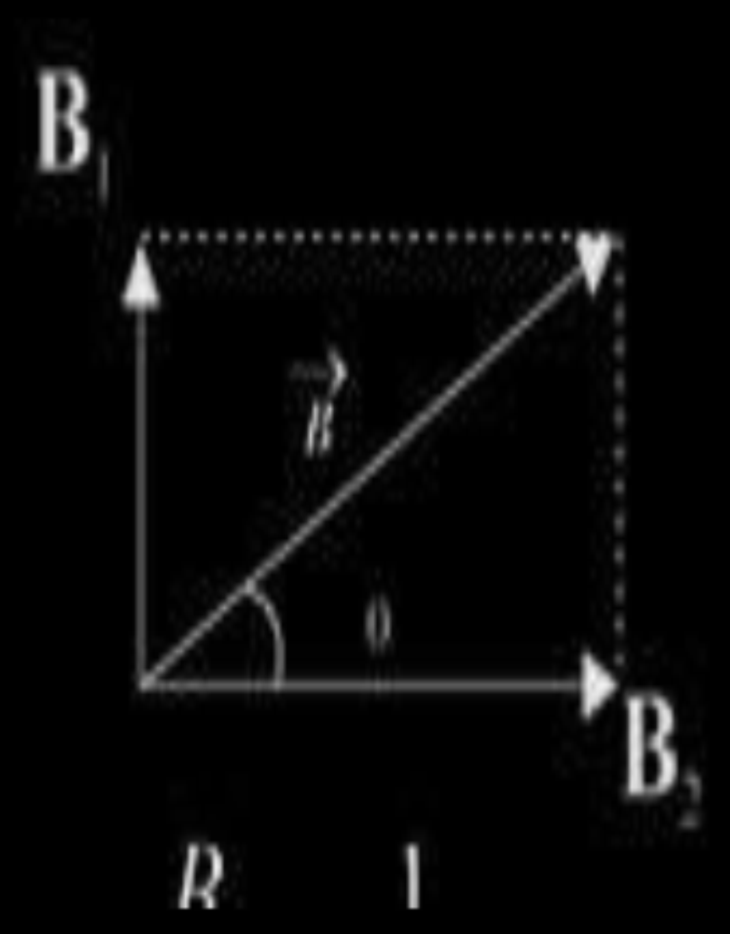
$$B_2 = \frac{\mu o \sqrt{3} I}{2R}$$

$$B = \sqrt{B_1^{\,2} + B_2^{\,2}}$$ ½

$$= \sqrt{\left(\frac{\mu o I}{2R}\right)^2 + \left(\frac{\mu o \sqrt{3} I}{2R}\right)^2}$$

$$B = \frac{\mu o I}{R}$$ ½

For Direction



$$\tan\theta = \frac{B_1}{B_2} \quad \Rightarrow \theta = 30^0$$ ½

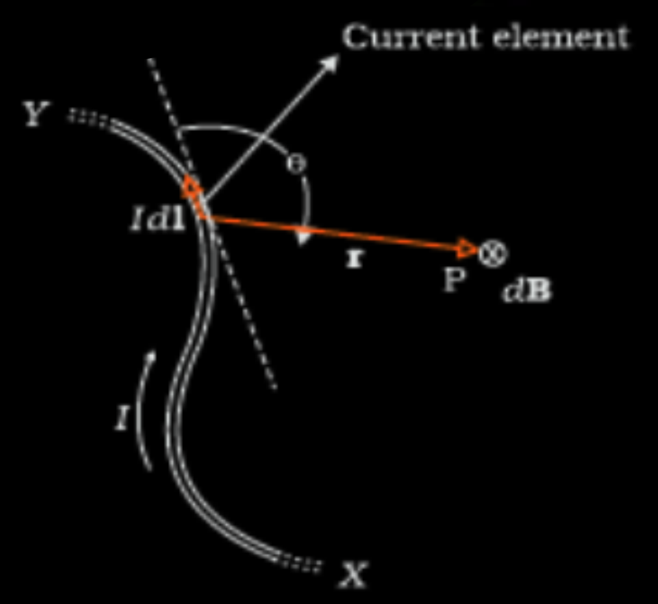
(a) State and explain the law used to determine magnetic field at a point due to a current element. Derive the expression for the magnetic field due to a circular current carrying loop of radius r at its centre.

(b) A long wire with a small current element of length 1 cm is placed at the origin and carries a current of 10 A along the X-axis. Find out the magnitude and direction of the magnetic field due to the element on the Y-axis at a distance 0·5 m from it.

a. Statement of the law and expression for the magnetic field - 1+2
b. Finding the magnitude and direction of magnetic field - 1½ +½

a. According to Biot Savart law, the magnetic field due to a current element is given by

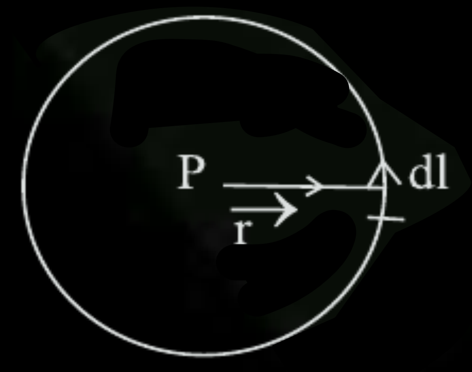
$$\vec{dB} = \frac{\mu_o}{4\pi} I \frac{\vec{dl} \times \vec{r}}{[\vec{r}]^3}$$



1

Alternatively award this 1 mark if a student makes statement of Biot Savart law.

Derivation of magnetic field



½

Magnetic field due to current element dl

$$\vec{dB} = \frac{\mu_o}{4\pi} I \frac{\vec{dl} \times \hat{r}}{[\vec{r}]^2}$$ where $\hat{r}$ is a unit vector along $\vec{r}$

$$\vec{r} \perp \vec{dl}$$

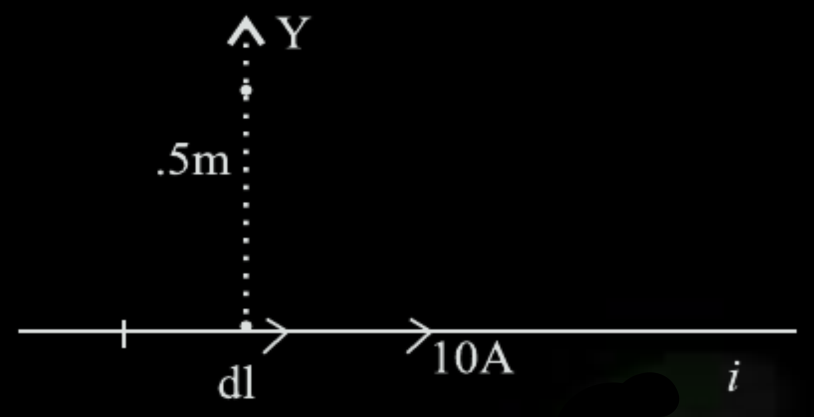
Direction of $\vec{dB}$ is perpendicular, pointing outward. ½

$\therefore$ Field due to the whole loop

$$\left|\vec{B}\right| = \int dB = \frac{\mu_o I}{4\pi r^2} \int dl = \frac{\mu_o I}{4\pi r^2} \times 2\pi r$$ ½

$$\left|\vec{B}\right| = \frac{\mu_o I}{2r}$$ ½

b.



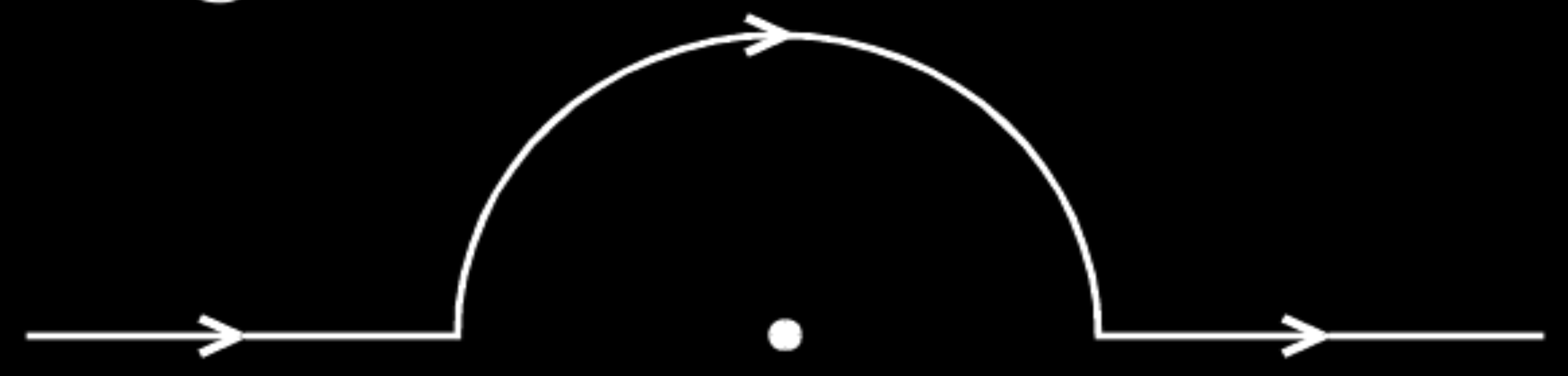
$$\left|\vec{dB}\right| = \frac{\mu_o I}{4\pi} \frac{dl \sin\theta}{[\vec{r}]^2}$$ ½

$$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{10 \times (1 \times 10^{-2}) \times \sin 90^0}{(0.5)^2}$$ 1

$$= 4 \times 10^{-8} T$$ ½

(a) Derive the expression for the magnetic field due to a current carrying coil of radius r at a distance x from the centre along the X-axis.

(b) A straight wire carrying a current of 5 A is bent into a semicircular arc of radius 2 cm as shown in the figure. Find the magnitude and direction of the magnetic field at the centre of the arc.



a)	Derivation of expression with diagram	3
b)	Calculation of magnitude of magnetic field at the center of the arc.	1½
	Direction of field	½

(a) State Biot – Savart law and express it in the vector form.

(b) Using Biot – Savart law, obtain the expression for the magnetic field due to a circular coil of radius r, carrying a current I at a point on its axis distant x from the centre of the coil.

(a) State Biot – Savart law and express this law in the vector form.

(b) Two identical circular coils, P and Q each of radius R, carrying currents 1 A and $\sqrt{3}$ A respectively, are placed concentrically and perpendicular to each other lying in the XY and YZ planes. Find the magnitude and direction of the net magnetic field at the centre of the coils.

(a) Statement of Biot Savart law	1
Expression in vector form	½
(b) Magnitude of magnetic field at centre	1
Direction of magnetic field	½

(a) It states that magnetic field strength, $d\boldsymbol{B}$, due to a current element, $Id\vec{\boldsymbol{l}}$, at a point, having a position vector **r** relative to the current element, is found to depend (i) directly on the current element, (ii) inversely on the square of the distance $\lvert\mathbf{r}\rvert$, (iii) directly on the sine of angle between the current element and the position vector **r**.	1
In vector notation, $$\overrightarrow{d\boldsymbol{B}} = \frac{\mu_0}{4\pi}\frac{Id\vec{\boldsymbol{l}} \times \vec{\boldsymbol{r}}}{\lvert\vec{\boldsymbol{r}}\rvert^3}$$ **Alternatively**, $$\left(d\vec{\boldsymbol{B}} = \frac{\mu_0}{4\pi}\frac{Id\vec{\boldsymbol{l}} \times \hat{r}}{\lvert\vec{r}\rvert^2}\right)$$	½
(b) $B_p = \frac{\mu_0 \times 1}{2R} = \frac{\mu_0}{2R}$ (along z – direction) $B_Q = \frac{\mu_0 \times \sqrt{3}}{2R} = \frac{\mu_0\sqrt{3}}{2R}$ (along x – direction)	½
$$\therefore B = \sqrt{{B_p}^2 + {B_Q}^2} = \frac{\mu_0}{R}$$	½
This net magnetic field **B**, is inclined to the field $\mathbf{B_p}$, at an angle Θ, where $$\tan\theta = \sqrt{3}$$ $$\left(/\,\theta = \tan^{-1}\sqrt{3} = 60^0\right)$$ (in XZ plane)	½

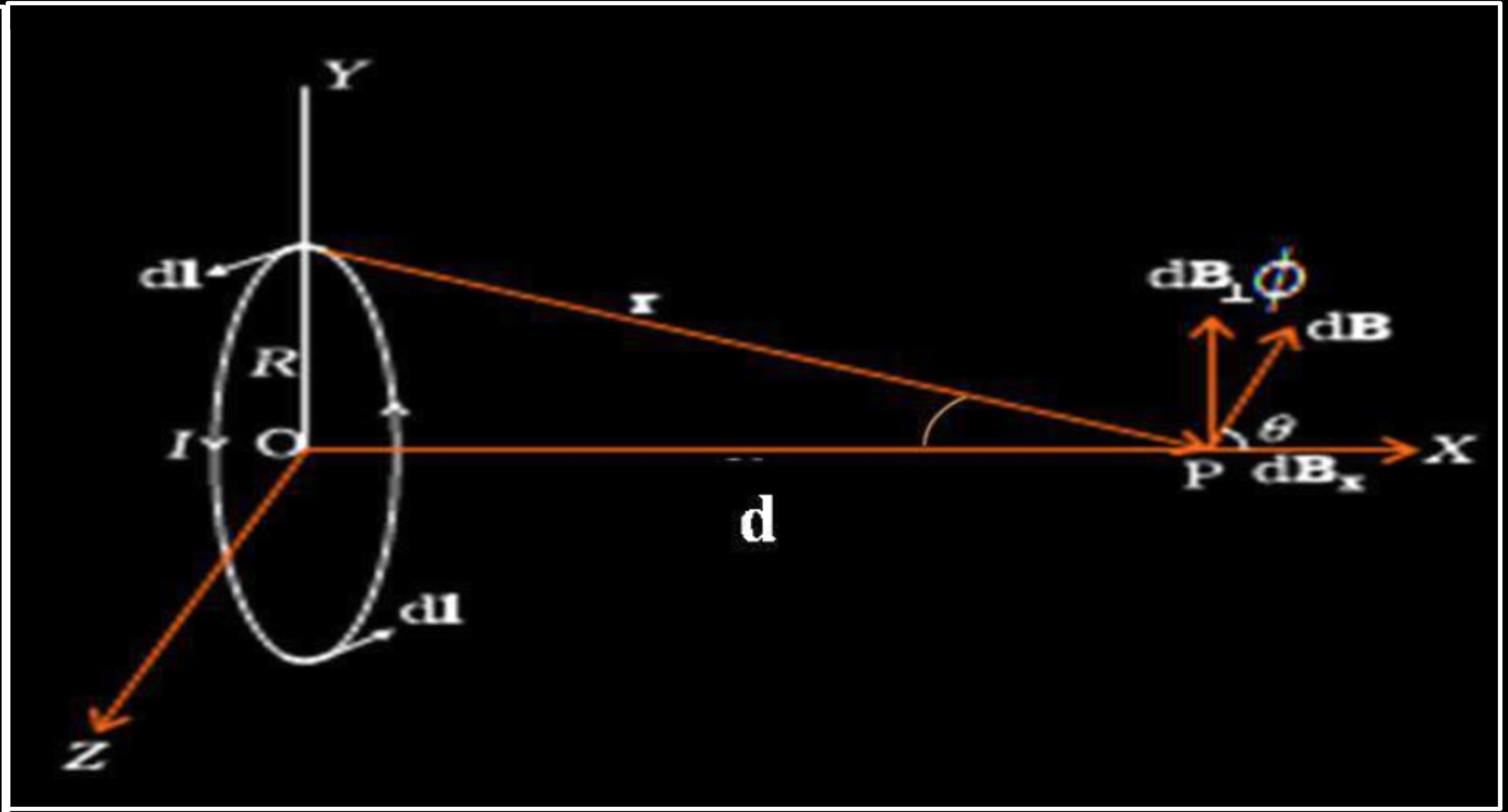
(i) Express Biot-Savart law in the vector form.

(ii) Use it to obtain the expression for the magnetic field at an axial point, distance d from the centre of a circular coil of radius R carrying current I.

(iii) Also, find the ratio of the magnitudes of the magnetic field of this coil at the centre and at an axial point for which $x = R\sqrt{3}$.

(i) Biot-Savart law in vector form	1
(ii) Deriving an expression for the magnetic field at a point on the axial line of current carrying coil	3
(iii) Ratio of magnetic field at the centre and given outside point	1

(i) $\overrightarrow{dB} = \frac{\mu_o I \overrightarrow{d\ell}\times\hat{r}}{4\pi r^2} = \frac{\mu_o I \overrightarrow{d\ell}\times\vec{r}}{4\pi r^3}$	1
(ii) $dB = \frac{\mu_0 Idl\sin\theta}{4\pi r^2}$ here θ =90; $dB = \frac{\mu_0 Idl}{4\pi r^2}$	½
$= dB\ Sin\phi = \frac{\mu_o I d\ell}{4\pi r^2}\sin\phi$	½
$B = \int_0^R \frac{\mu_o Idl}{4\pi r^2}\sin\varphi = \frac{\mu_o I(2\pi R^2)}{4\pi r^3}$	½+ ½
$B = \frac{\mu_o NI(R^2)}{2r^3} = \frac{\mu_o NIR^2}{2(R^2+d^2)^{\frac{3}{2}}}$	½
(i) Magnetic field at the centre of the coil $B_1 = \frac{\mu_o NI}{2R}$	
Magnetic field at the outside point $B_2 = \frac{\mu_o NIR^2}{2[R^2+3R^2]^{\frac{3}{2}}} = \frac{\mu_o NIR^2}{2[4R^2]^{\frac{3}{2}}} = \frac{\mu_o NI}{2*8R}$	½
$\frac{B_1}{B_2} = 8$	½
[Note : If the student takes r= $\sqrt{3}$R, the ratio of B centre to B axial would be 3 $\sqrt{3}$: 1. Award 1 mark in this case also.]	2



Use Biot-Savart law to derive the expression for the magnetic field on the axis of a current carrying circular loop of radius R.

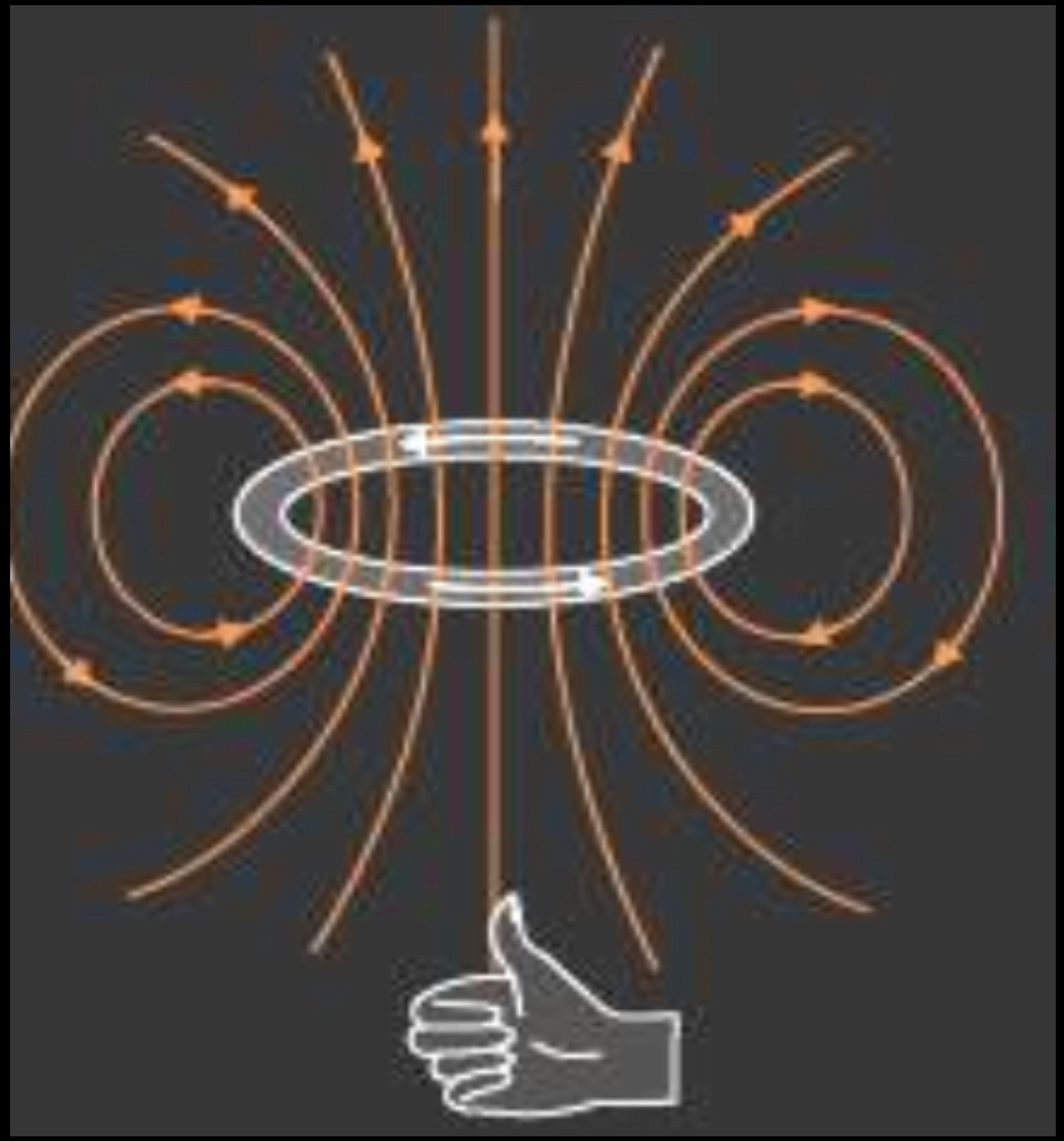
Draw the magnetic field lines due to a circular wire carrying current I.

Explanation for magnetic field on the axis of current loop	2
Drawing- magnetic field lines	1

Use Biot-Savart law to derive the expression for the magnetic field on the axis of a current carrying circular loop of radius R.

Draw the magnetic field lines due to a circular wire carrying current I.

(ii)



Explanation for magnetic field on the axis of current loop	2
Drawing- magnetic field lines	1